Deep Learning with Neural Networks

Autoregressive models

Alejandro Lancho Serrano, alancho@ing.uc3m.es

A quick recap on probability theory

- Two main rules that we need to remember

Marginalization rule: $p(x) = \sum_y p(x, y)$

Product rule: $p(x, y) = p(y \mid x)p(x) = p(x \mid y)p(y)$ ← Bayes Theorem!

- How to model **any** joint distribution over observed data?

Product rule!

$$p(\mathbf{x}) = p(x_1) \prod_{d=2}^{D} p(x_d \mid \mathbf{x}_{<d})$$

where $\mathbf{x}_{<d} = [x_1, \ldots, x_{d-1}]^T$

Deep belief networks (a.k.a. deep autoregressive models)

- Explicit model conditional probabilities

$$p_{\text{model}}(\boldsymbol{x}) = p_{\text{model}}(x_1) \prod_{i=2}^{n} p_{\text{model}}(x_i | x_1, \ldots, x_{i-1})$$

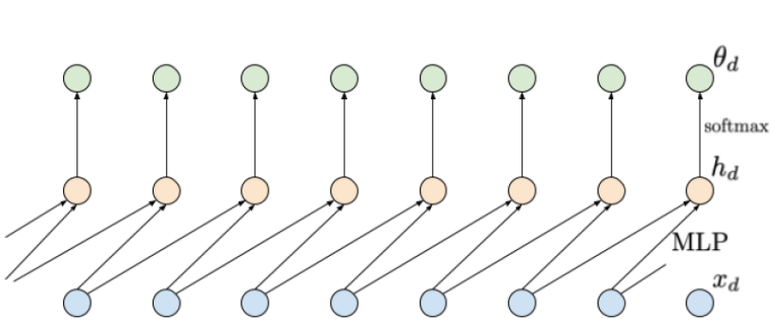
Each conditional can be a neural network

- Maximum likelihood training

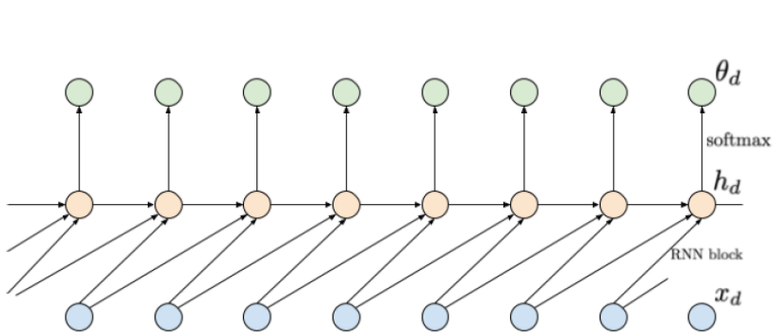
$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \log p_{\text{model}}(\mathbf{x}_n)$$

Modeling all conditional probabilities separately can be problematic!

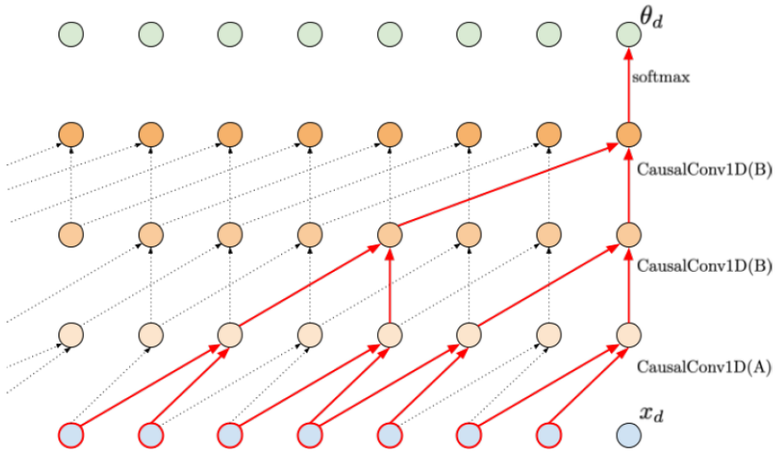
Possible ways to model conditional probabilities efficiently



Finite memory



Infinite memory (RNNs)



Infinite memory (CNNs)

More up to date, **Transformers**!

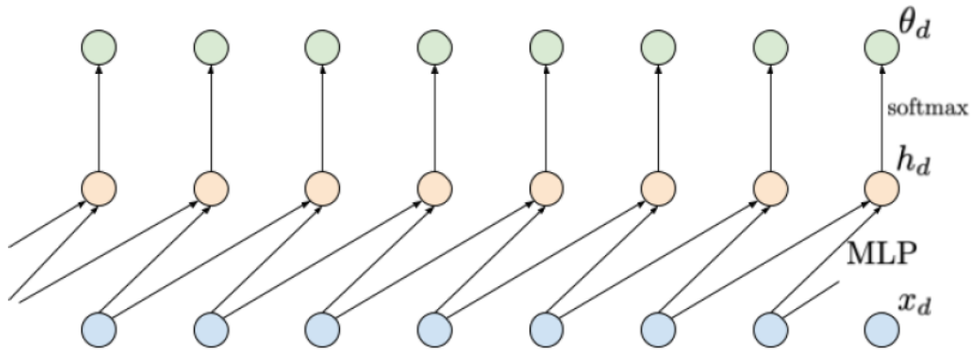
Finite memory

- We can express conditionals assuming **finite memory**

  Finite memory: e.g., 2 variables

  A share parameterization: e.g., an single MLP

  Why one?
  - **Weight sharing** across time
  - **Generalization** and **efficiency**
  - **Sequential dependency** modeling



Then, an example would be as follows:

$$p(x_d \mid x_{d-1}, x_{d-2}) = \text{Cat}(\text{softmax}(\text{NN}(x_{d-1}, x_{d-2})))$$

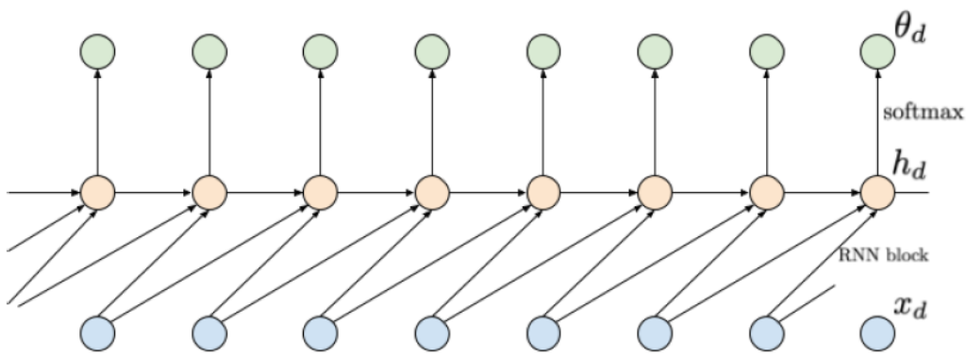
But, limiting as we need to decide for the length of the memory

Infinite memory: RNNs

- We can **model infinite memory** by using **RNNs**

  Infinite memory by **sharing context**

  A shared parameterization through an RNN



For example:

$$p(x_d | x_{d-1}, x_{d-2}) = \text{Cat}(\text{softmax}(\text{RNN}(x_{d-1}, h_{d-1})))$$

No need to define the memory length

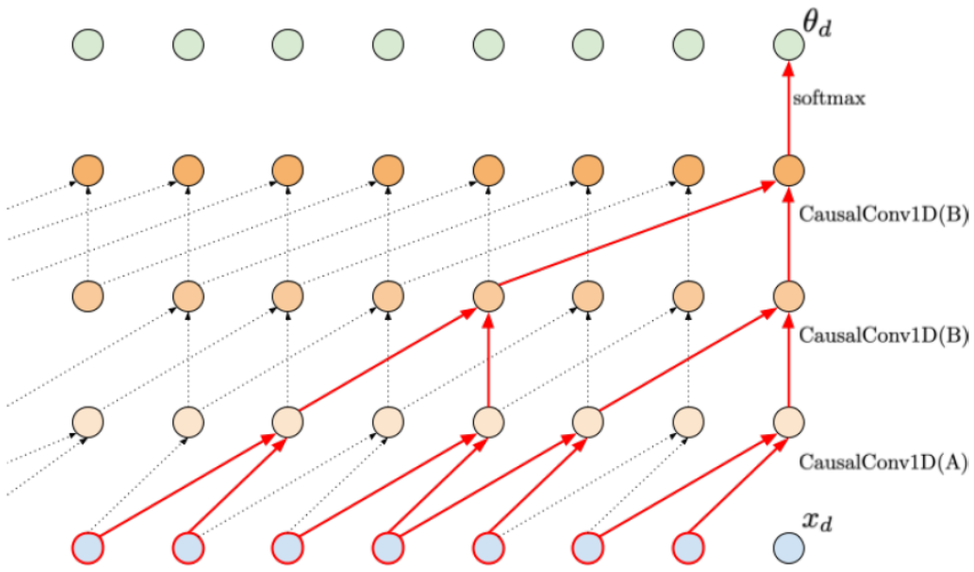
Training RNNs is sometimes problematic

Infinite memory: CNNs

- We can **model infinite memory** by using **CNNs**

  Infinite memory by **applying causal convolution**

  A shared parameterization through a CNN



For example:

$$p(x_d \mid x_{d-1}, x_{d-2}) = \text{Cat}(\text{softmax}(\text{CNN}(\mathbf{x}_{<d})))$$

No need to define the memory length

Easy training!

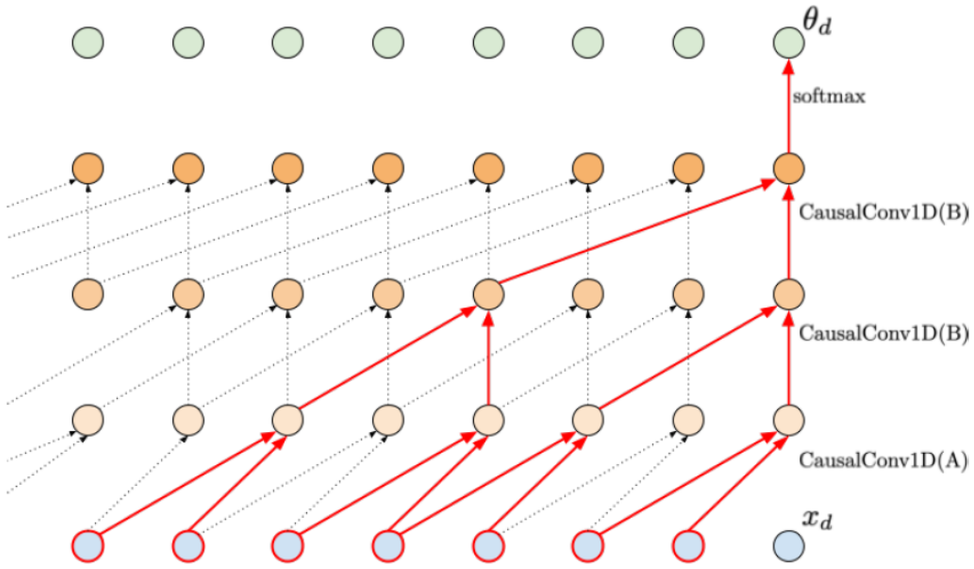
Infinite memory: CNNs

- **How to build** an autoregressive model using CNNs?

Consider $\mathbf{x} = [x_1, \ldots, x_D]^T$

We use causal 1D convolutions:

- **Option A**: Do not use x_d
- **Option B**: Use all inputs
- **Never look into the future!**



Additional tip: Use **dilation** if you want to enlarge dependencies

Infinite memory: CNNs

- **Causal convolutions**

```
class CausalConv1d(nn.Module):
    def __init__(self, in_channels, out_channels, kernel_size, dilation, A=False):
        super(CausalConv1d, self).__init__()

        self.kernel_size = kernel_size
        self.dilation = dilation  # important to get larger dependencies
        self.A = A  # option A (A=True) or option B (B=False)
        self.padding = (kernel_size - 1) * dilation + A * 1

        self.conv1d = nn.Conv1d(in_channels, out_channels,
                                kernel_size, stride=1, padding=0,
                                dilation=dilation)  # we pad later

    def forward(self, x):
        x = nn.functional.pad(x, (self.padding, 0))  # padding only on the left
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Infinite memory: CNNs

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Option A:
kernel size = 2
dilation = 1
A = True

Pad first

Infinite memory: CNNs

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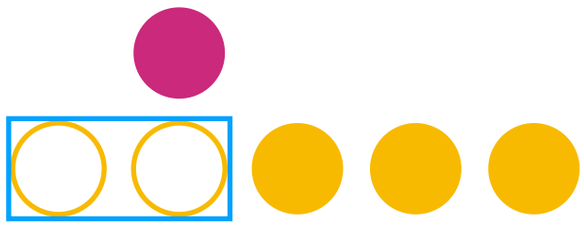

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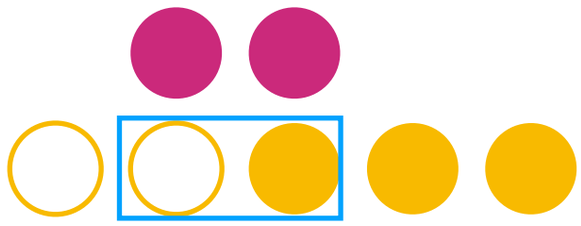

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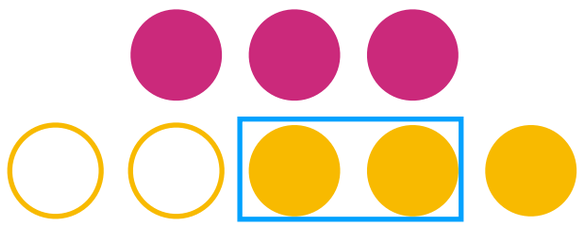

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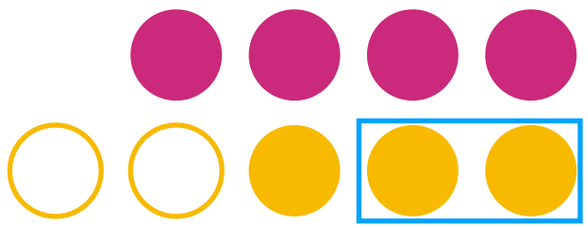

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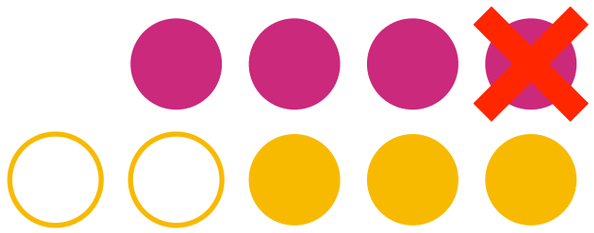

Since it's **Option A**,
we skip the last element from the output

Infinite memory: CNNs

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Option A:
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Option B:
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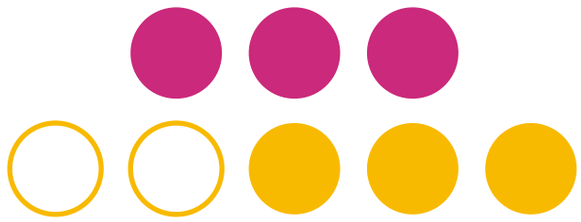
A = True



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Option A:
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A = True

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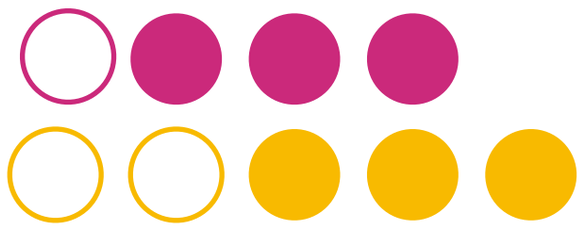

It's **Option B** now
We pad again from the left

Infinite memory: CNNs

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Option A:
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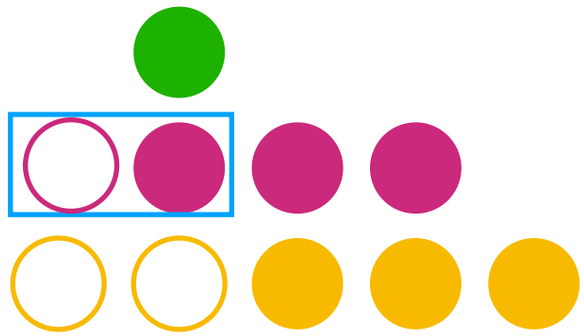

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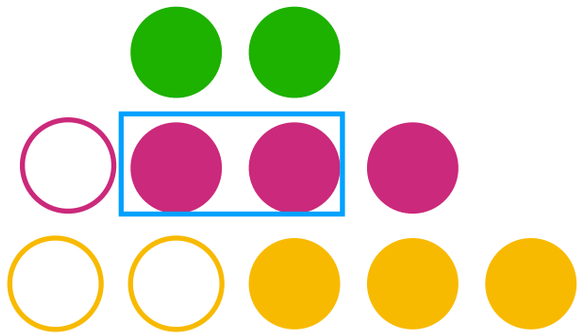

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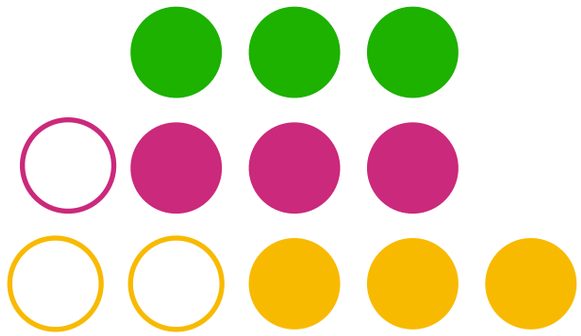

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Option A:
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Option B:
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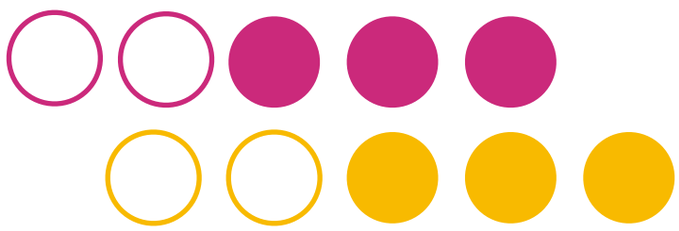
dilation = 2
A = True

Pad again



It's **Option B** now

with dilation 2!

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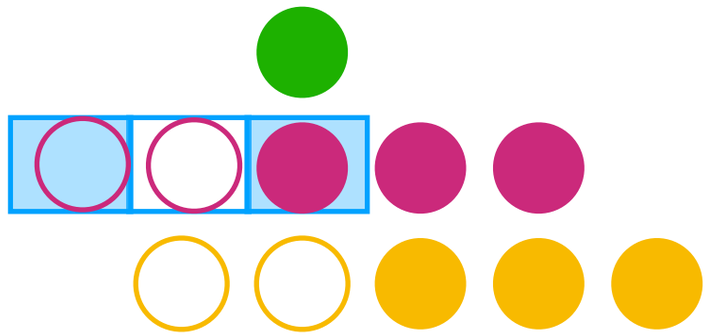
        self.conv1d = nn.Conv1d(in_channels, out_channels,
                                kernel_size, stride=1, padding=0,
                                dilation=dilation)  # we pad later

def forward(self, x):
    x = nn.functional.pad(x, (self.padding, 0))  # padding only on the left
    conv1d_out = self.conv1d(x)
    if self.A:
        return conv1d_out[:, :, : -1]  # NO dependency on the current element!
    else:
        return conv1d_out
```

Option A:
kernel size = 2
dilation = 1
A = True

Option B:
kernel size = 2
dilation = 2
A = True



It's **Option B** now

with dilation 2!

Infinite memory: CNNs

- **Causal convolutions**

```
class CausalConv1d(nn.Module):
    def __init__(self, in_channels, out_channels, kernel_size, dilation, A=False):
        super(CausalConv1d, self).__init__()

        self.kernel_size = kernel_size
        self.dilation = dilation  # important to get larger dependencies
        self.A = A  # option A (A=True) or option B (B=False)
        self.padding = (kernel_size - 1) * dilation + A * 1

        self.conv1d = nn.Conv1d(in_channels, out_channels,
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dilation = 1
A = True

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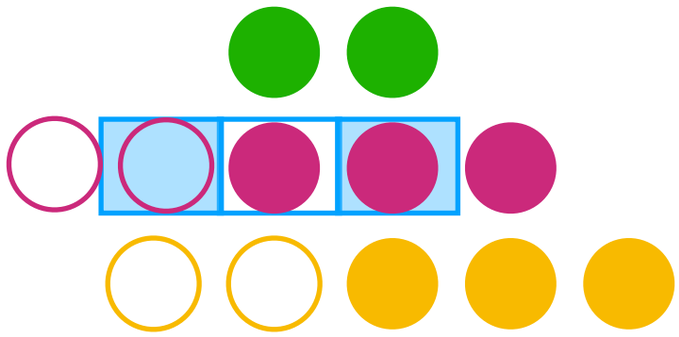
kernel size = 2
dilation = 2
A = True



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Infinite memory: CNNs

- **Causal convolutions**

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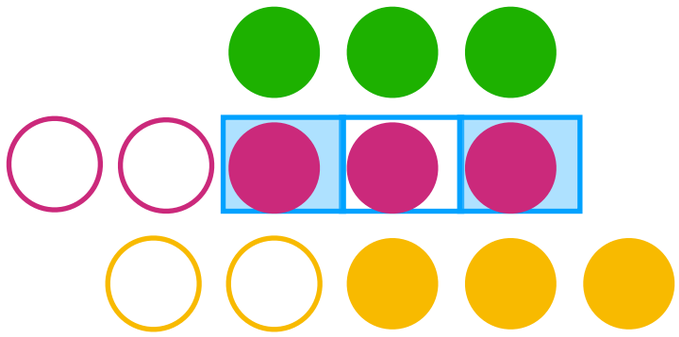
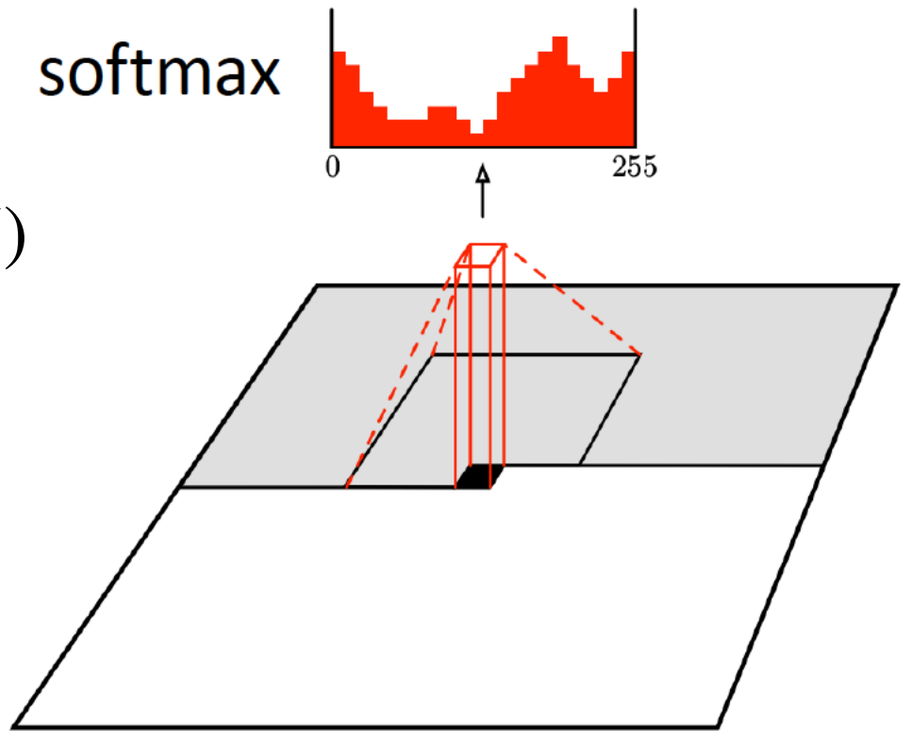


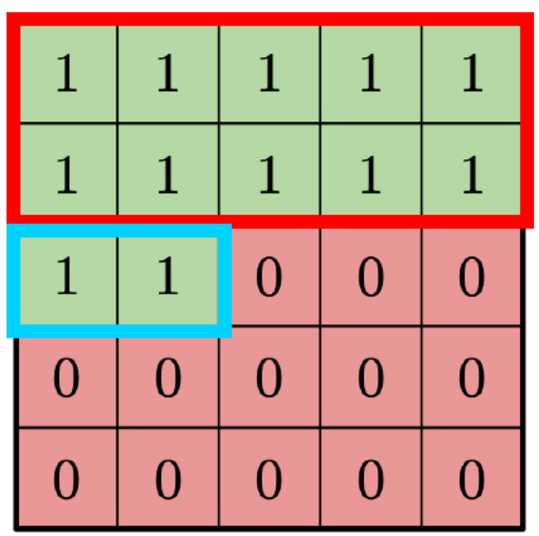
It's **Option B** now

with dilation 2!

We can use causal convolutions in 2D: PixelCNN

- We can use the very same idea with images and Conv2D
- Need to remember about **Causal Convolutions**
- We should remember about 3 dimensions $(C \times H \times W)$

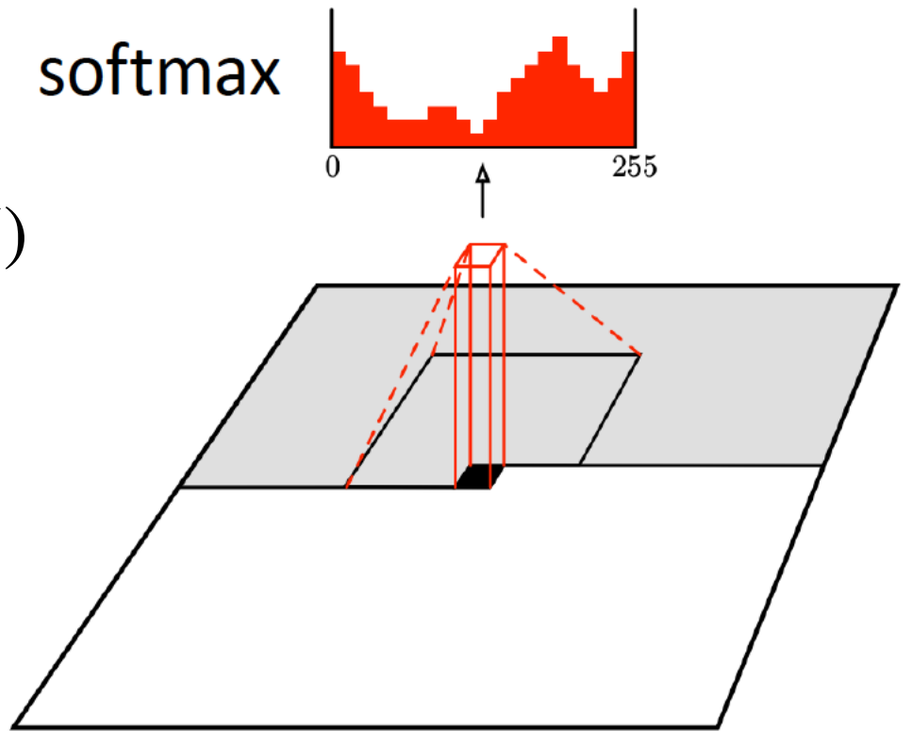




Van den Oord et al.,ICML2016
"Conditional image generation with PixelCNNdecoders"

We can use causal convolutions in 2D: PixelCNN

- We can use the very same idea with images and Conv2D
- Need to remember about **Causal Convolutions**
- We should remember about 3 dimensions $(C \times H \times W)$
- Accomplish this with two Conv2D layers:
  - The first Conv2D layer covers the **upper part**
  - The second Conv2D layer covers the **left part**



1	1	1	1	1
1	1	1	1	1
1	1	0	0	0
0	0	0	0	0
0	0	0	0	0

Van den Oord et al.,ICML2016
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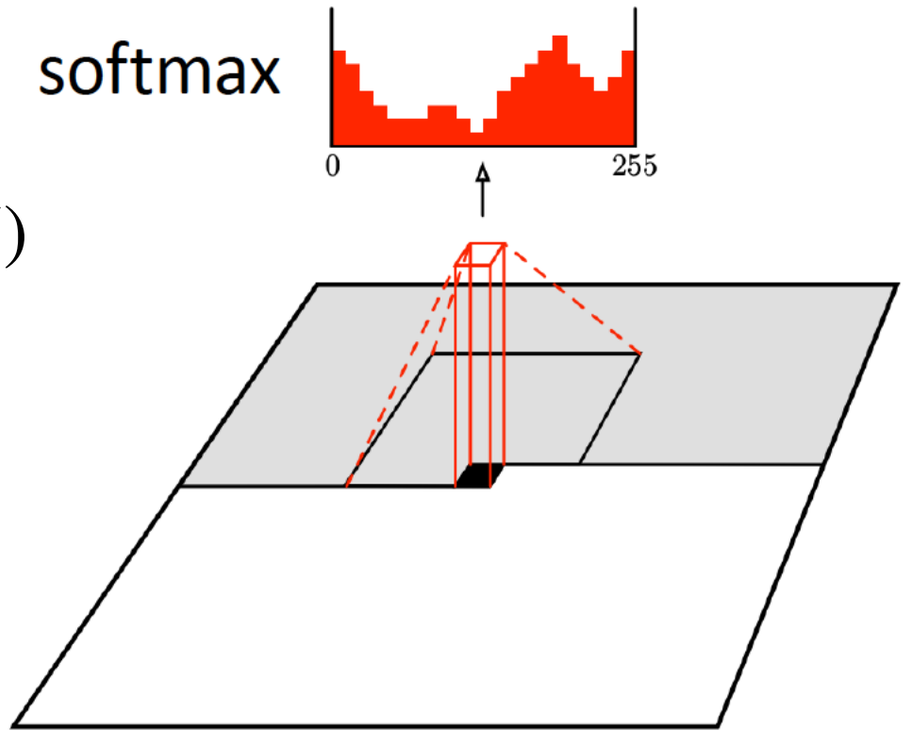
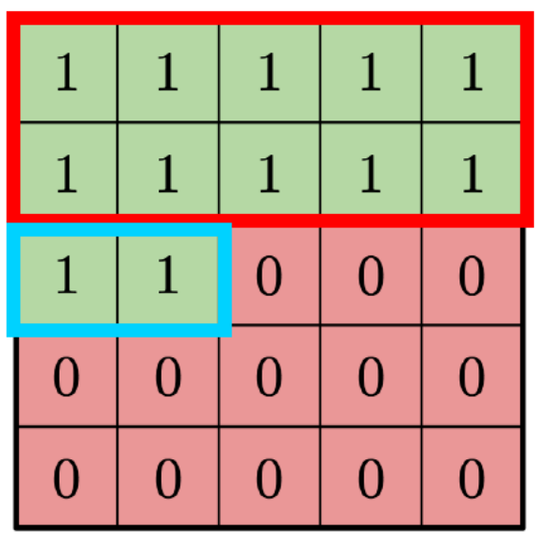
We can use causal convolutions in 2D: PixelCNN

- We can use the very same idea with images and Conv2D
- Need to remember about **Causal Convolutions**
- We should remember about 3 dimensions $(C \times H \times W)$
- Accomplish this with clever masking

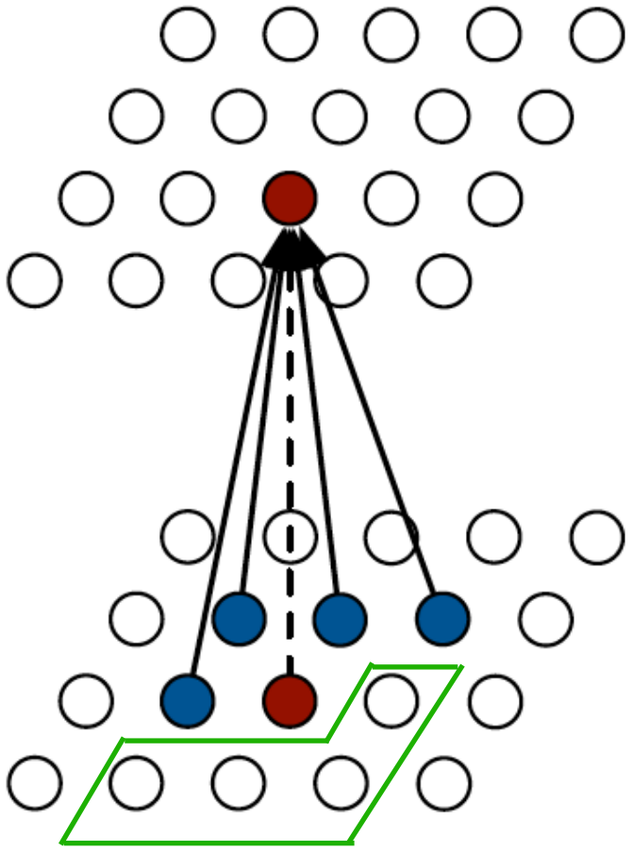




Van den Oord et al.,ICML2016
"Conditional image generation with PixelCNNdecoders"

Limitations of pixel CNN

- Although it can be trained quite fast
- The **receptive field is limited**
- Does not use all the available context



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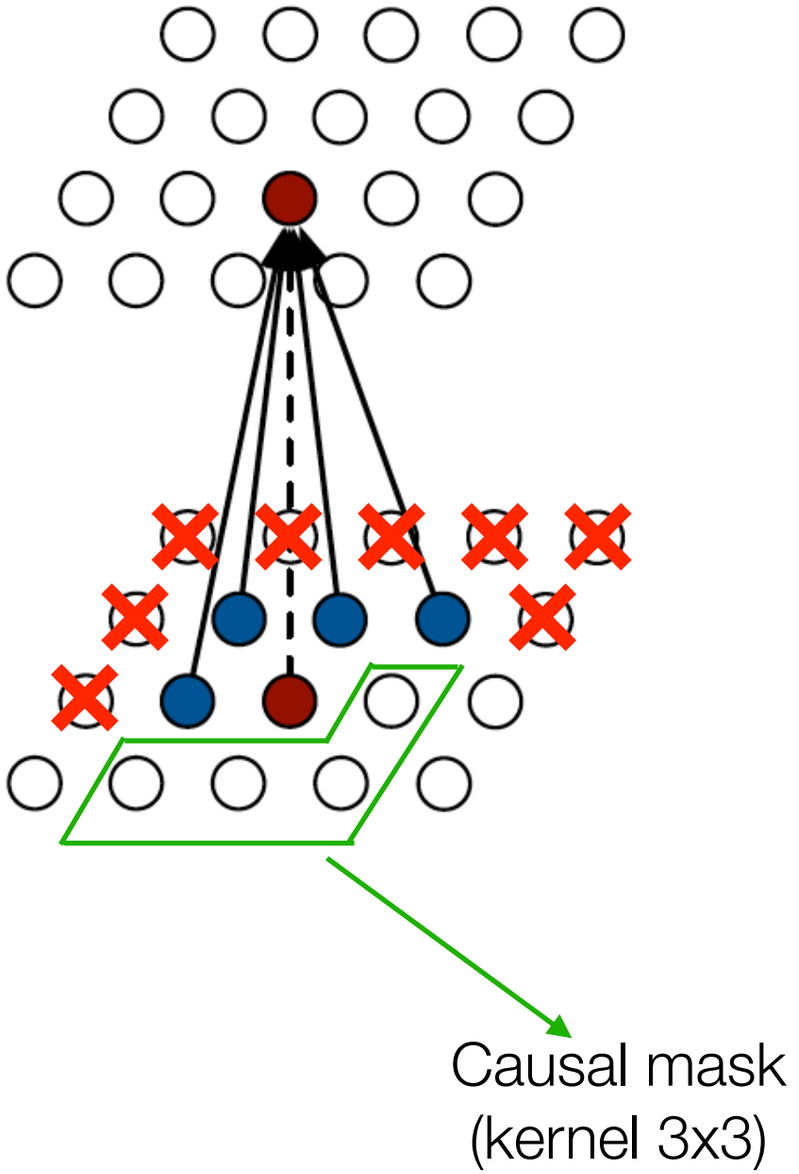

Van den Oord et al.,ICML2016
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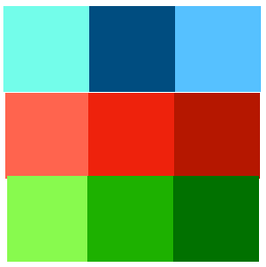
Recurrent Neural Language Model (Training)



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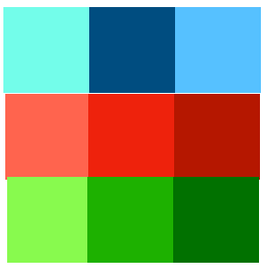

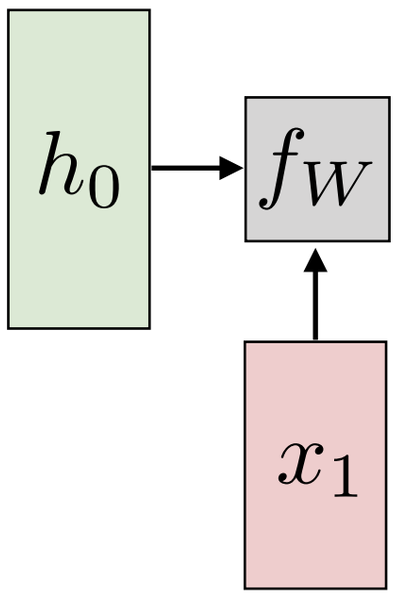

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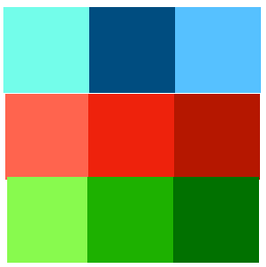

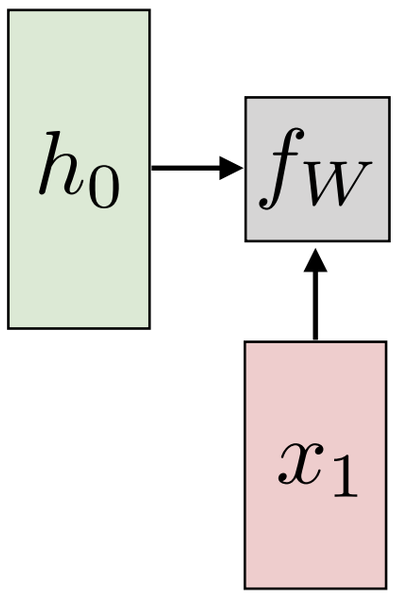

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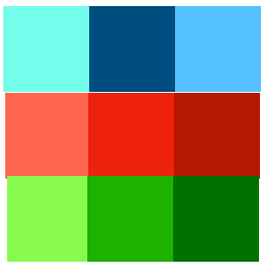

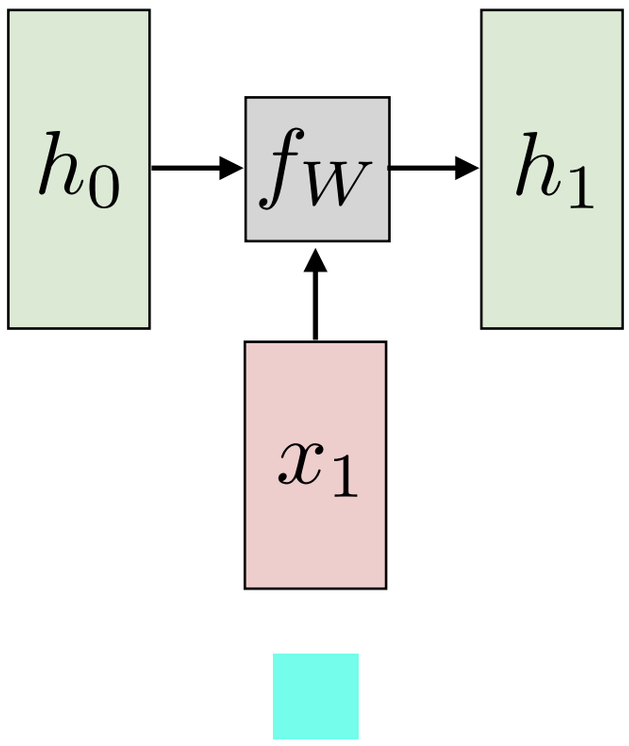

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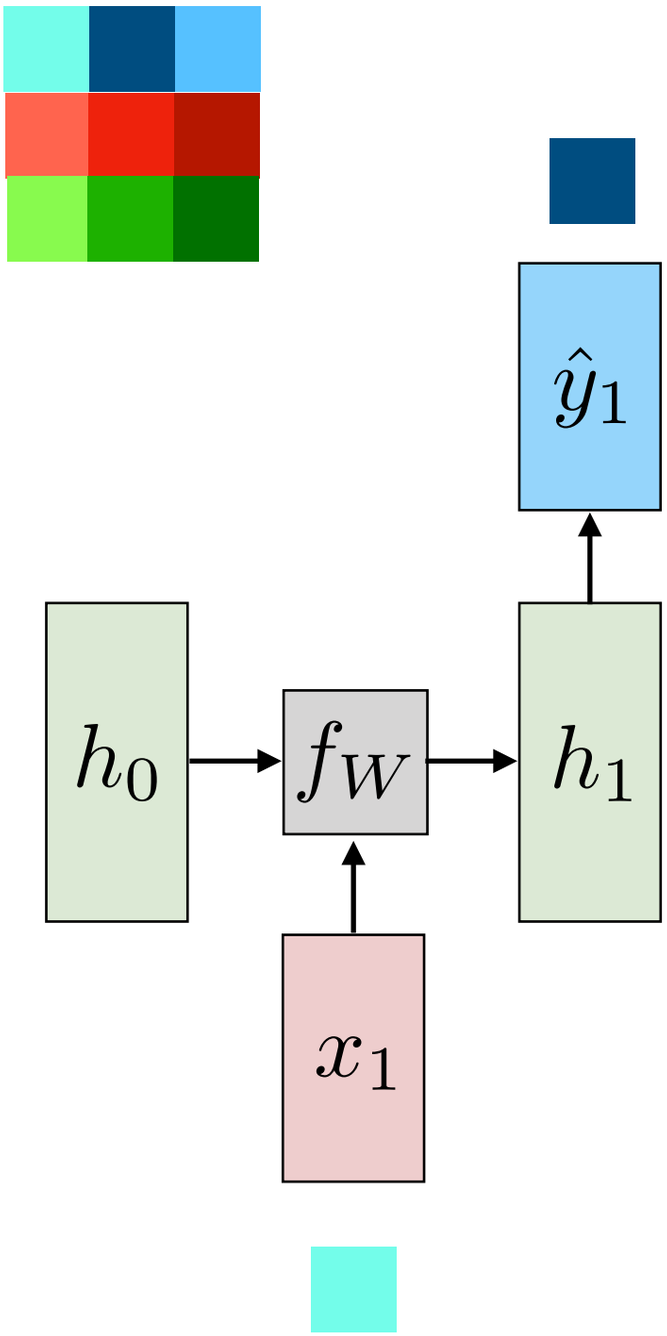

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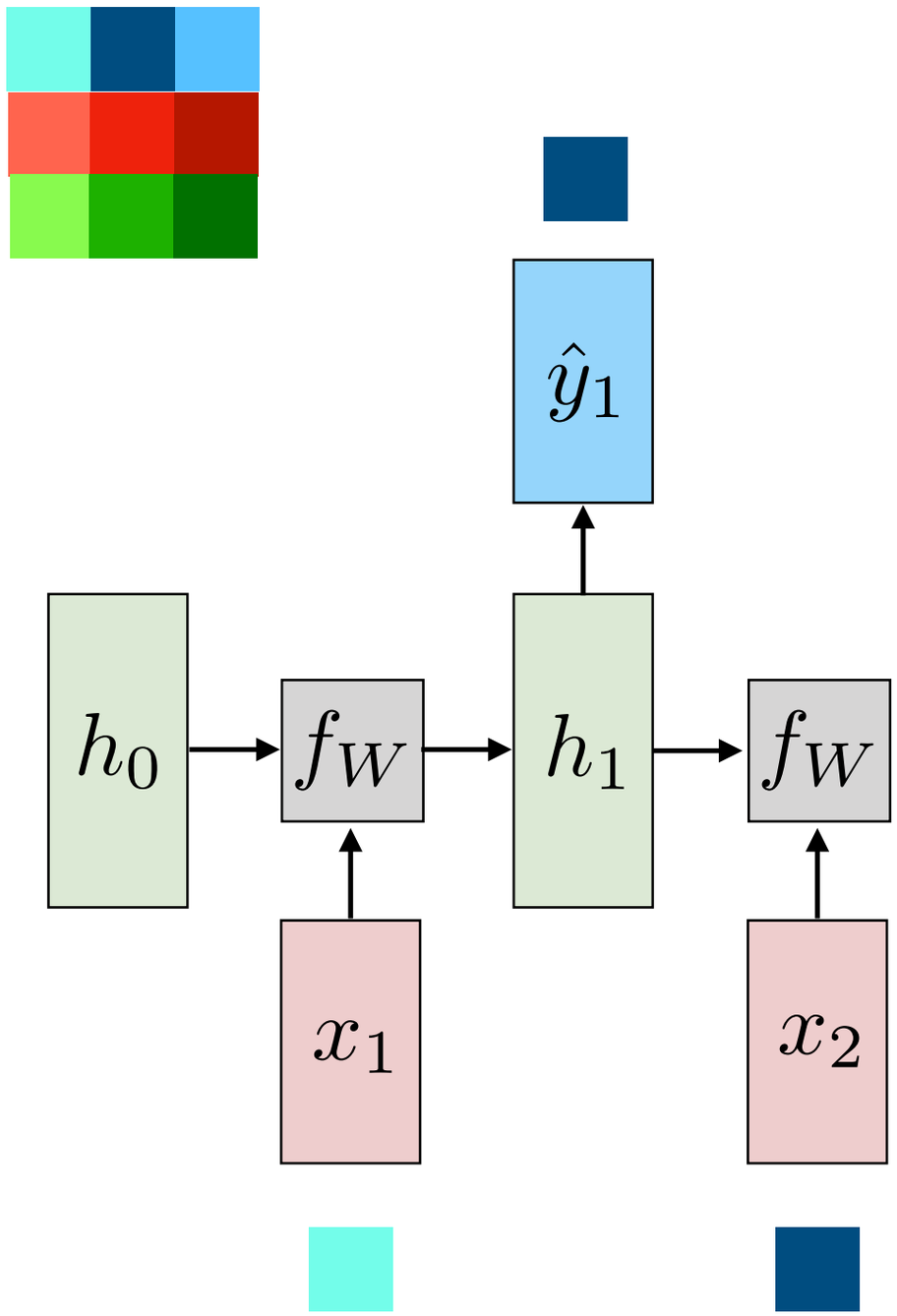

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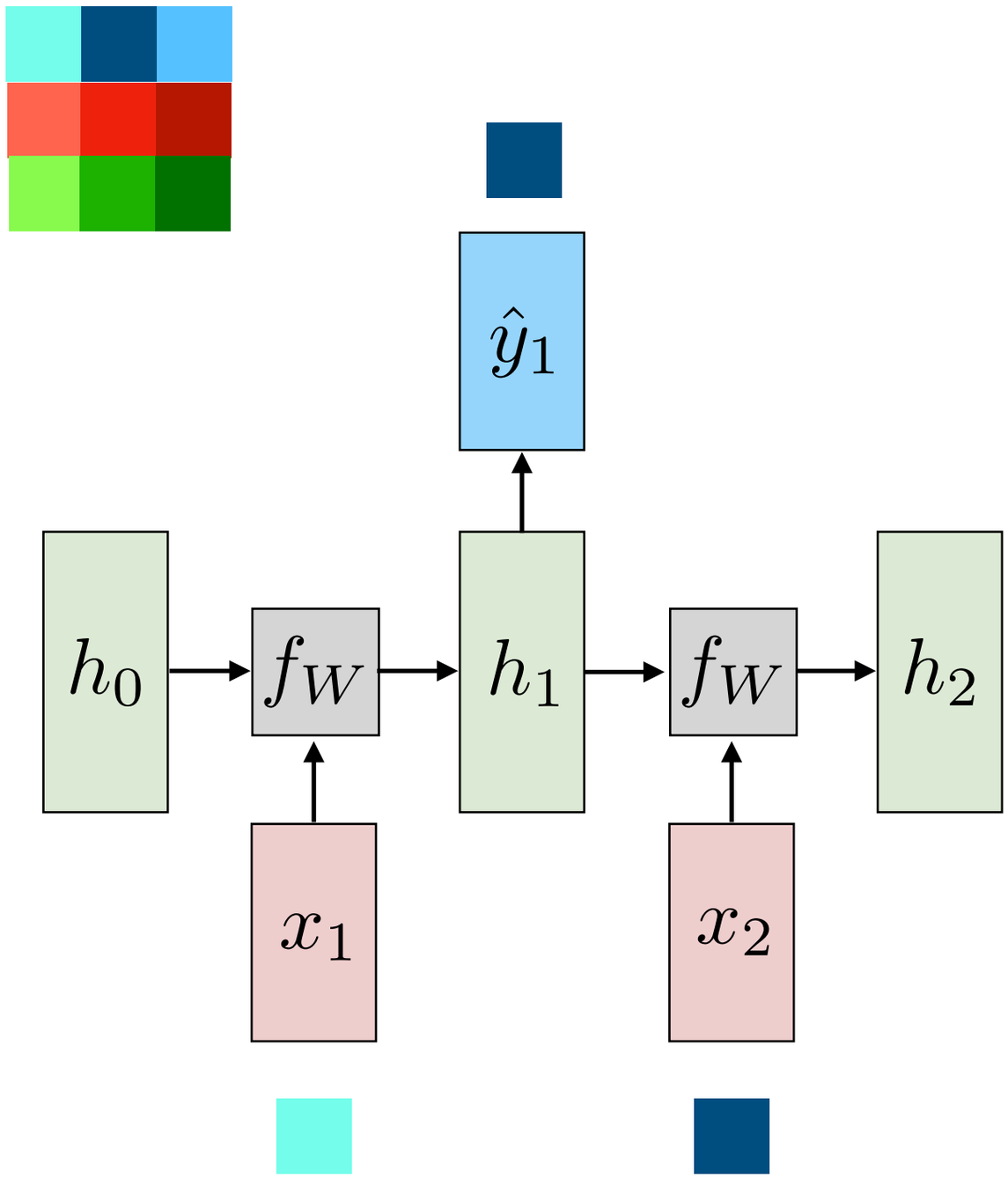

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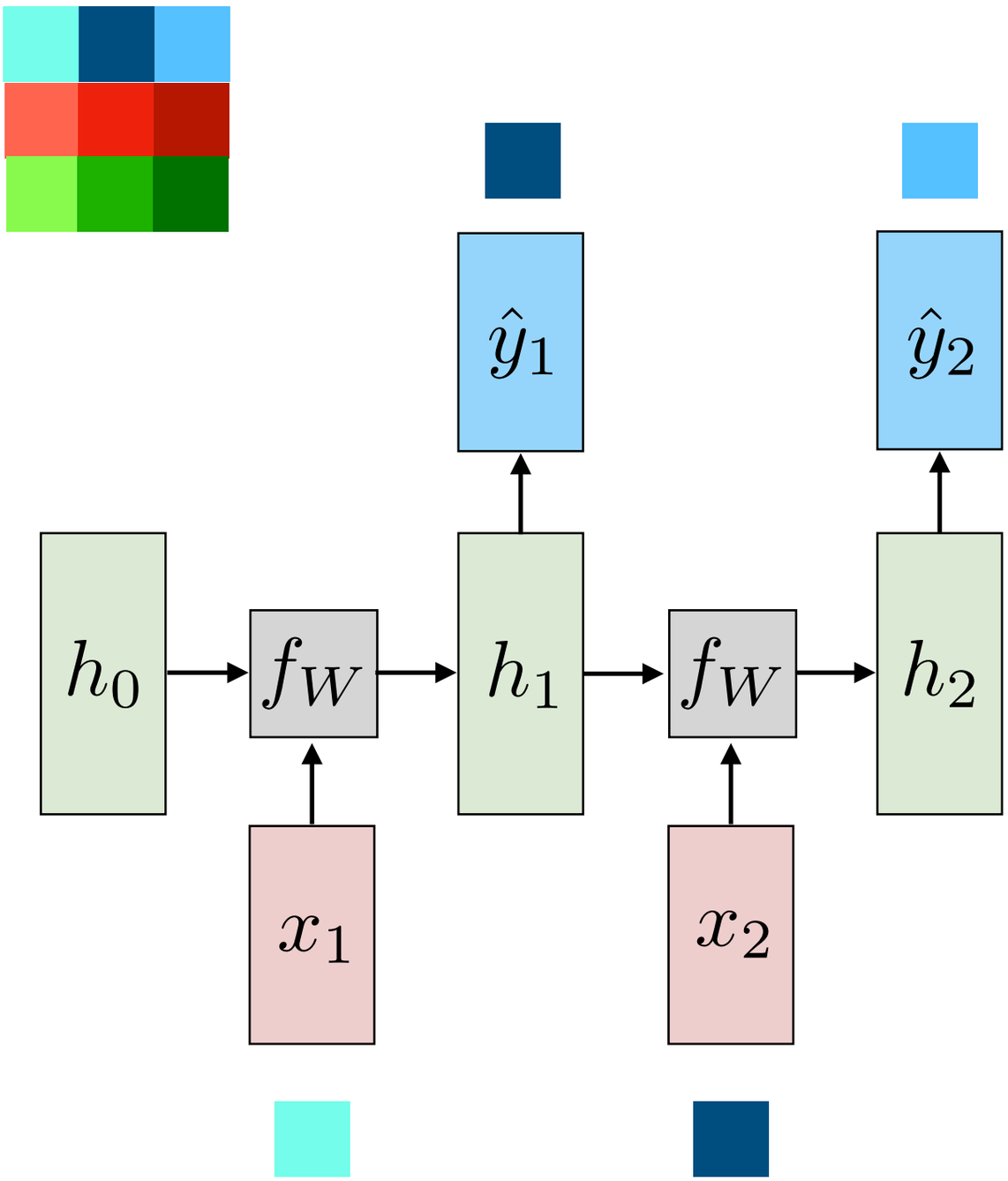

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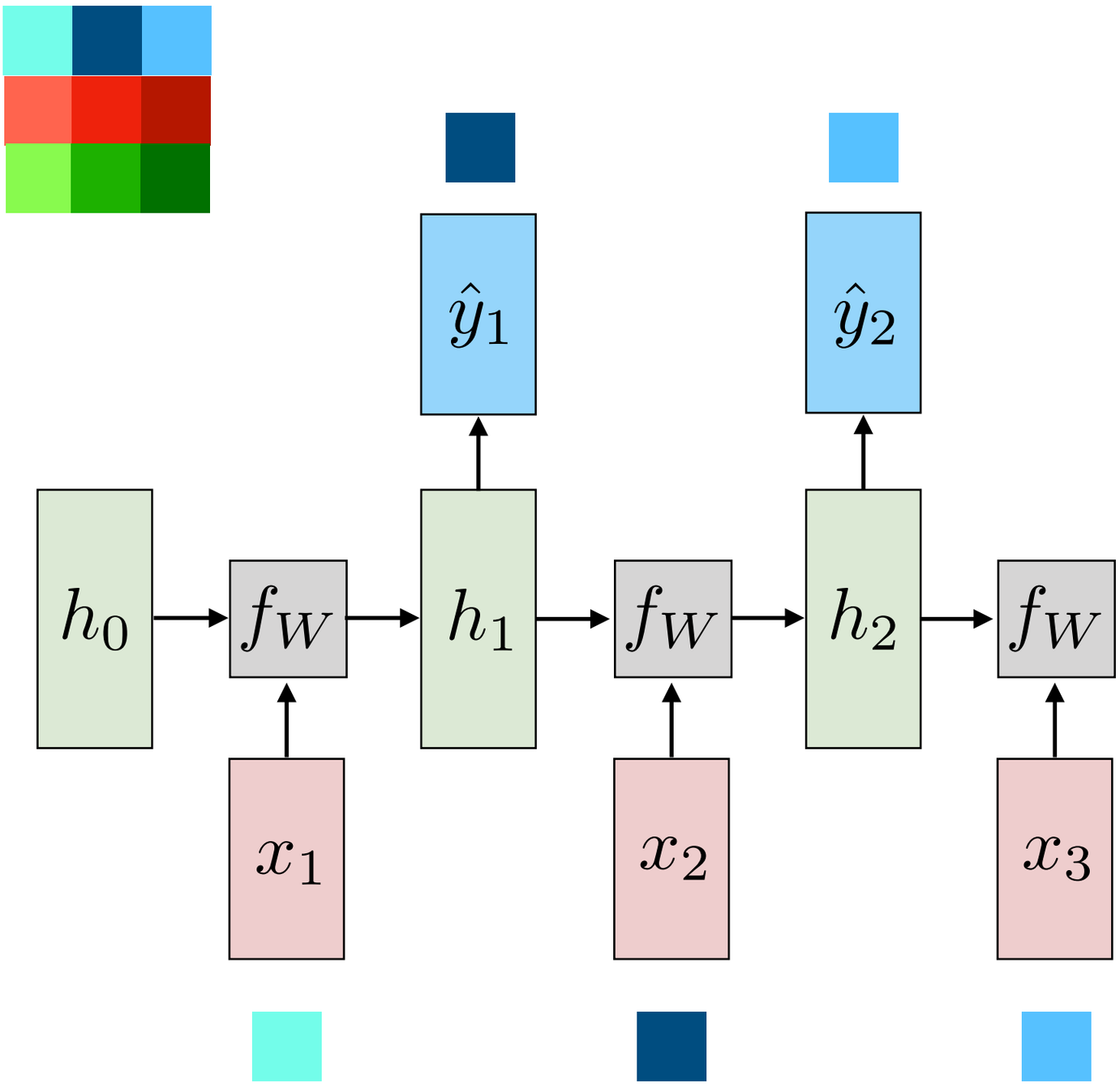

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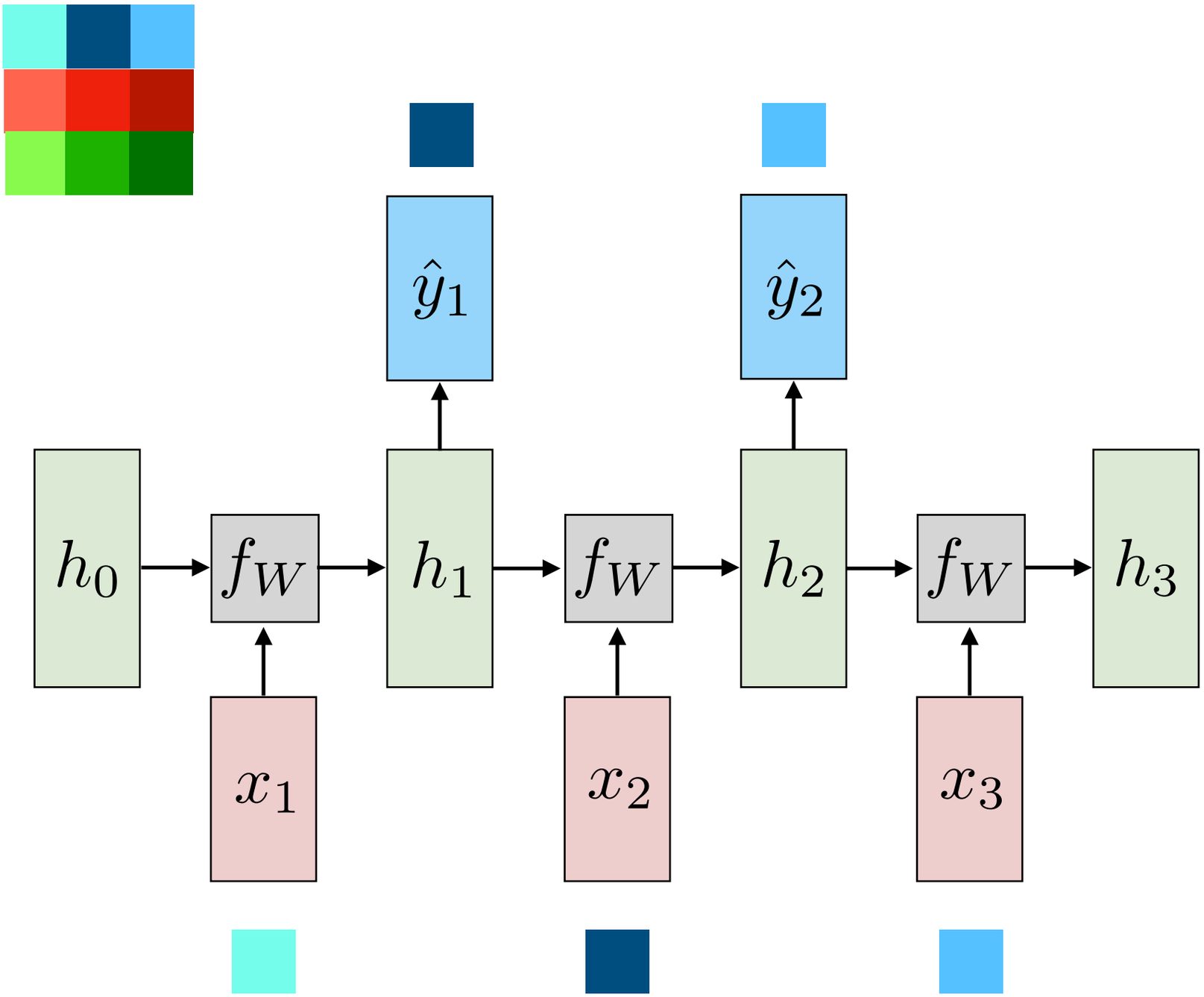

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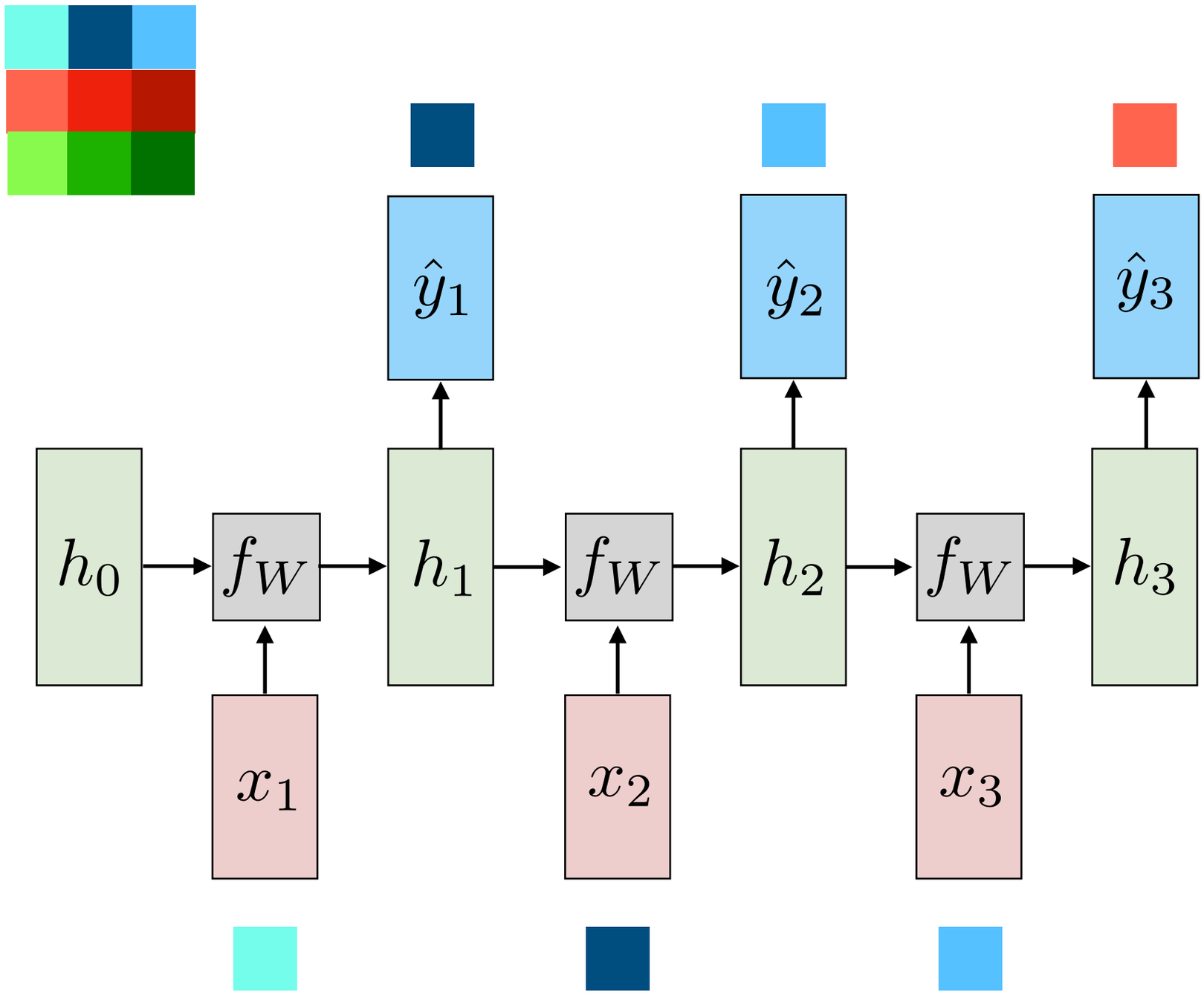

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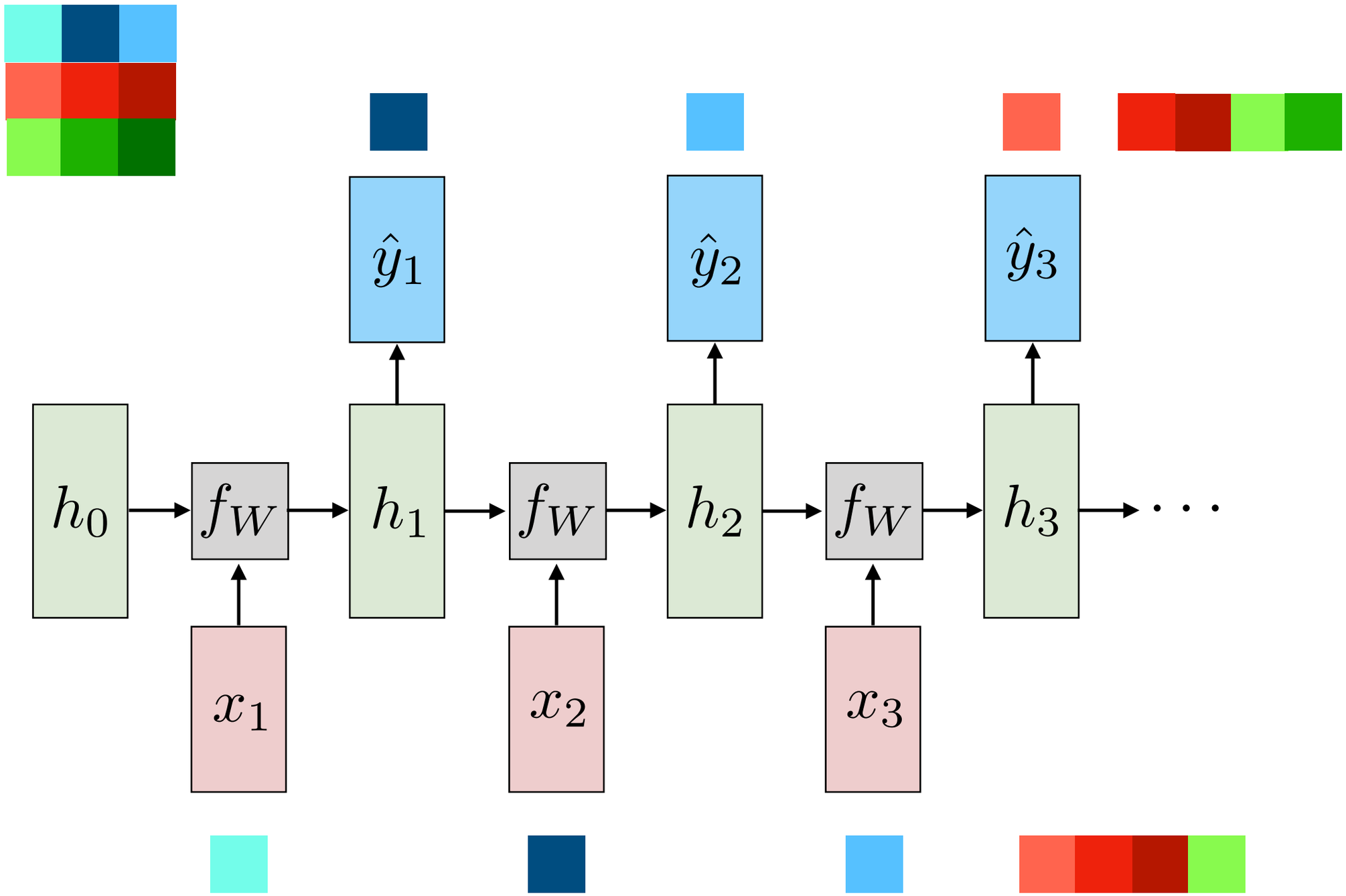

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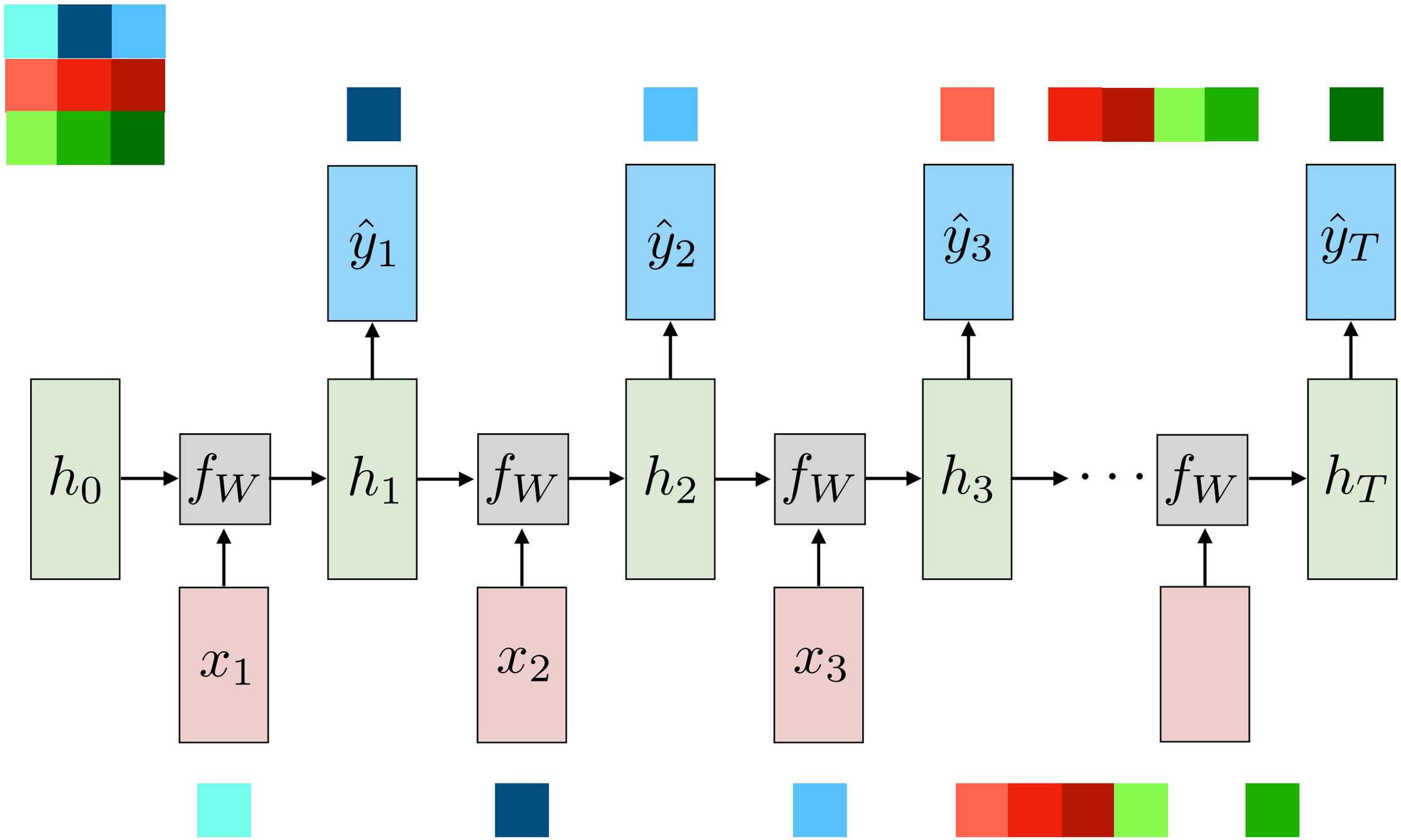

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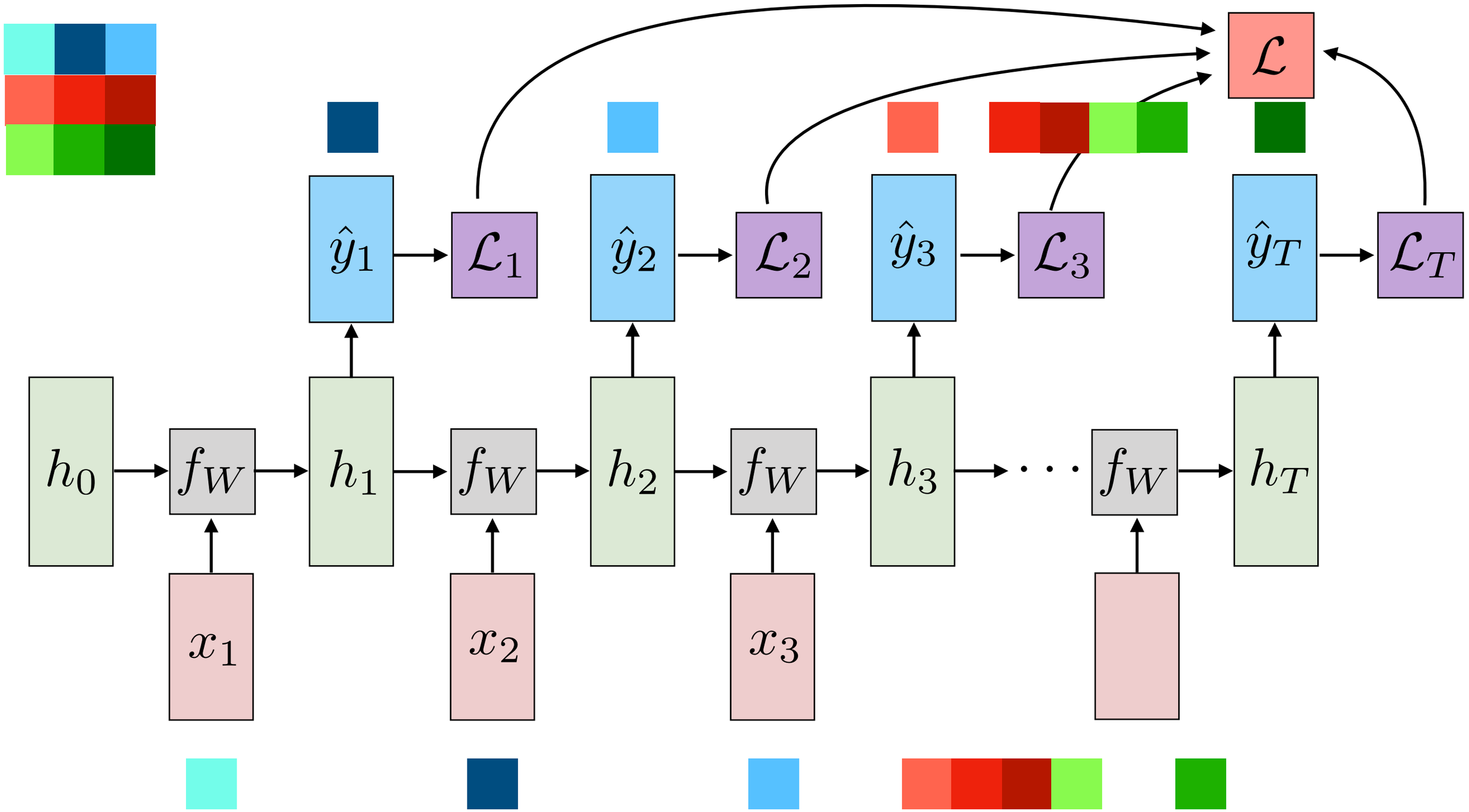

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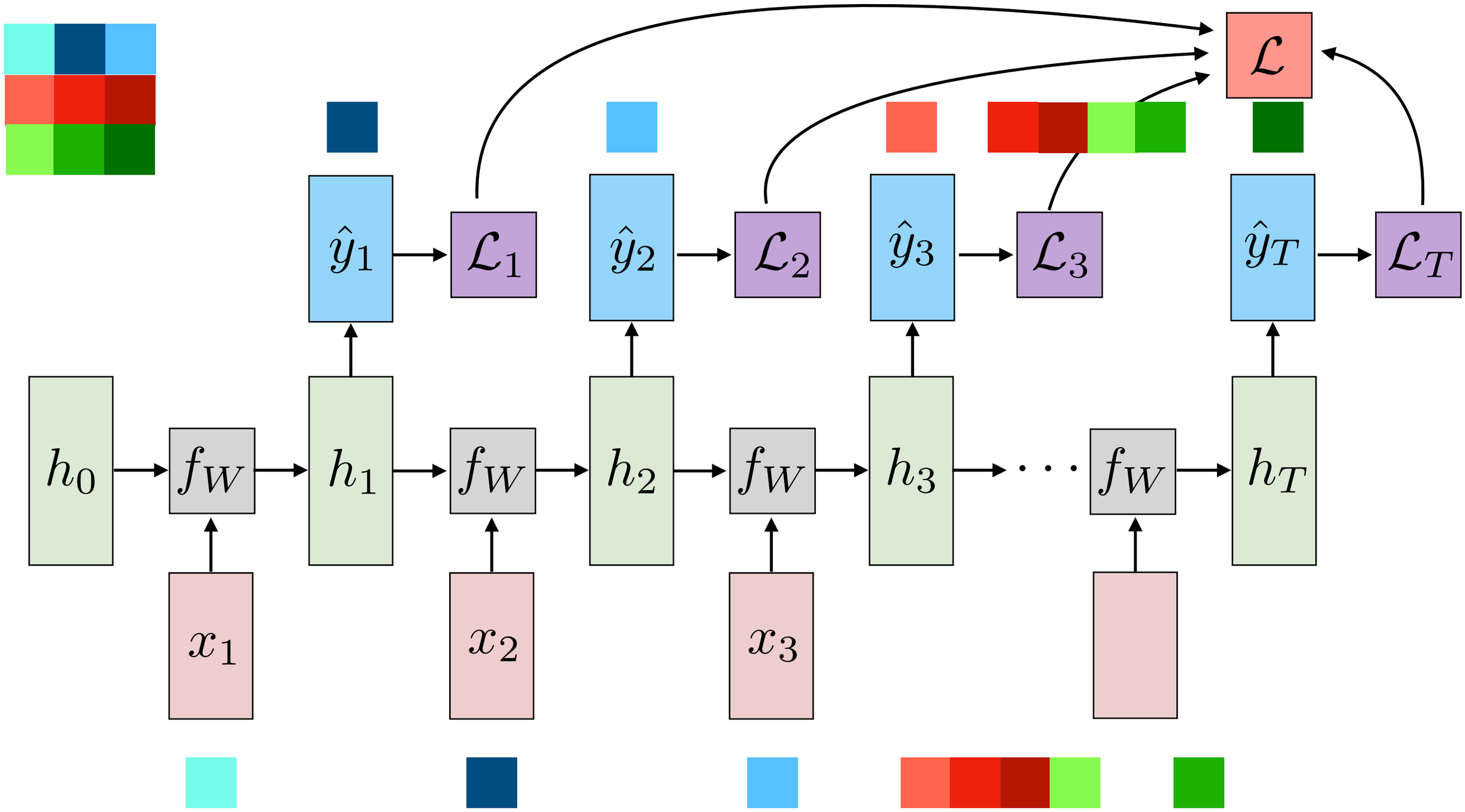
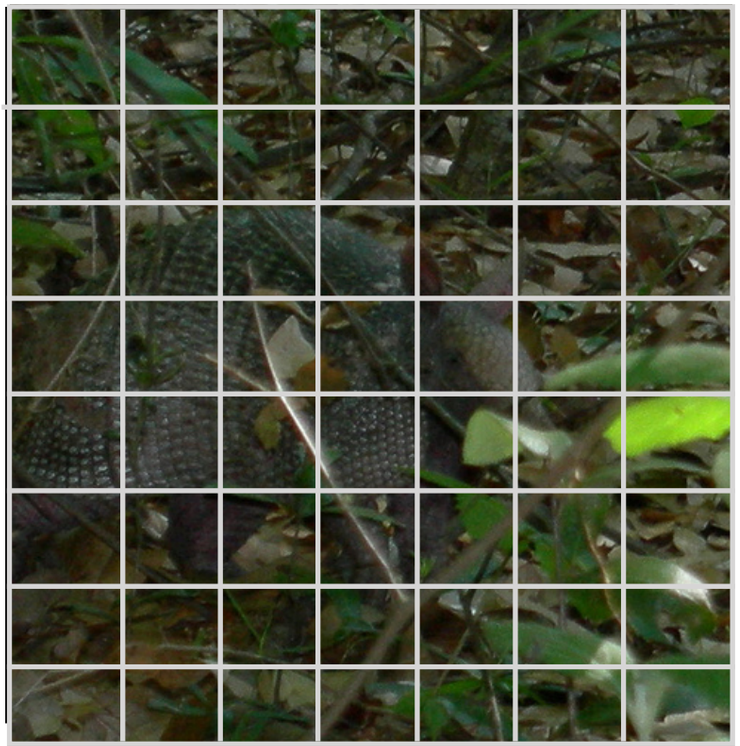


$$p(x_t) = \text{softmax}(\hat{y}_{t-1})$$

Maximum Likelihood Loss

Pixel RNN with row LSTM

$$p(\mathbf{x}) = \prod_{i=1}^{n^2} p\left(x_i | x_1, \ldots, x_{i-1}\right)$$



- Sequential model

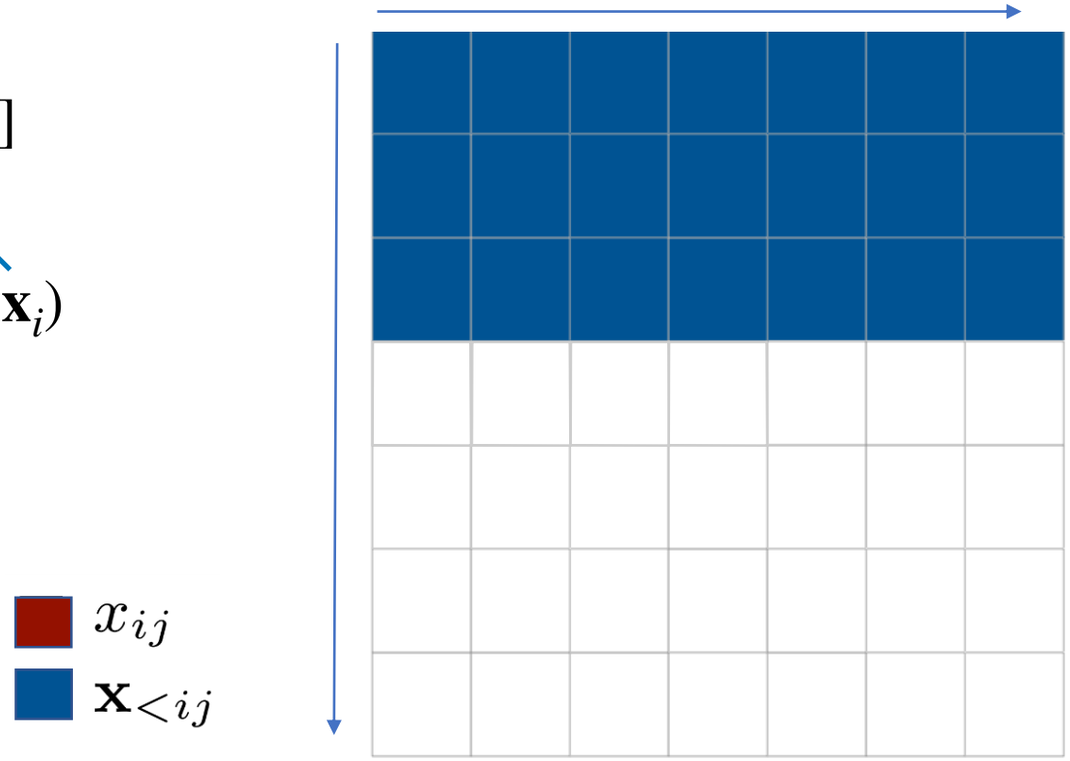
- Spatial Correlation ⟶ Spatial LSTMs

- Each row represents one state

$$[\mathbf{o}_i, \mathbf{f}_i, \mathbf{i}_i, \mathbf{g}_i] = \sigma(\mathbf{K}^{ss} \circledast \mathbf{h}_{i-1} + \mathbf{K}^{is} \circledast \mathbf{x}_i)$$

$[h \times n \times 1]$

$[4h \times n \times n]$



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"Conditional image generation with PixelCNNdecoders"

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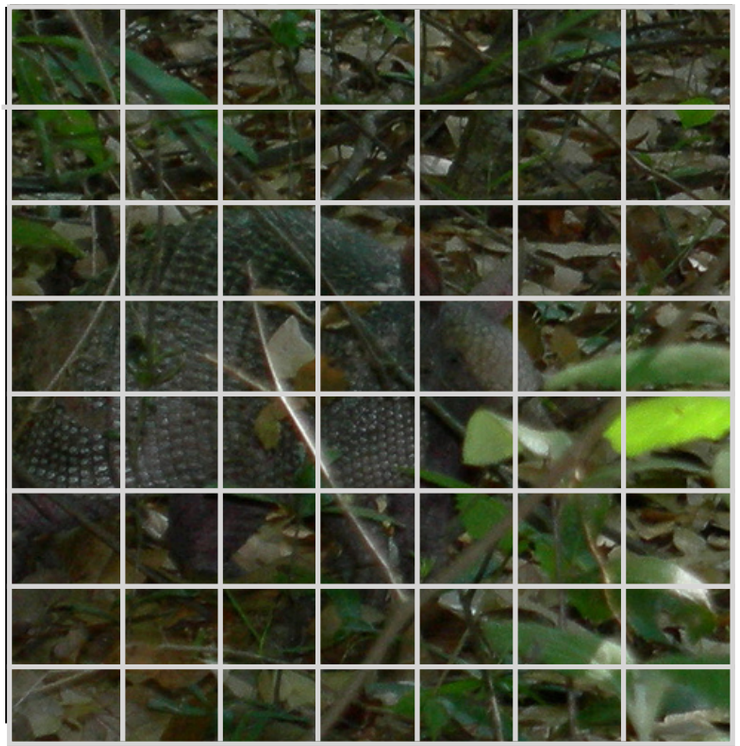

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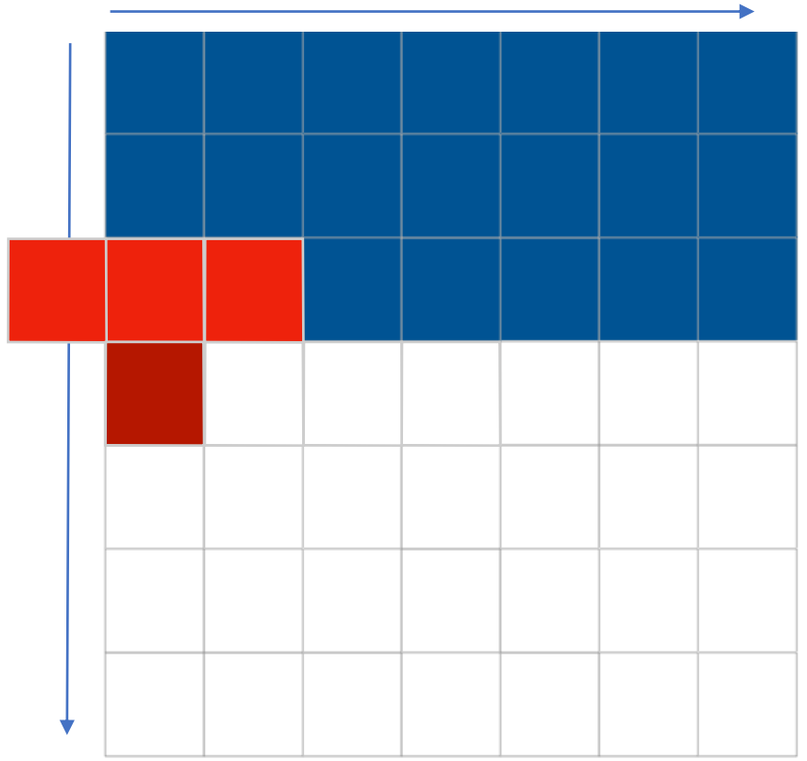

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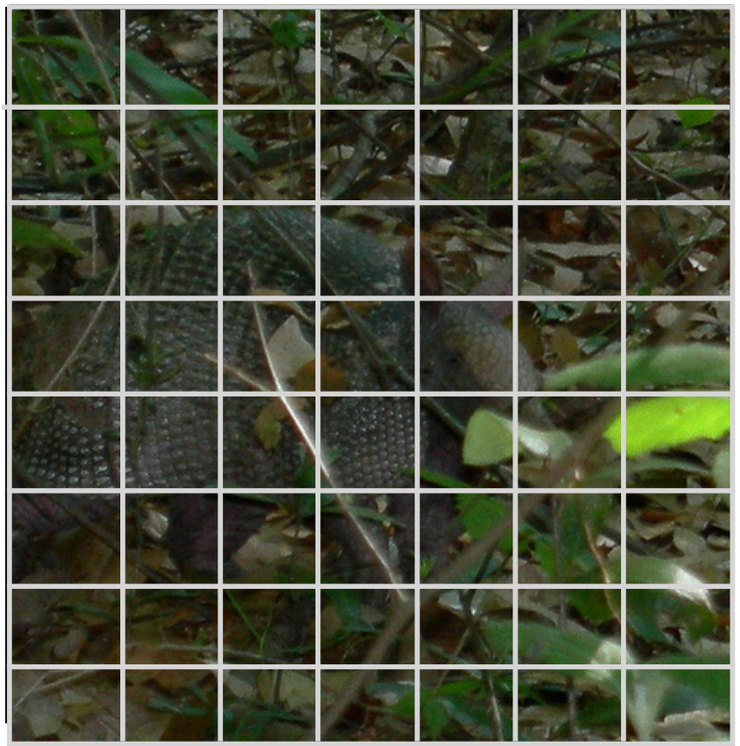

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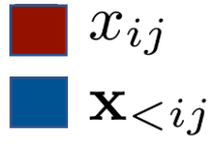

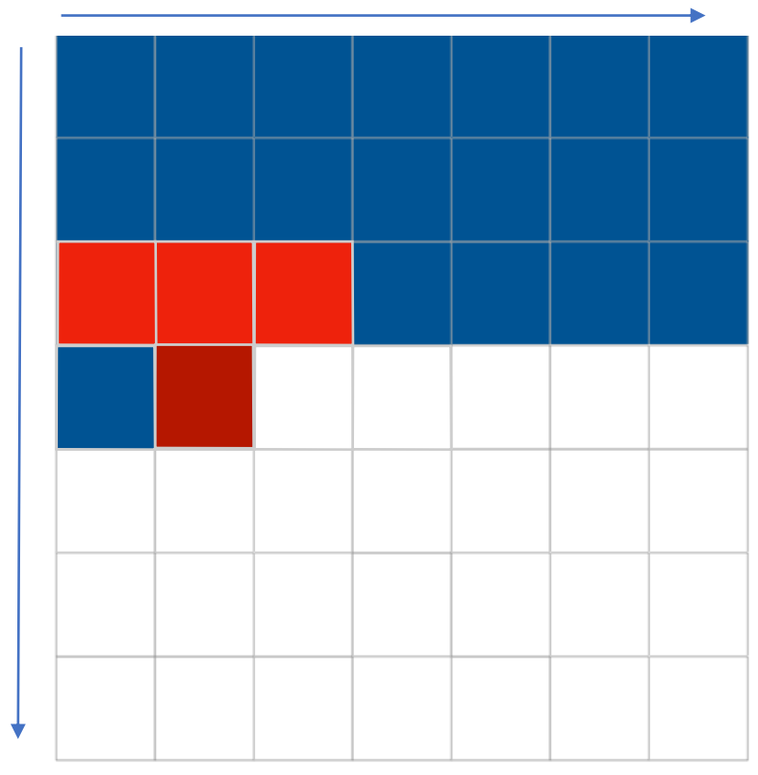

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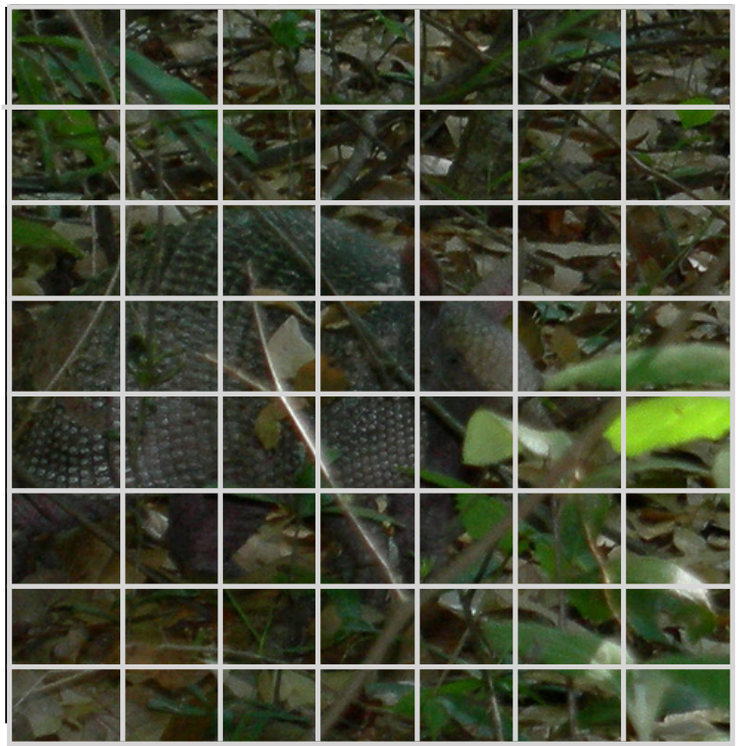

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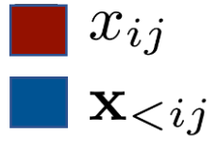

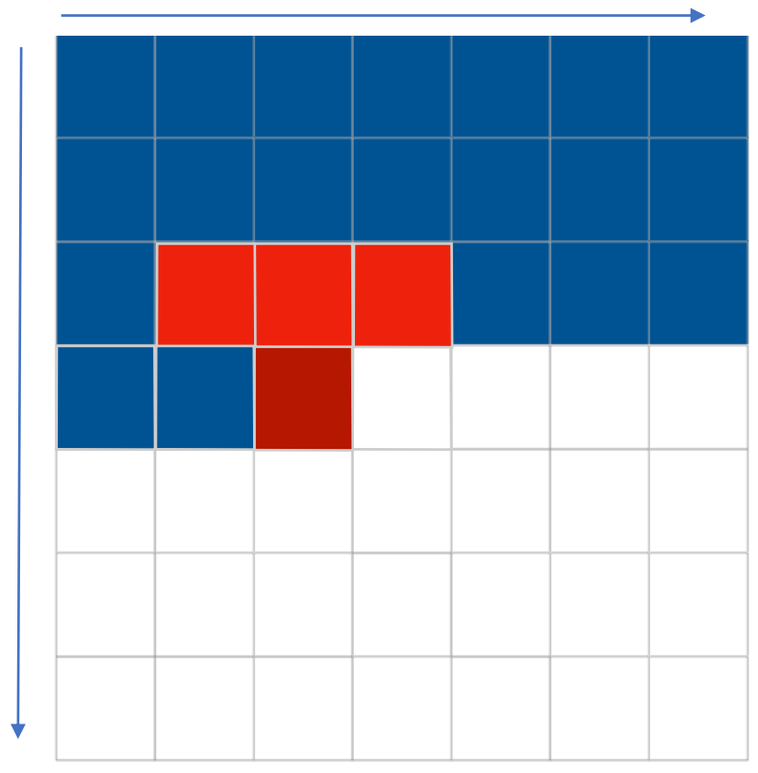

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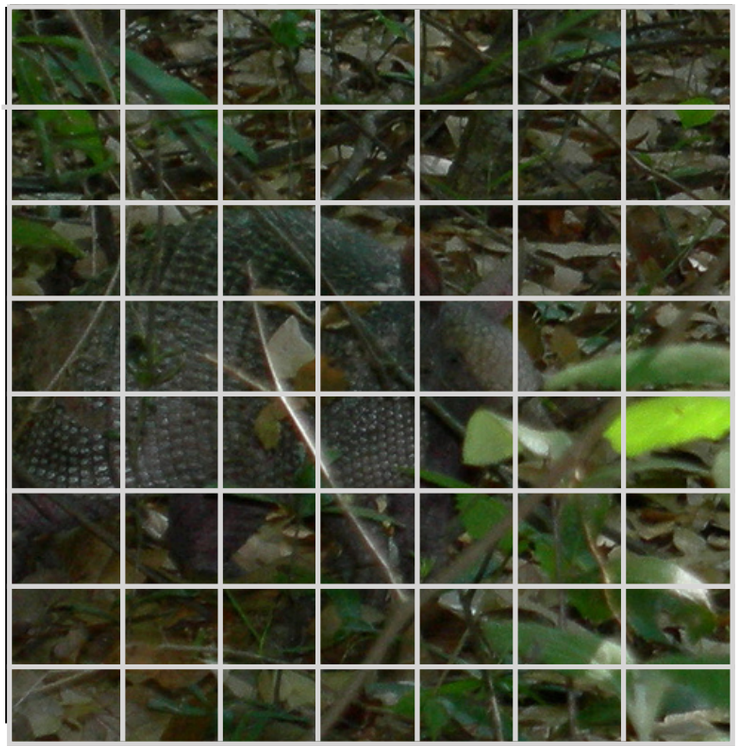


- Sequential model

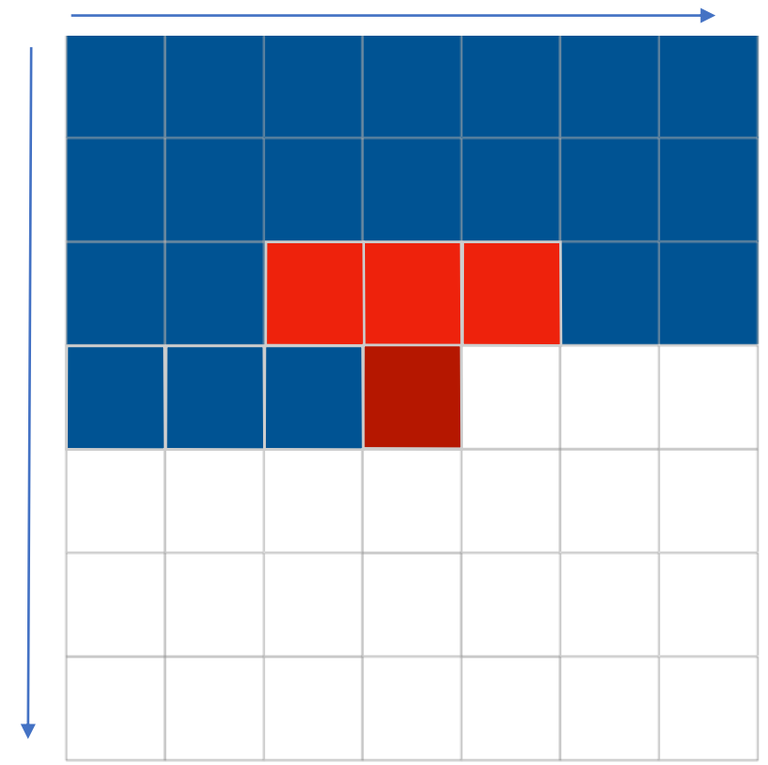
- Spatial Correlation ⟶ Spatial LSTMs

- Each row represents one state

$$[\mathbf{o}_i, \mathbf{f}_i, \mathbf{i}_i, \mathbf{g}_i] = \sigma(\mathbf{K}^{ss} \circledast \mathbf{h}_{i-1} + \mathbf{K}^{is} \circledast \mathbf{x}_i)$$

$[h \times n \times 1]$

$[4h \times n \times n]$



Van den Oord et al.,ICML2016
"Conditional image generation with PixelCNNdecoders"

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masked convolution

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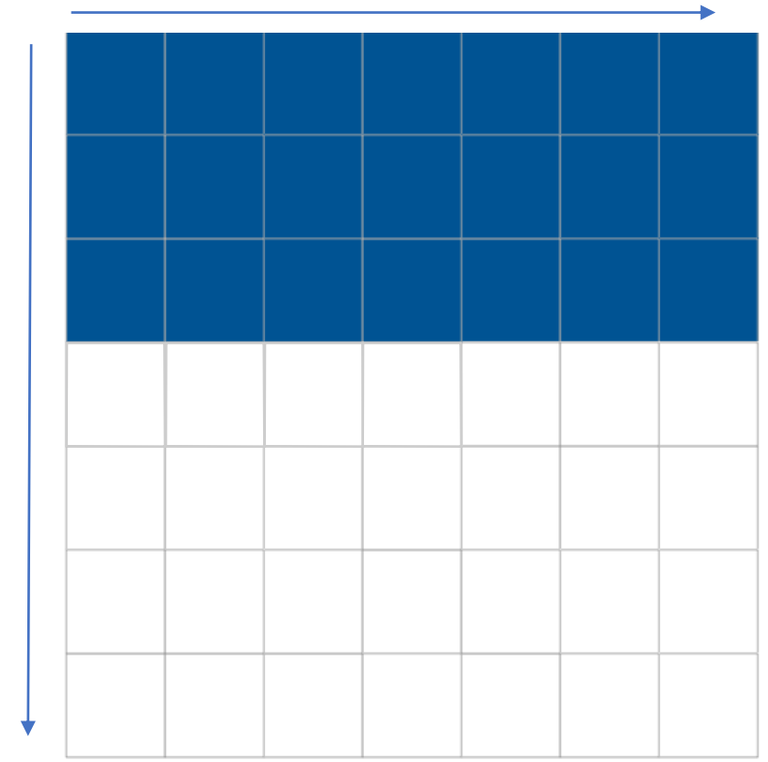

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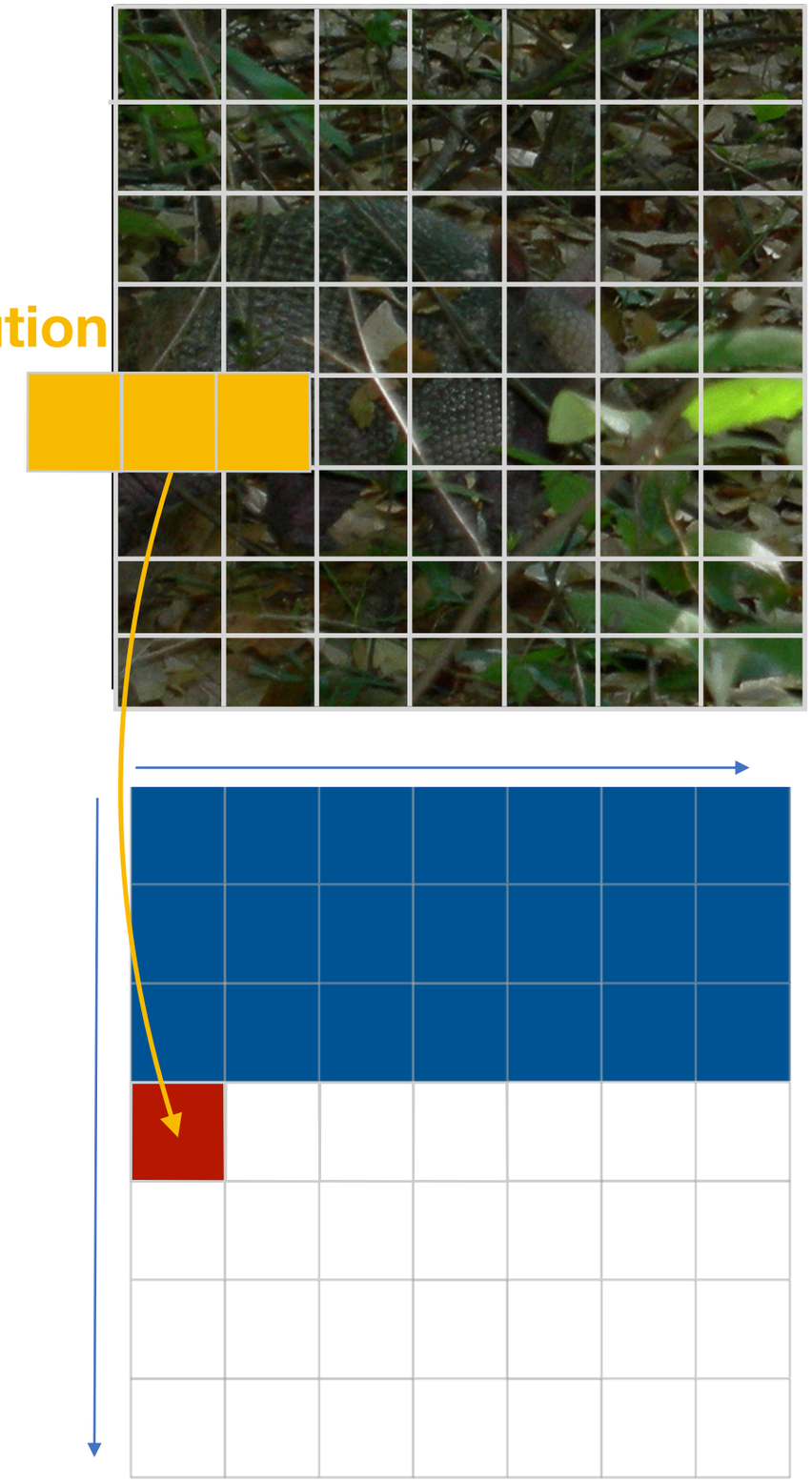

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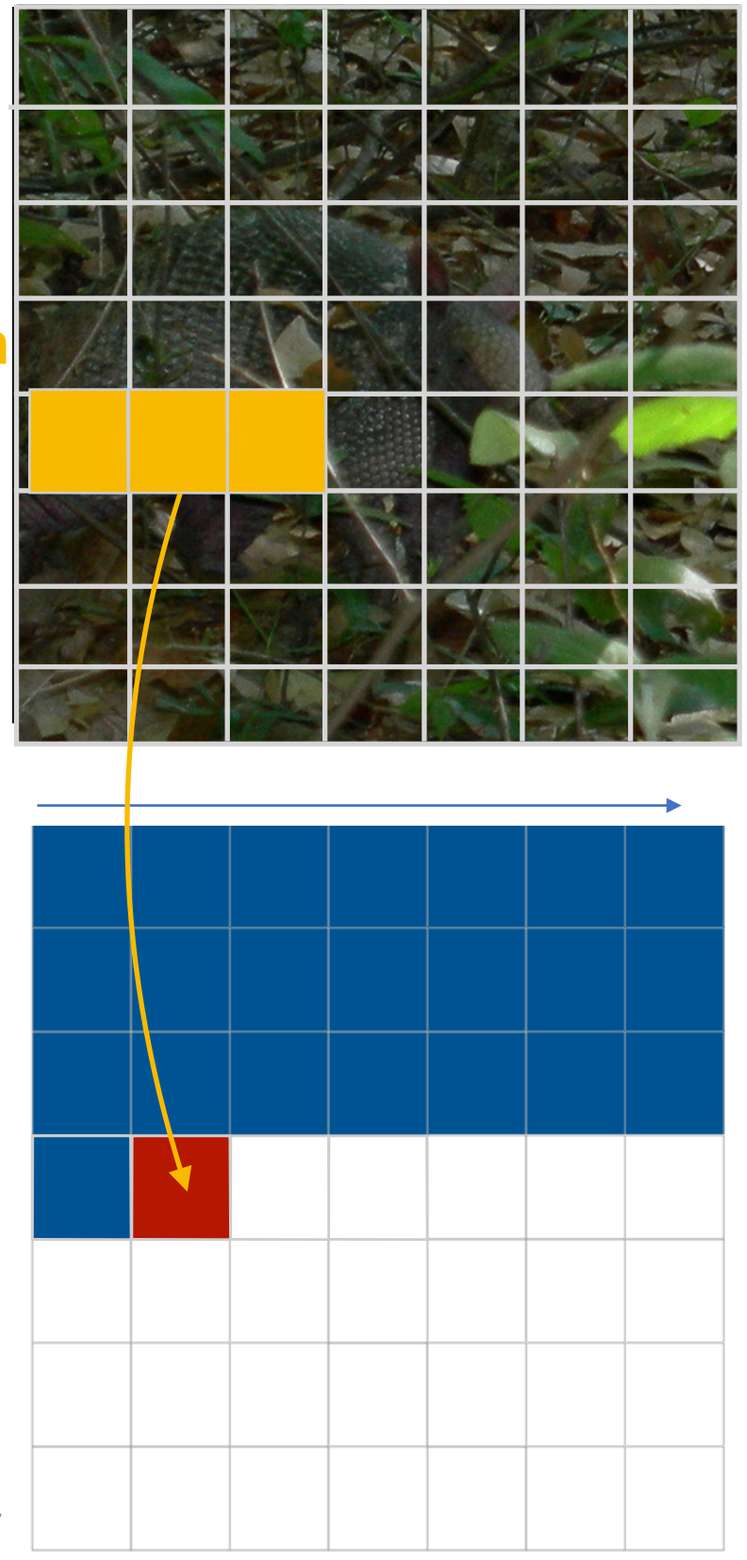

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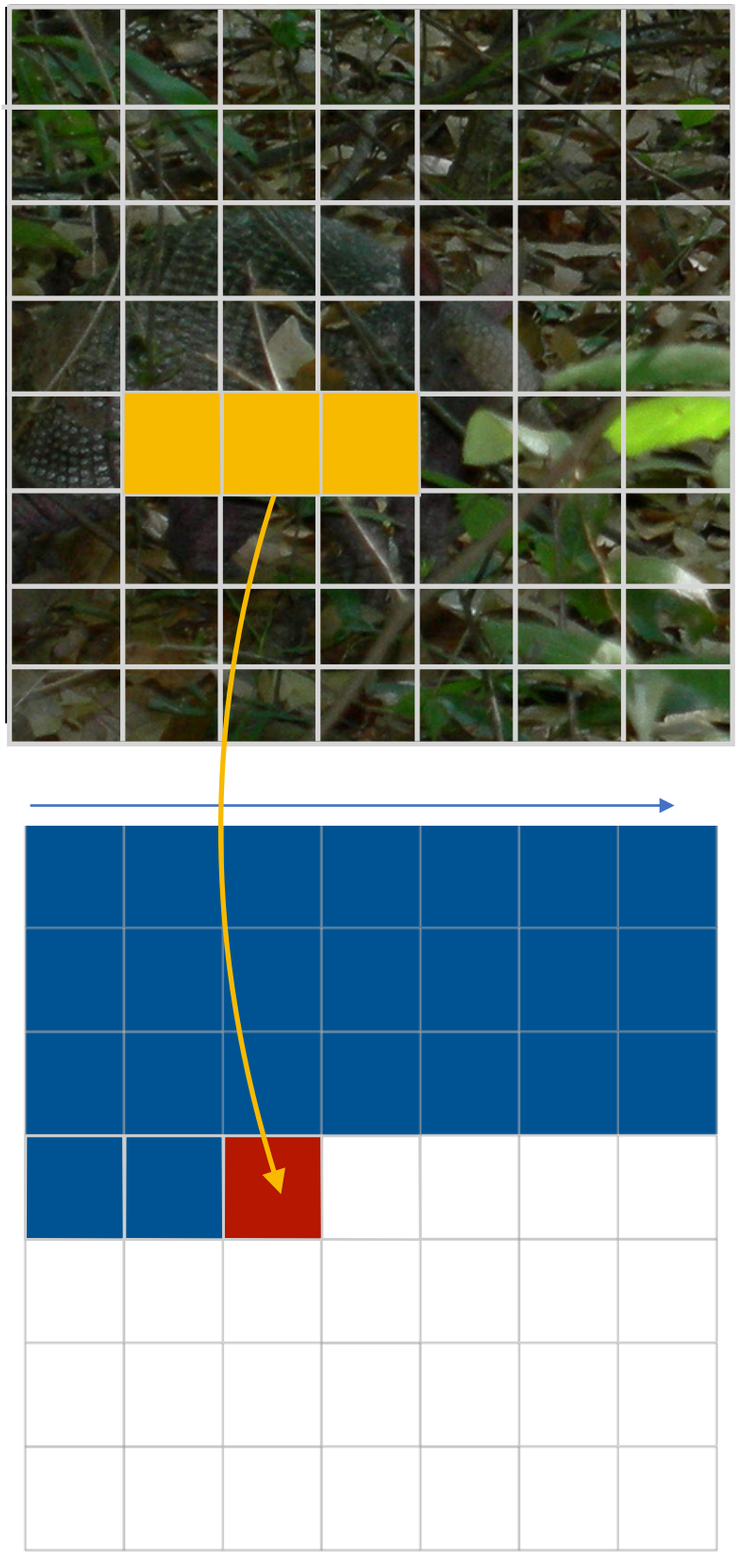

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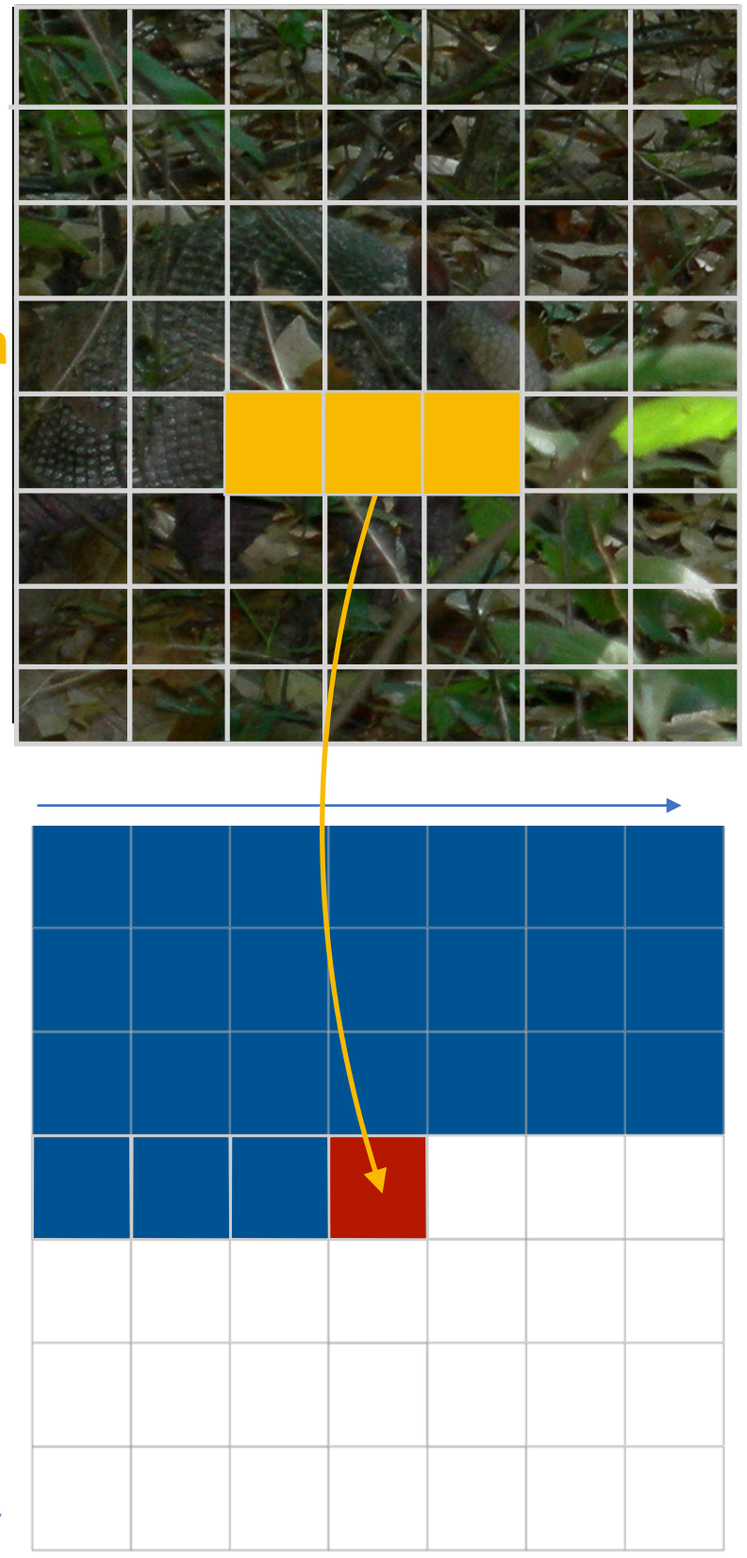
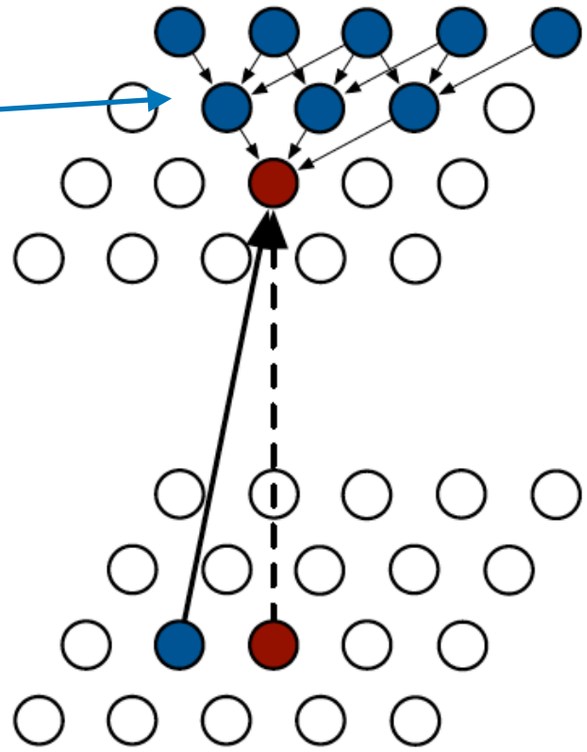


[Van den Oord et al.,ICML2016
"Conditional image generation with PixelCNNdecoders"]

Pixel RNN: pros and cons of row LSTM

- **Compute state for entire row**

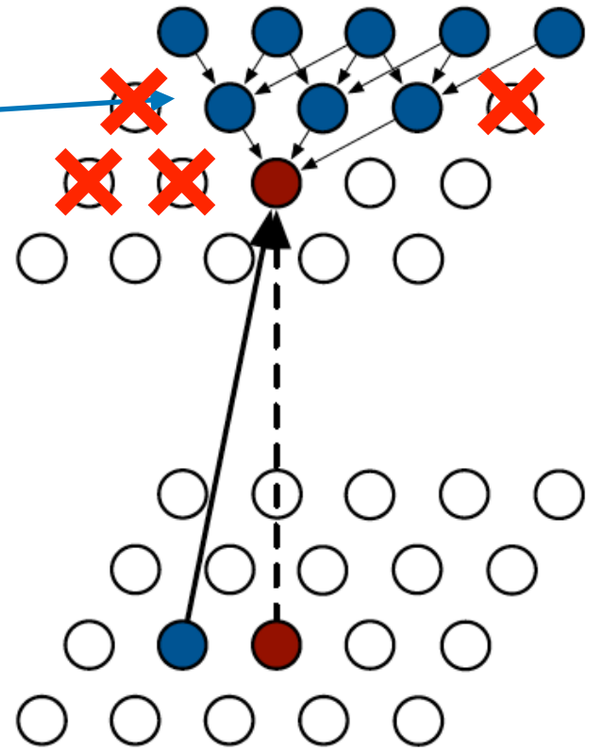
- Triangular receptive field



Van den Oord et al.,IMCL2016
"Conditional image generation with PixelCNNdecoders"

Pixel RNN: pros and cons of row LSTM

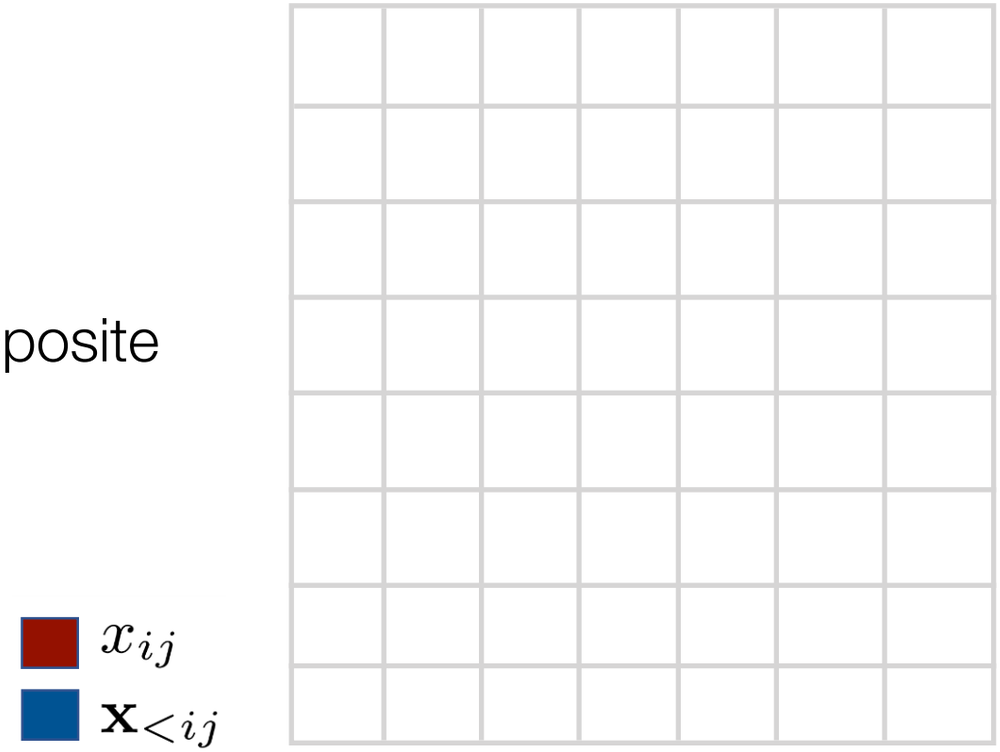
- **Compute state for entire row**
- Triangular receptive field
- **Does not use all available context**



Van den Oord et al.,IMCL2016
"Conditional image generation with PixelCNNdecoders"

Pixel RNN with diagonal biLSTM

- Capture the entire available context
- Each diagonal scan from one corner to the opposite



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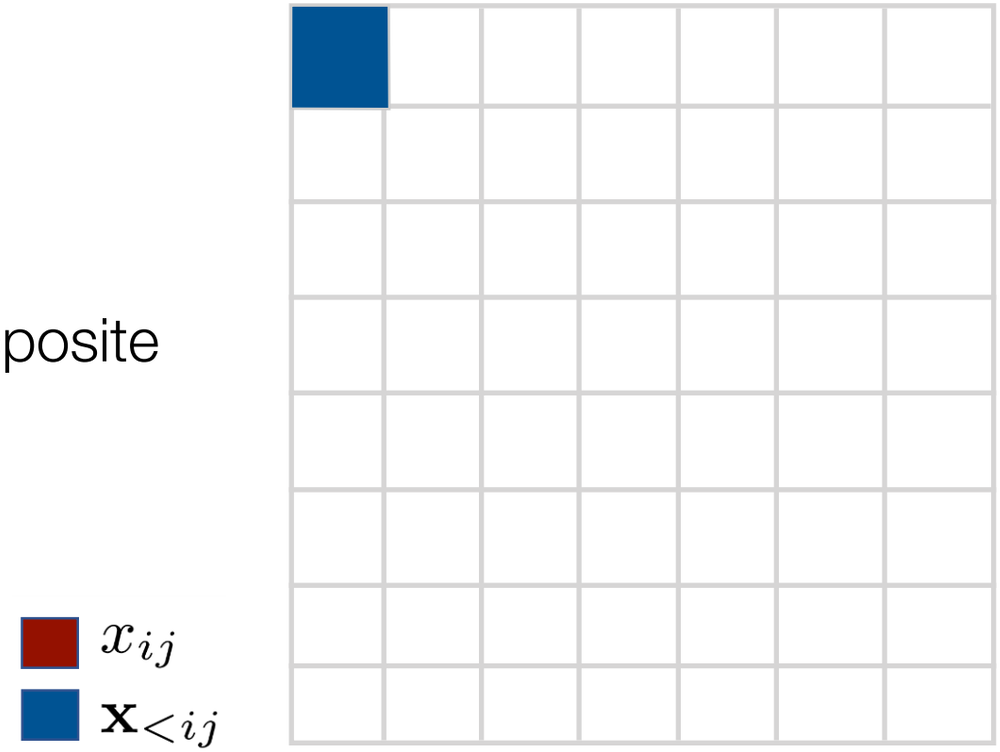

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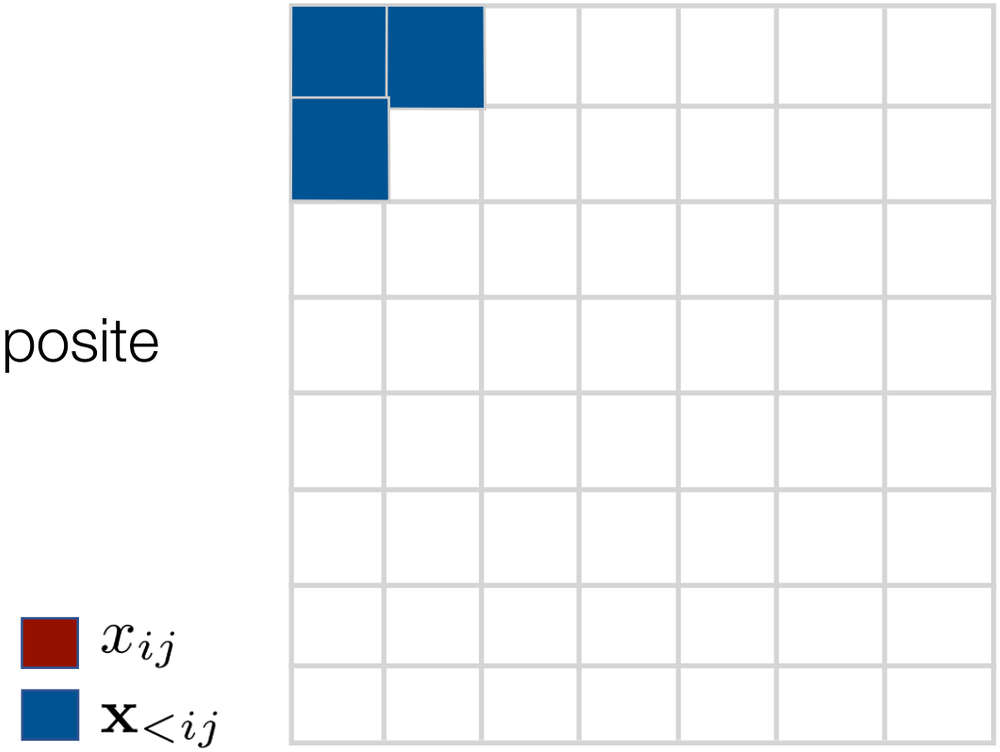

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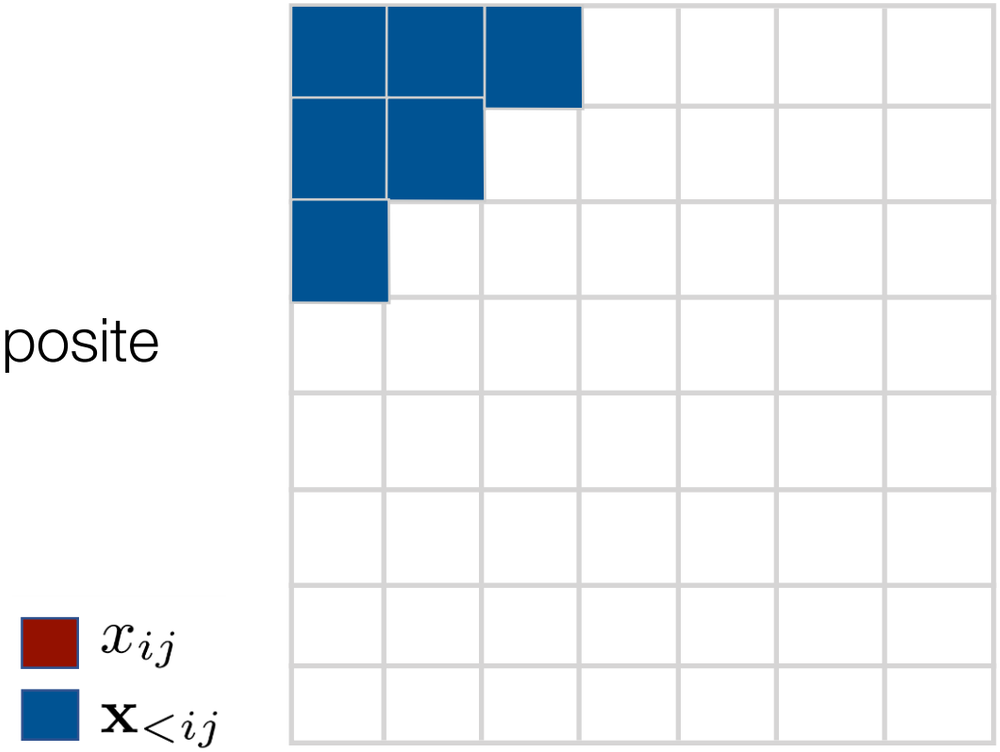

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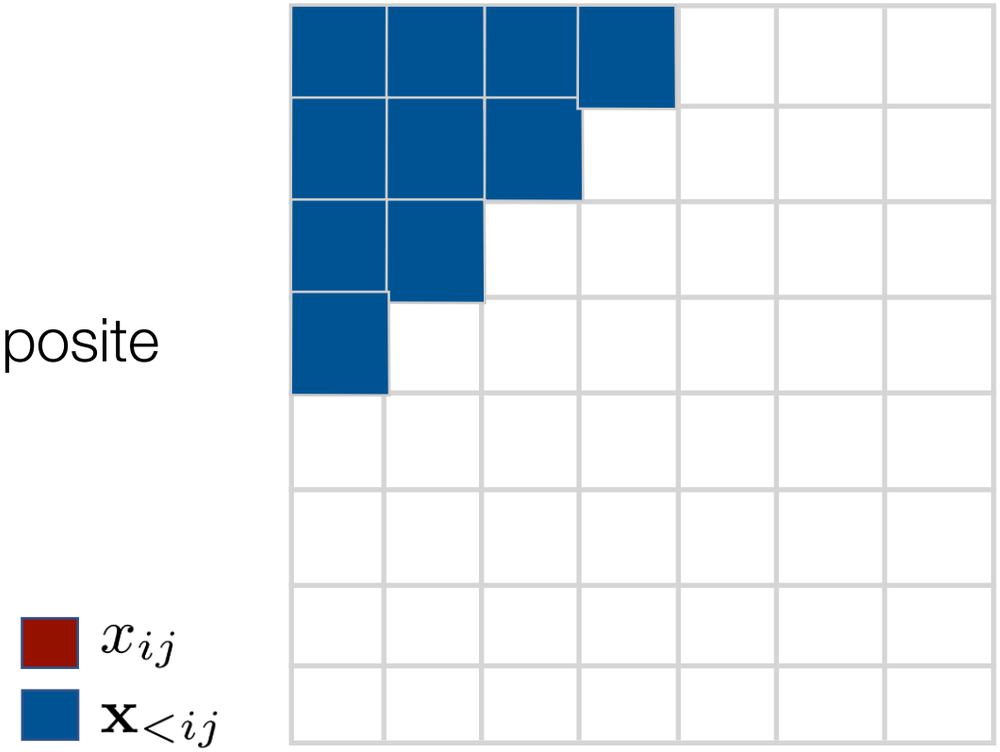

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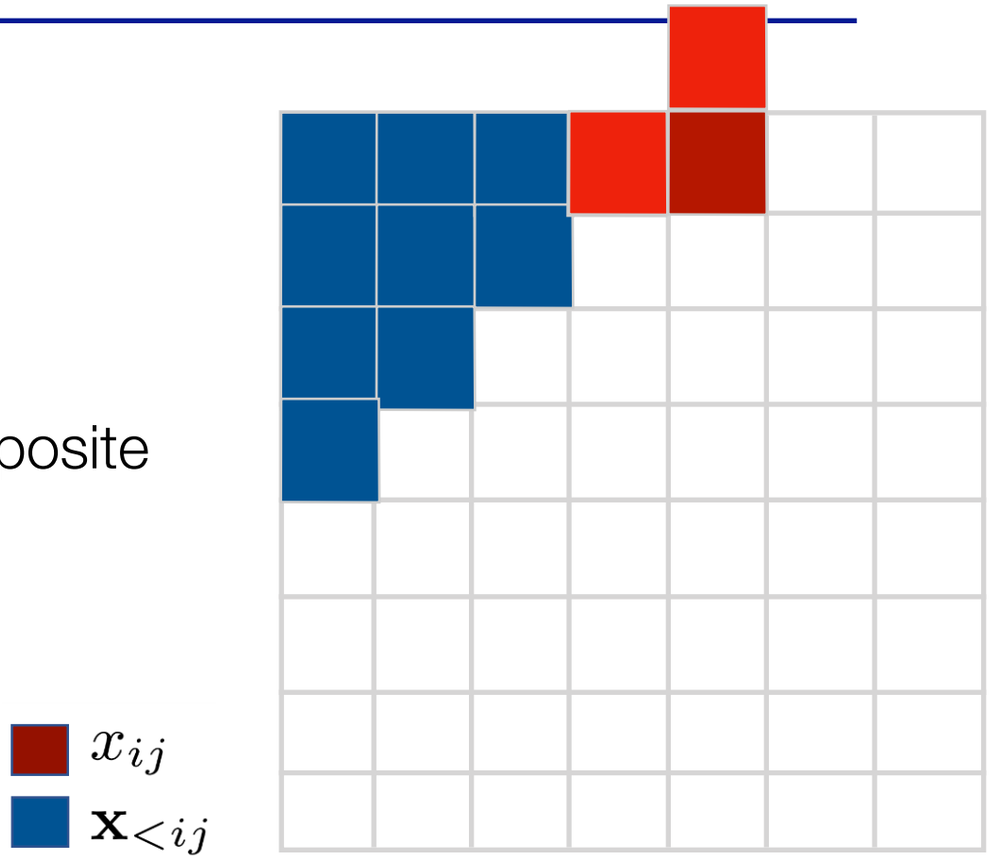

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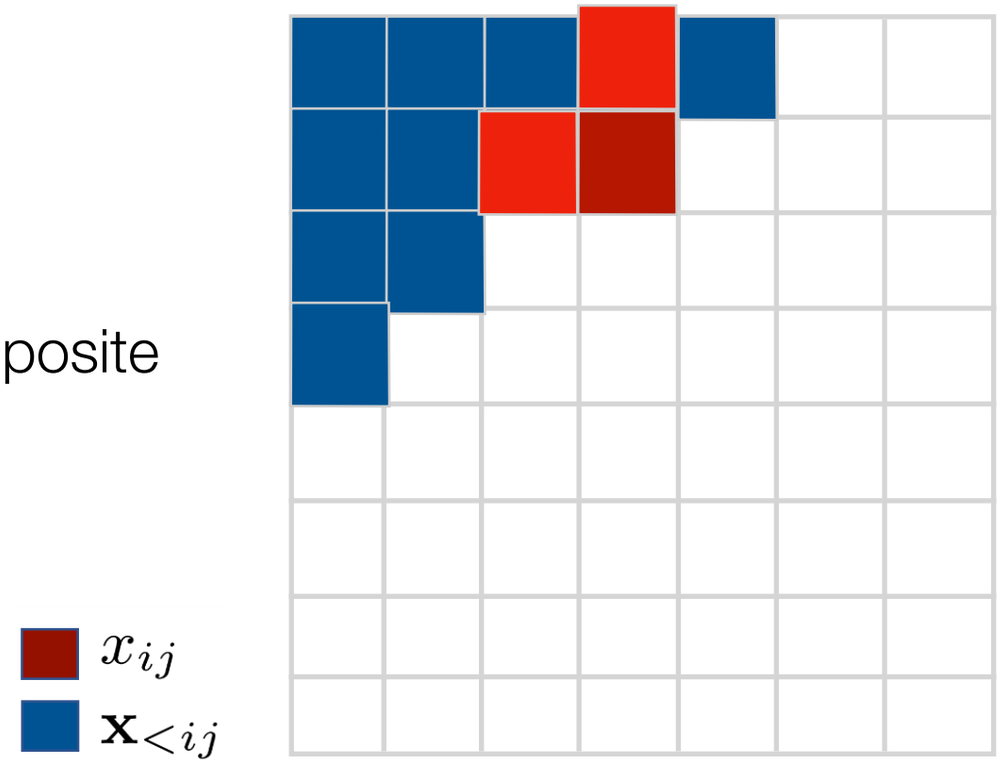

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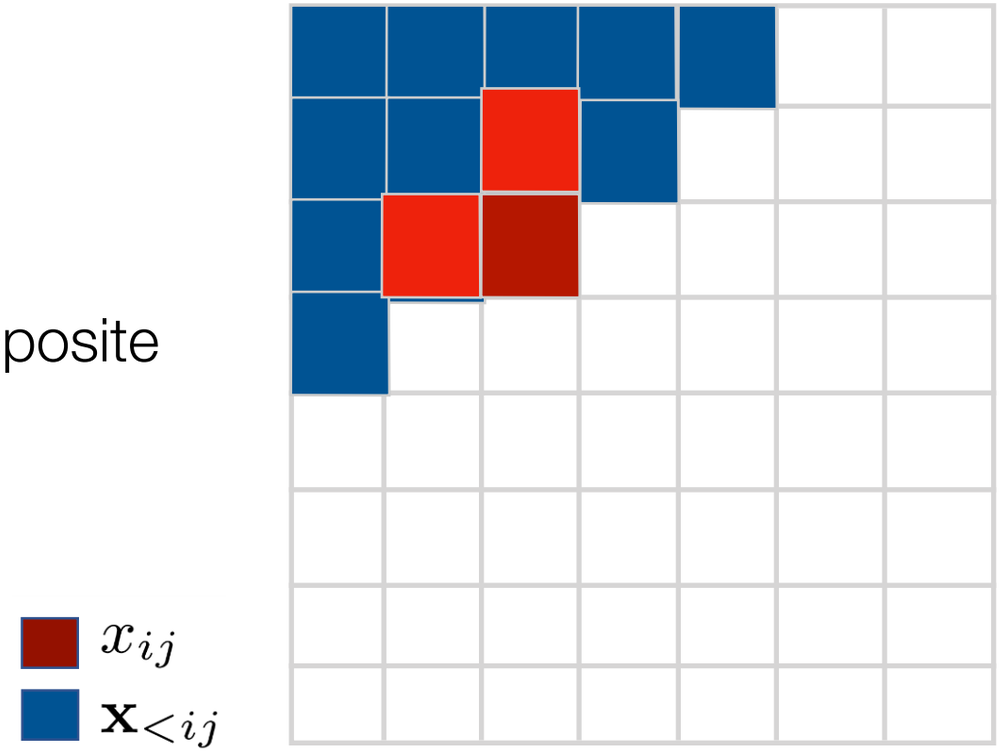

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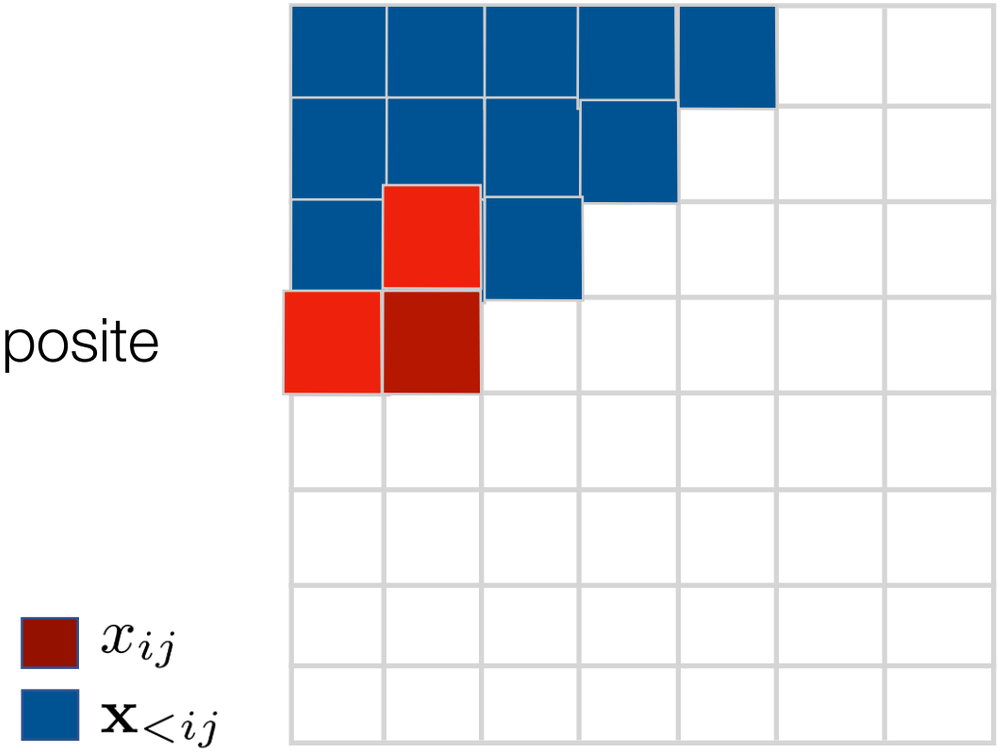

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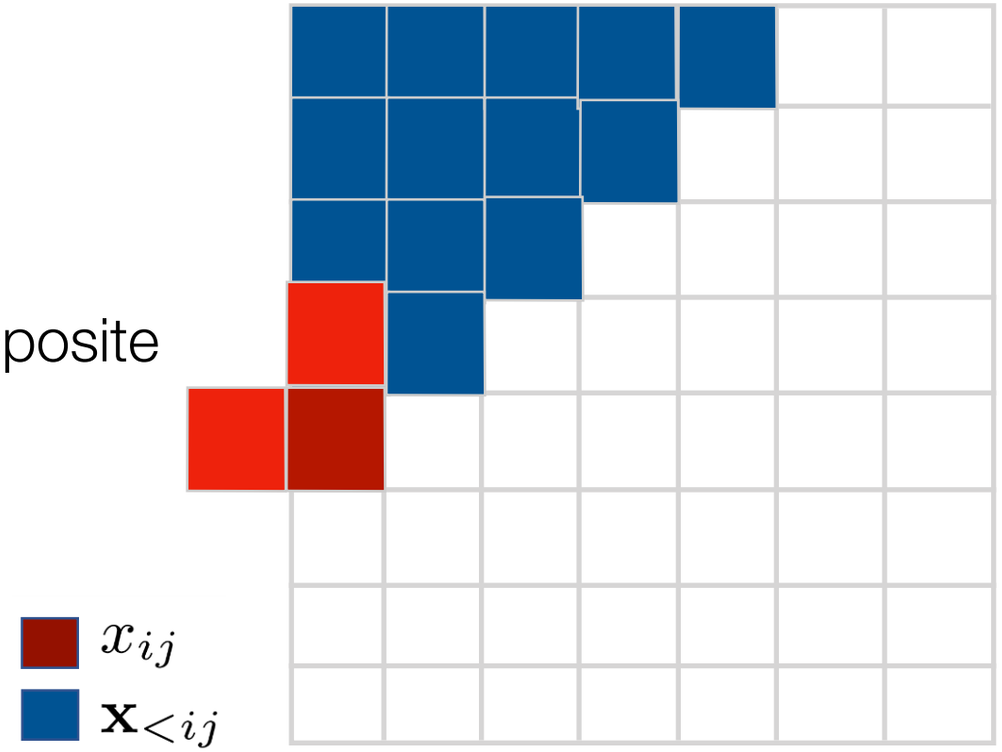

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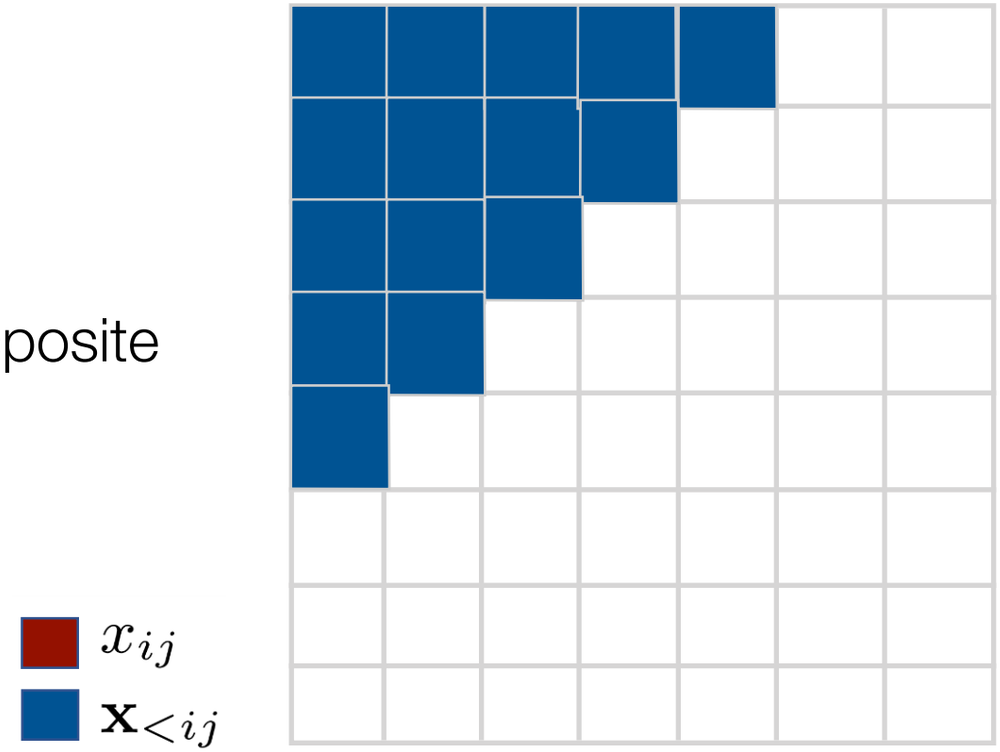

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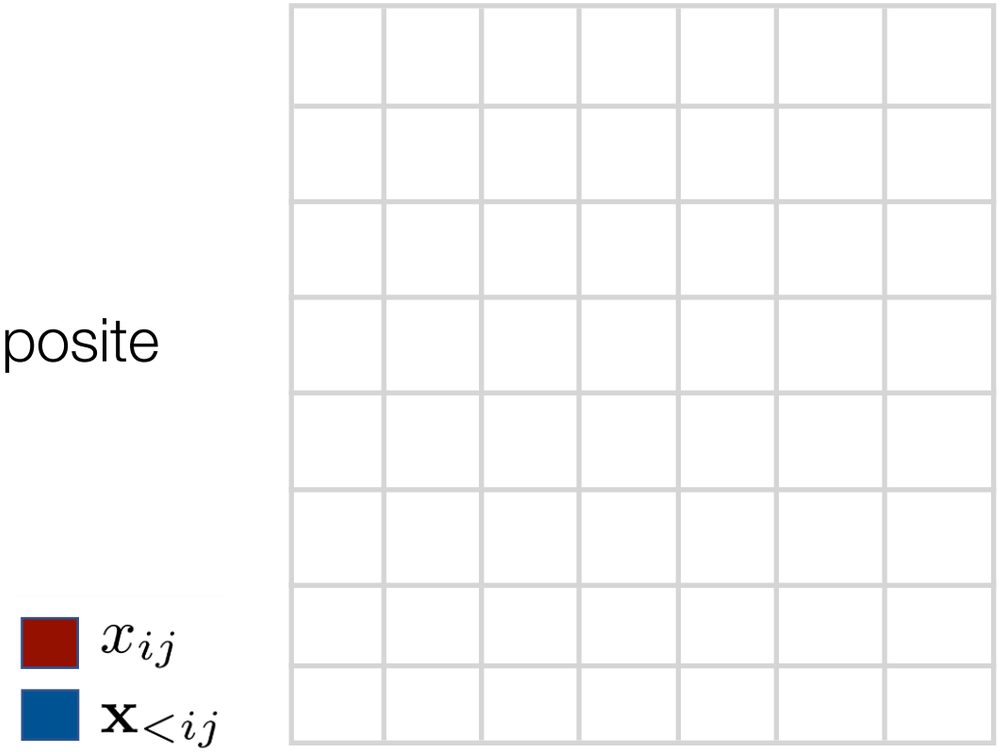

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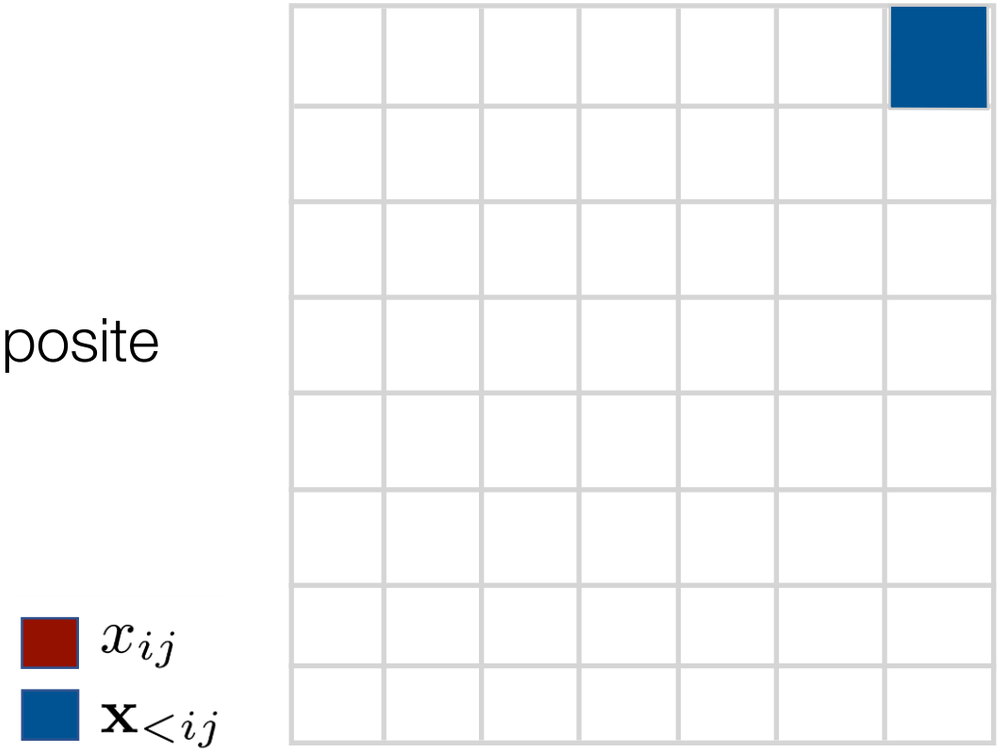

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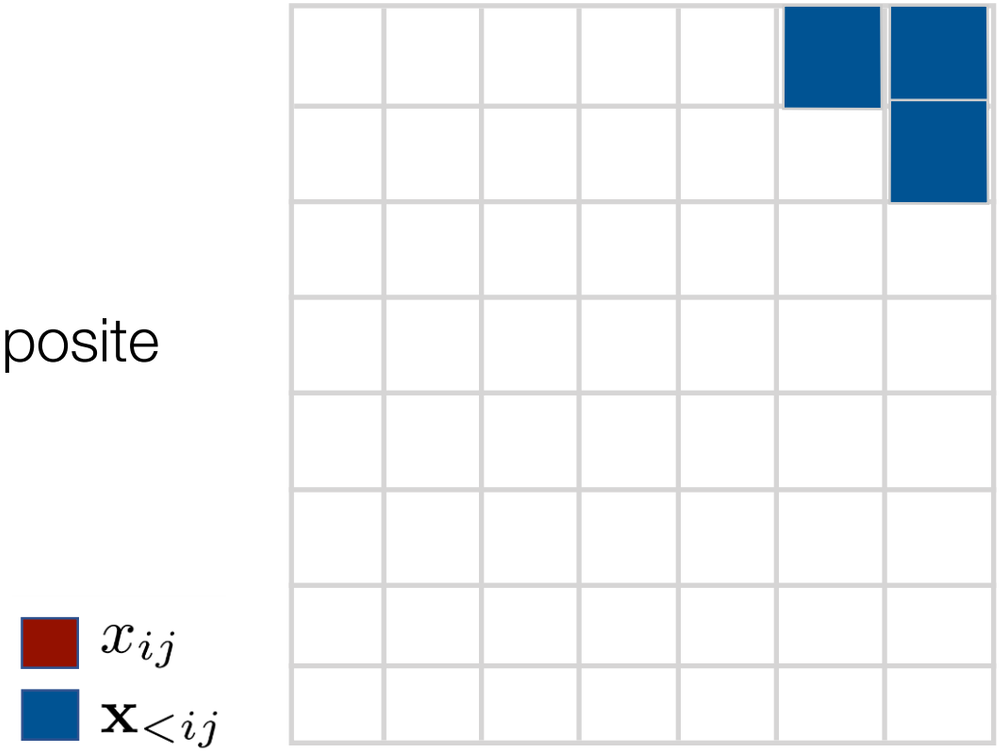

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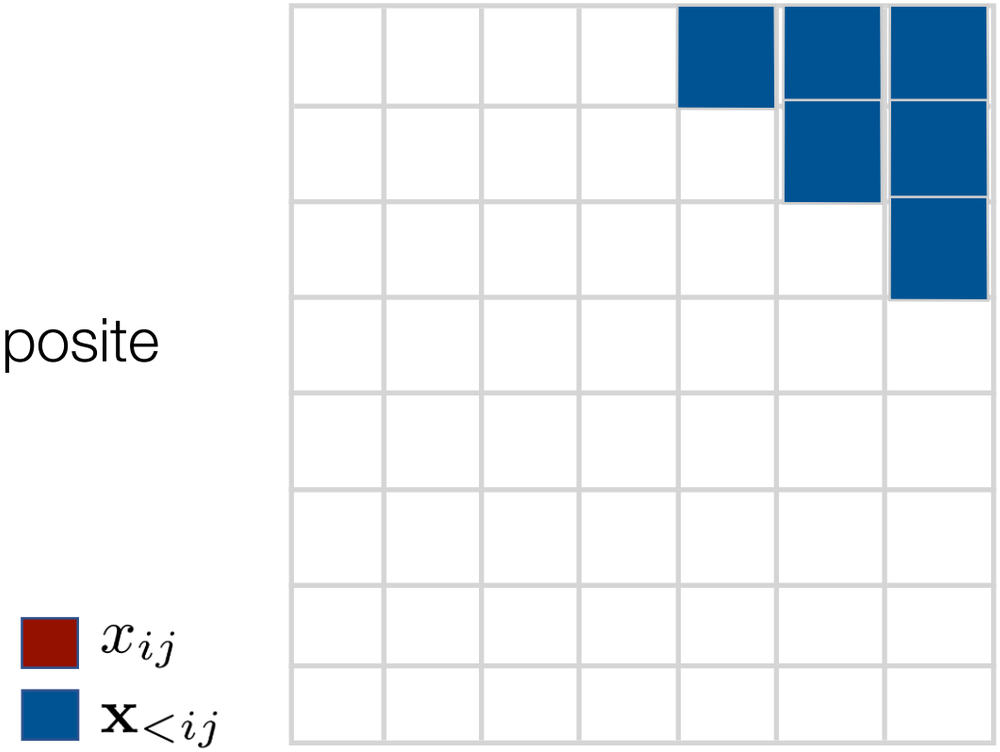

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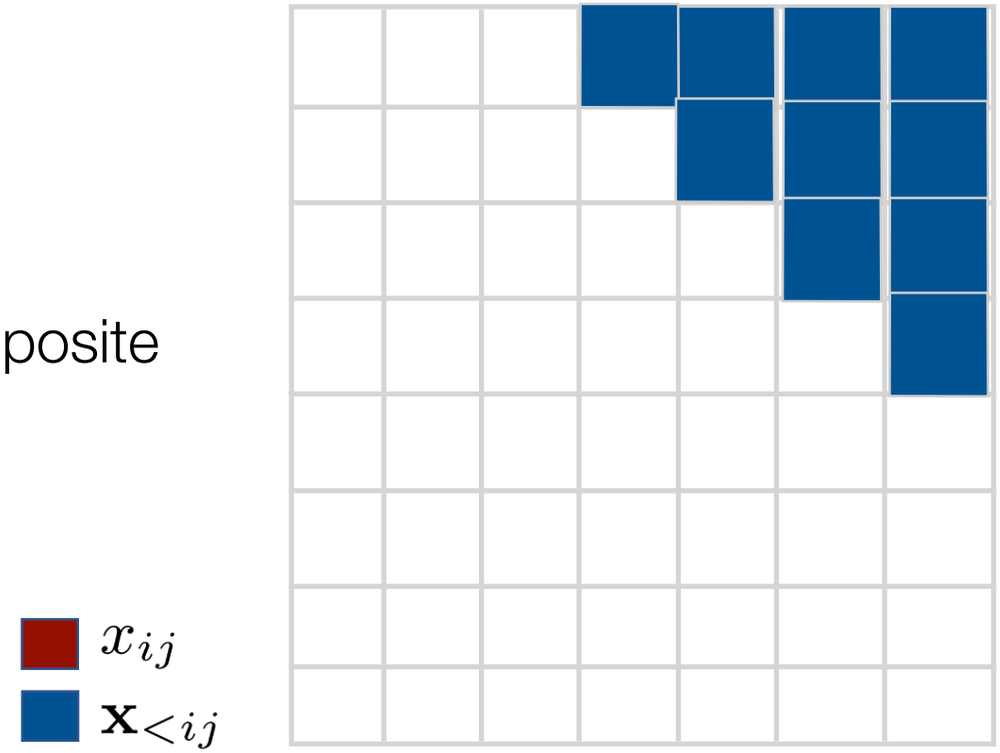

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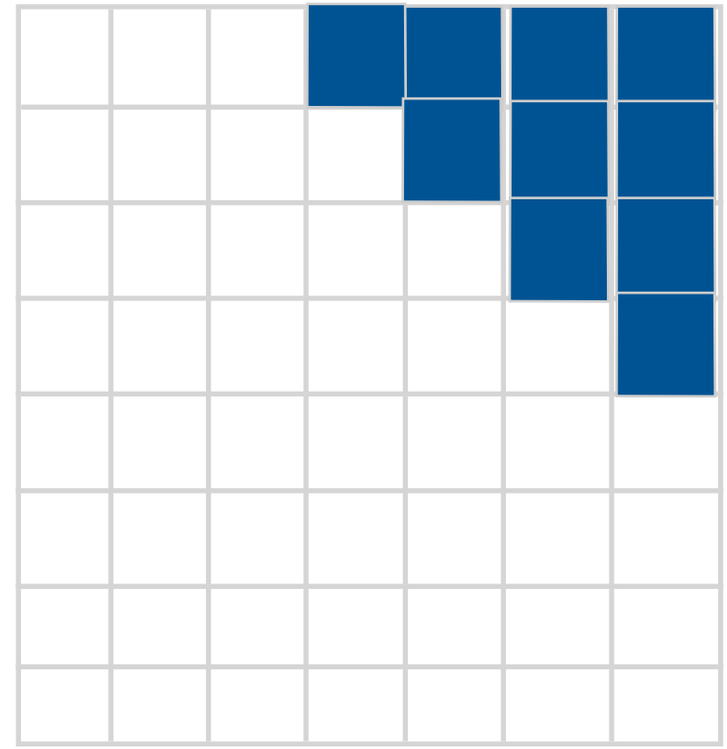

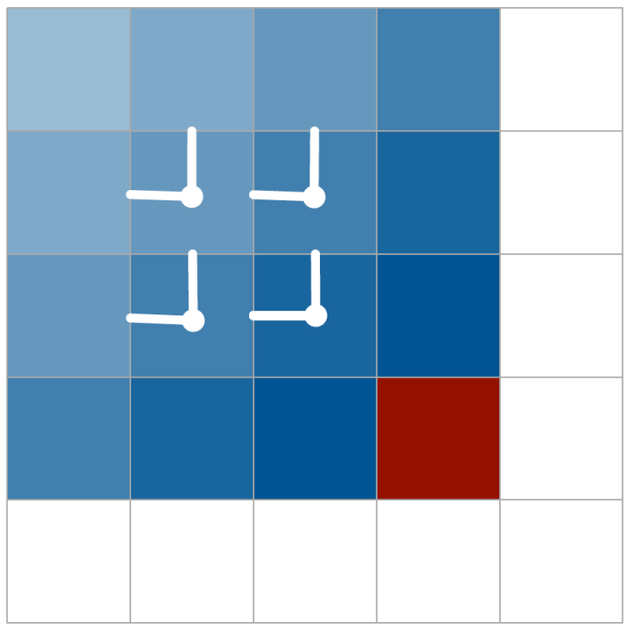

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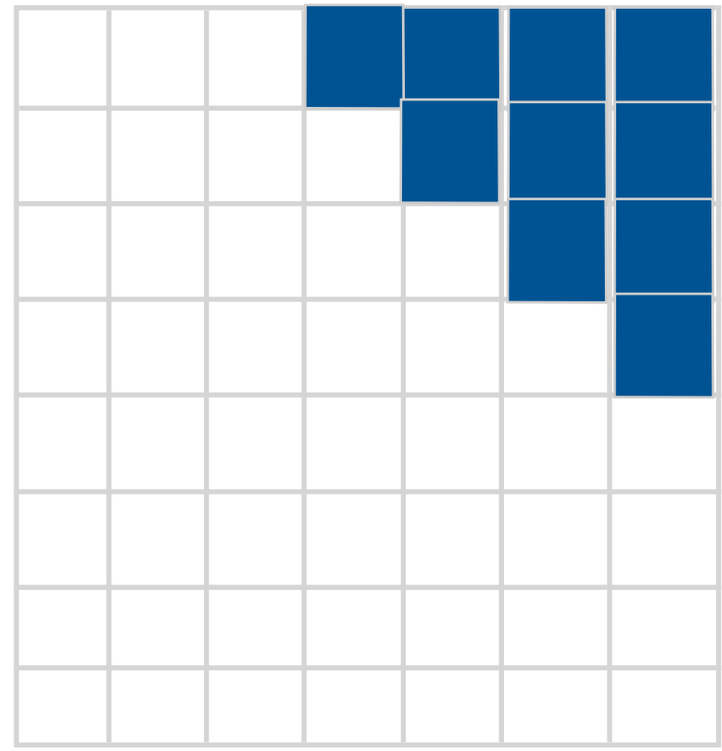

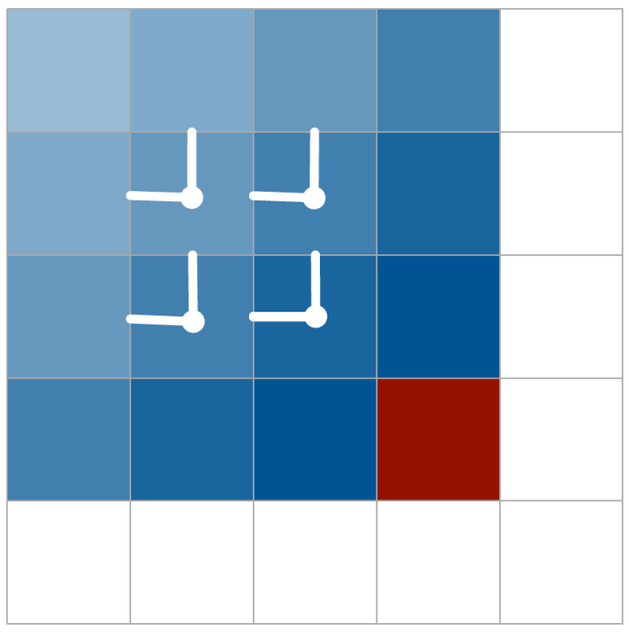

this diagonal convolution
computationally difficult

Trick: Skew pixels by one

Van den Oord et al.,ICML2016
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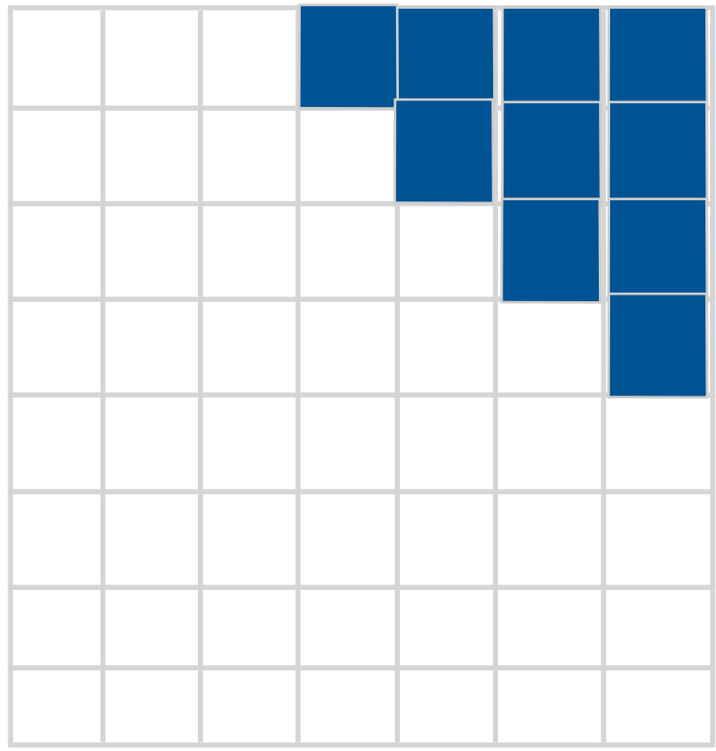

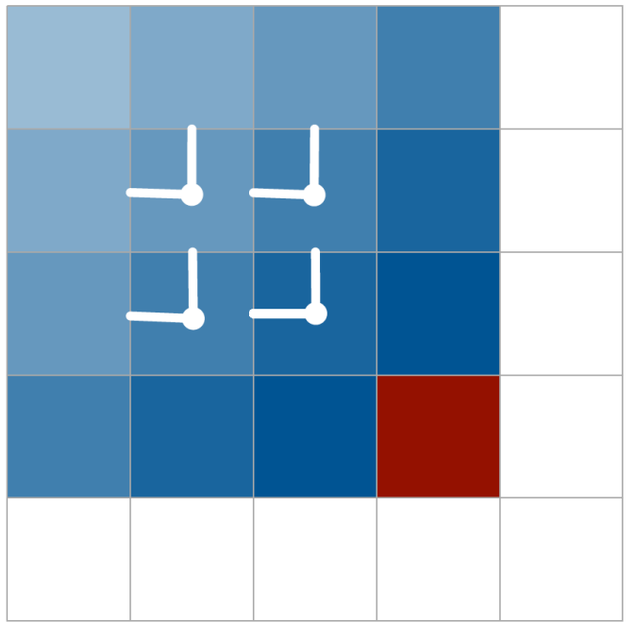

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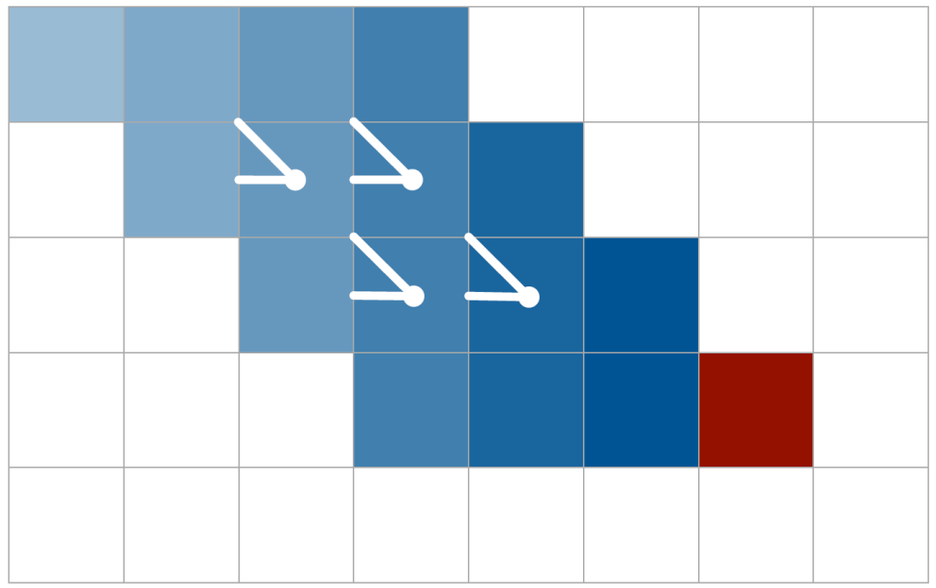
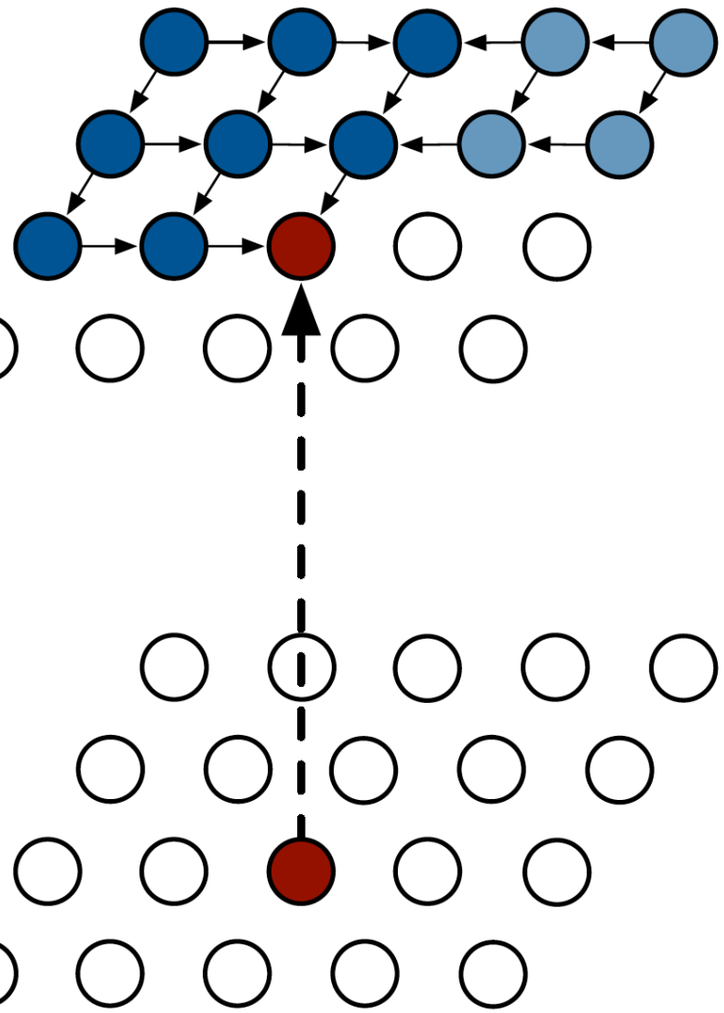


Van den Oord et al.,ICML2016
"Conditional image generation with PixelCNNdecoders"

Pixel RNN: pros and cons of diagonal biLSTM

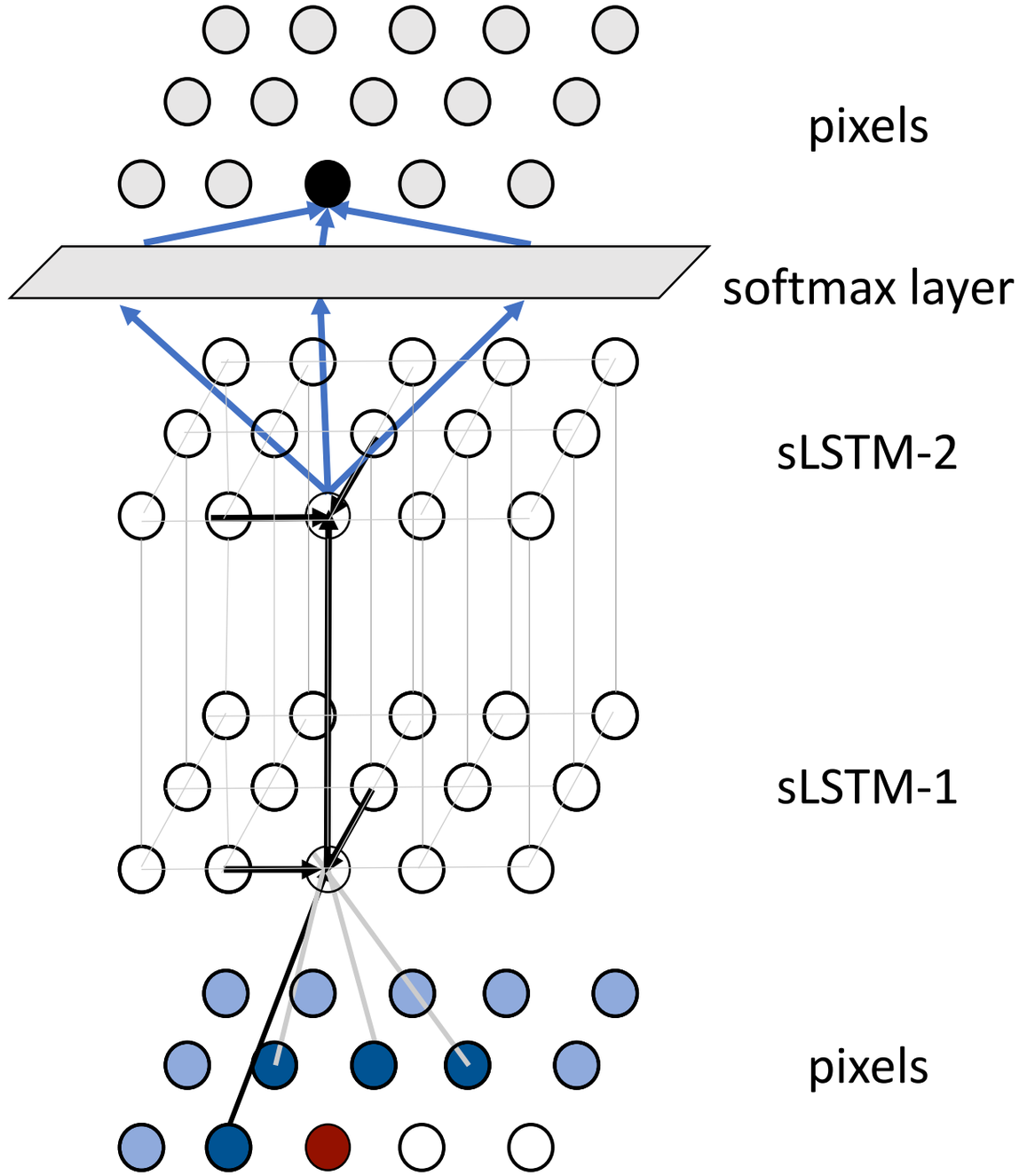
- Compute state for entire row
- Global receptive field => use of all available context



Van den Oord et al.,ICMLS2016
"Conditional image generation with PixelCNNdecoders"

Pixel recurrent neural networks (Pixel-RNN) (ICML 2016)

- Spatial LSTMs
- Treat pixels as discrete variables
- To estimate a pixel value, do classification in every channel (256 classes indicating pixel values 0-255)
- Implemented with a final softmax layer



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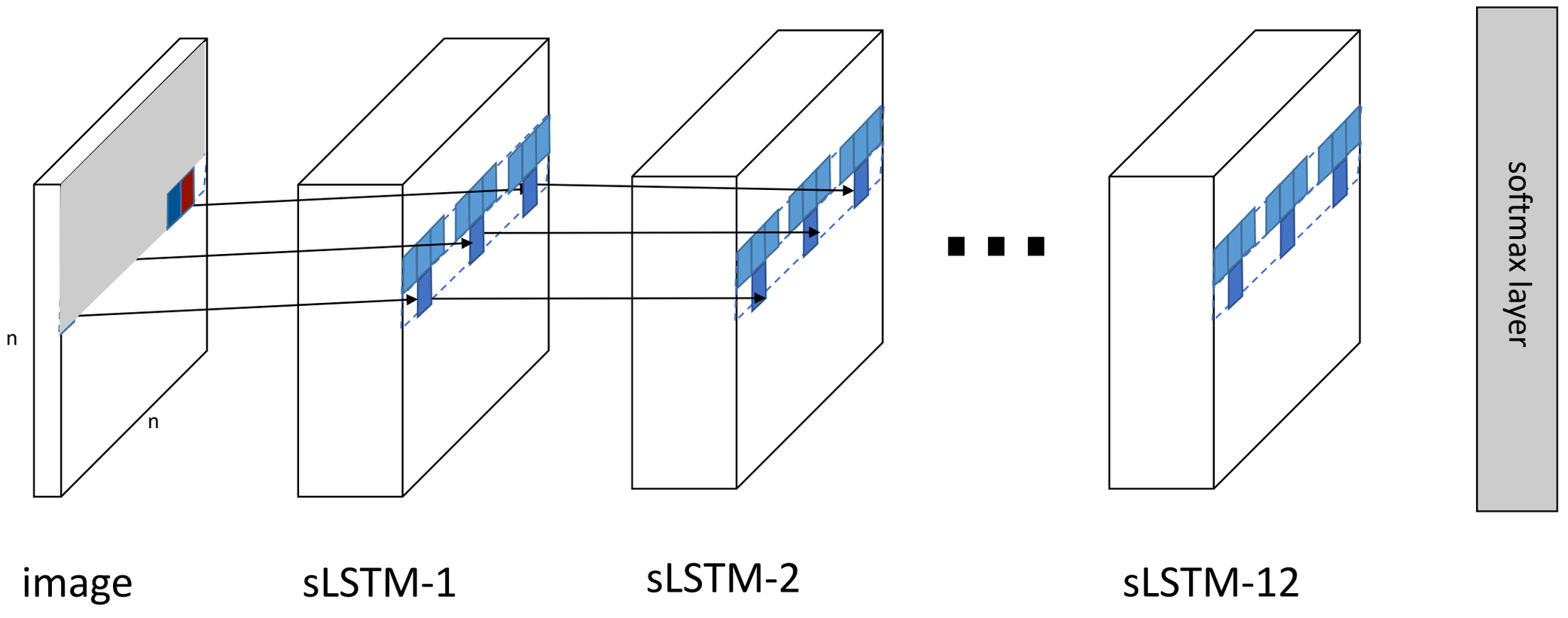

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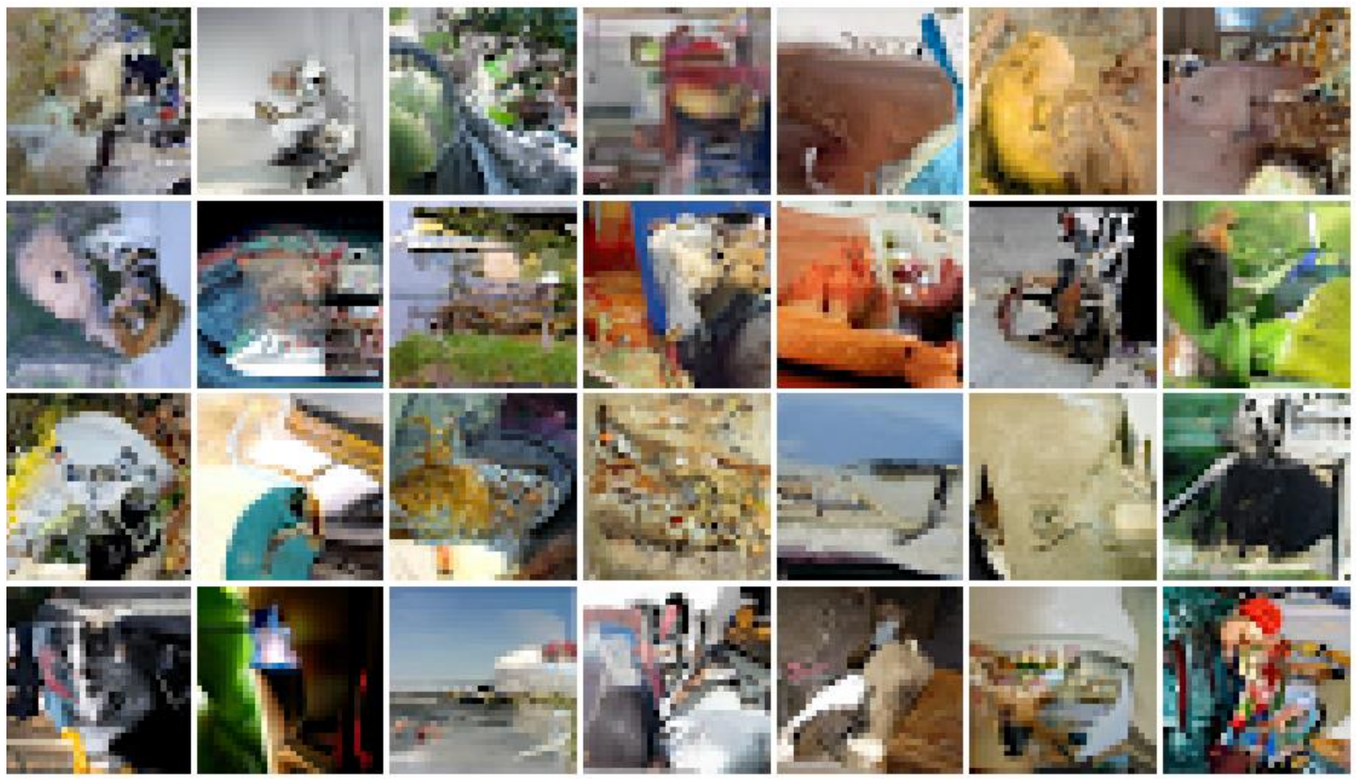
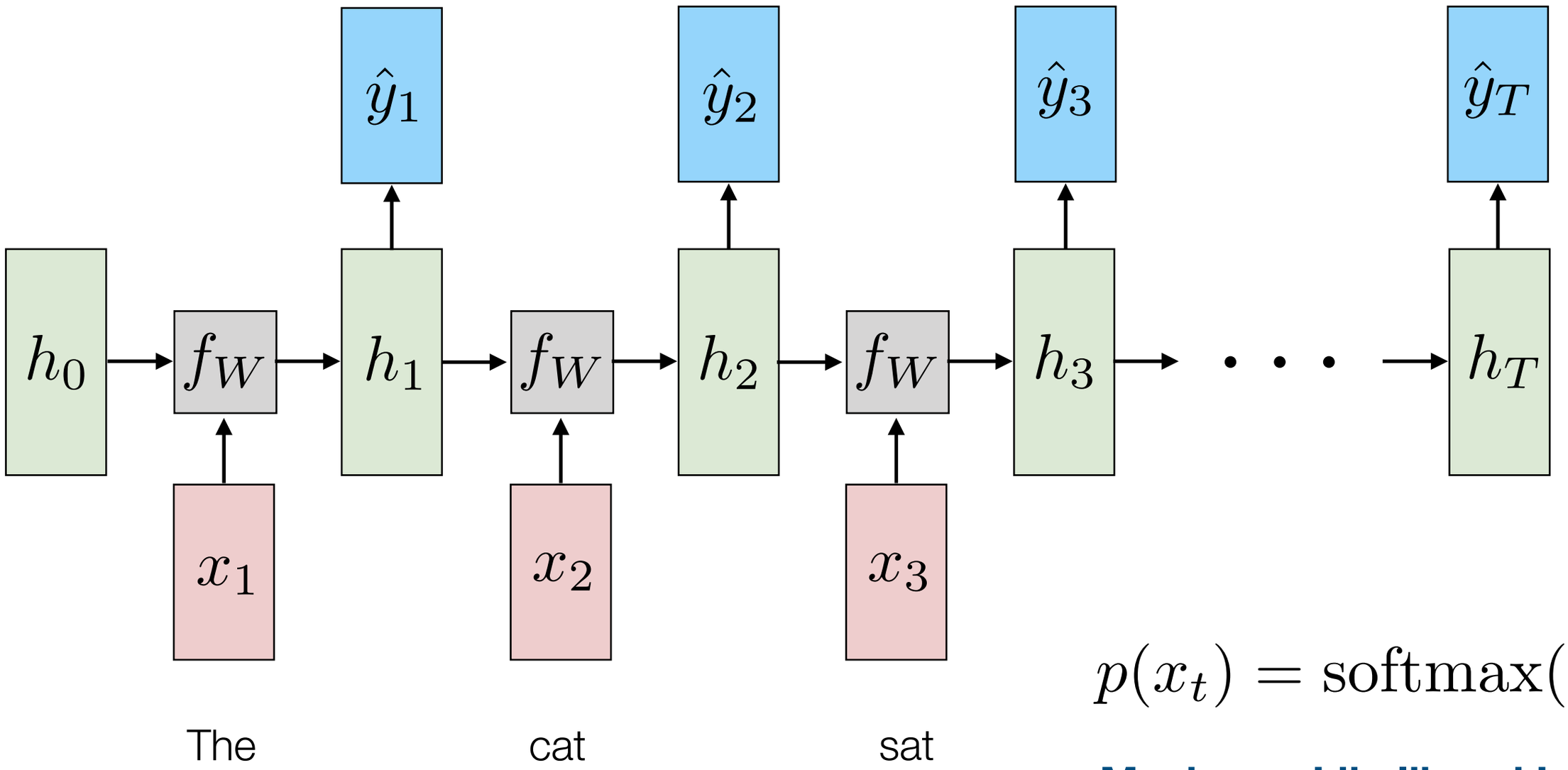


Figure: 32x32 ImageNet results from Diagonal BiLSTM model.

Van den Oord et al.,ICML2016
"Conditional image generation with PixelCNNdecoders"

Recurrent Neural Language Model (Training)

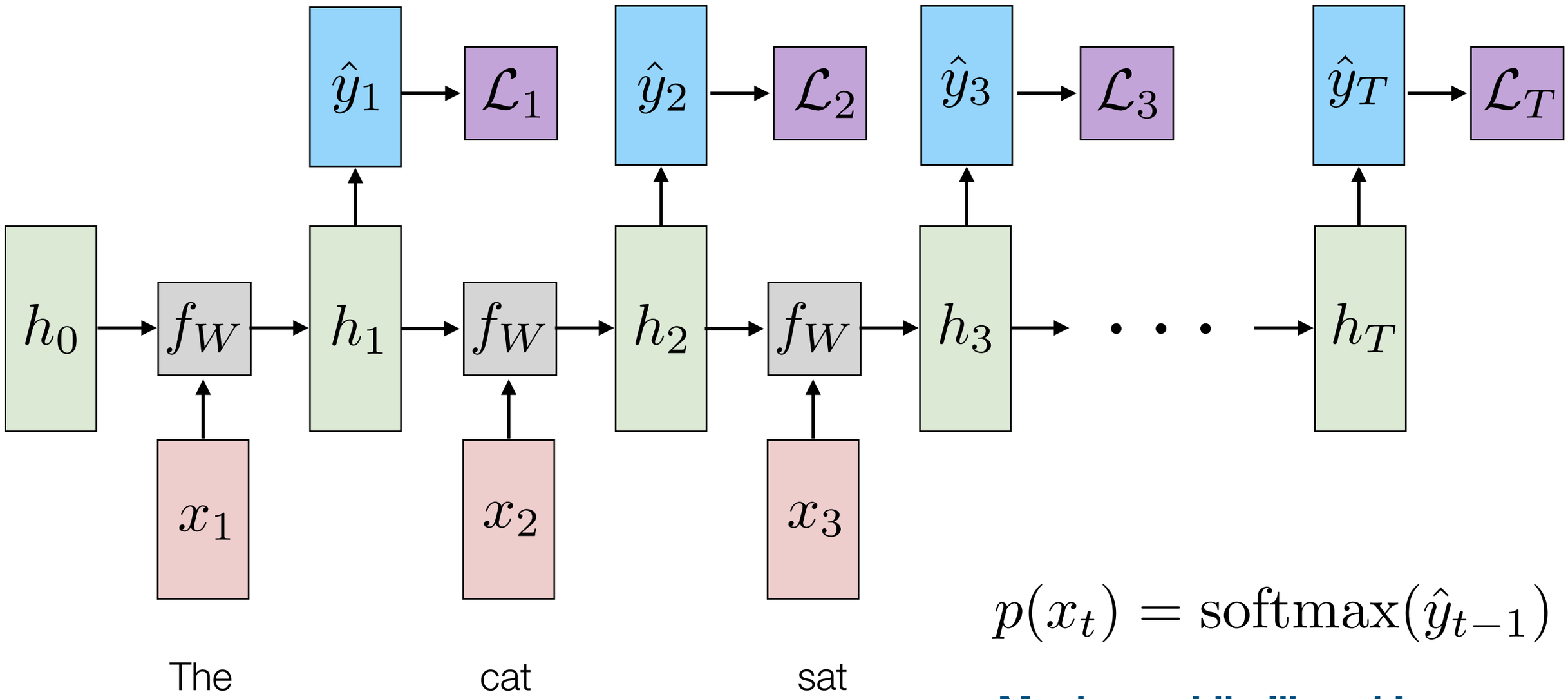


$$p(x_t) = \text{softmax}(\hat{y}_{t-1})$$

Maximum Likelihood Loss

Tomas Mikolov, Martin Karafiat, Lukas Burget, Jan Cernocky, and Sanjeev Khudanpur. 2010. Recurrent Neural Network Based Language Model. InProceedings of INTERSPEECH

Recurrent Neural Language Model (Training)



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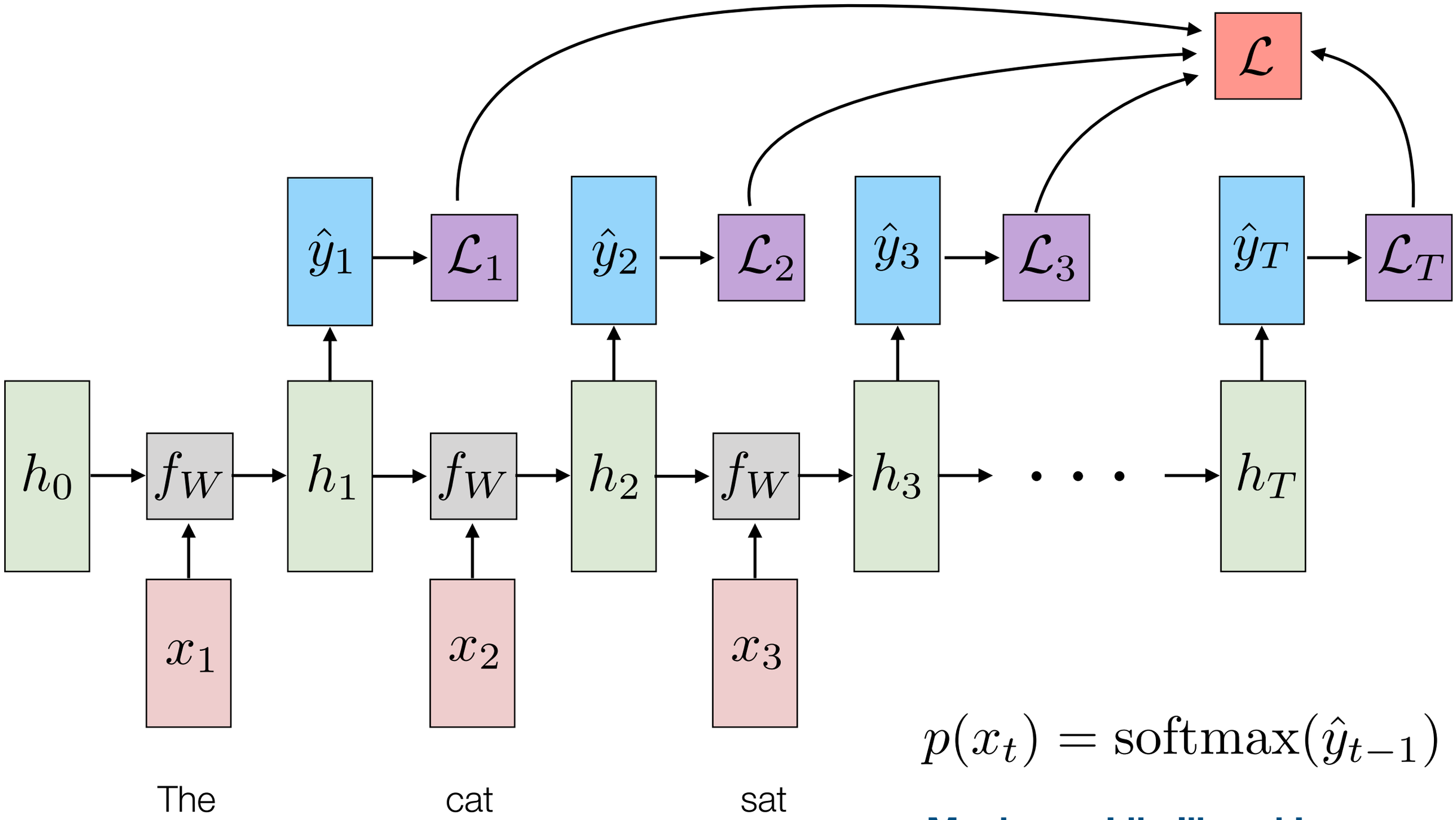
Recurrent Neural Language Model (Training)



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Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution

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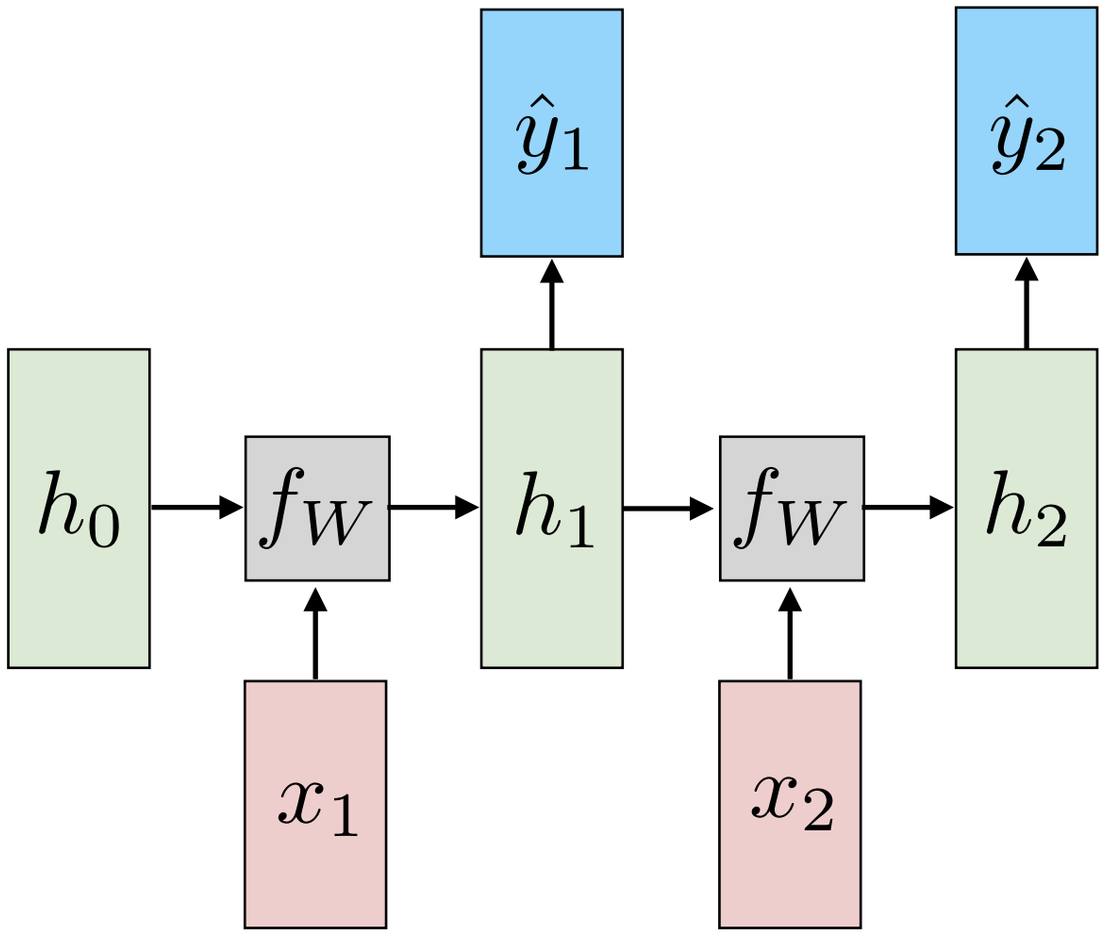
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution



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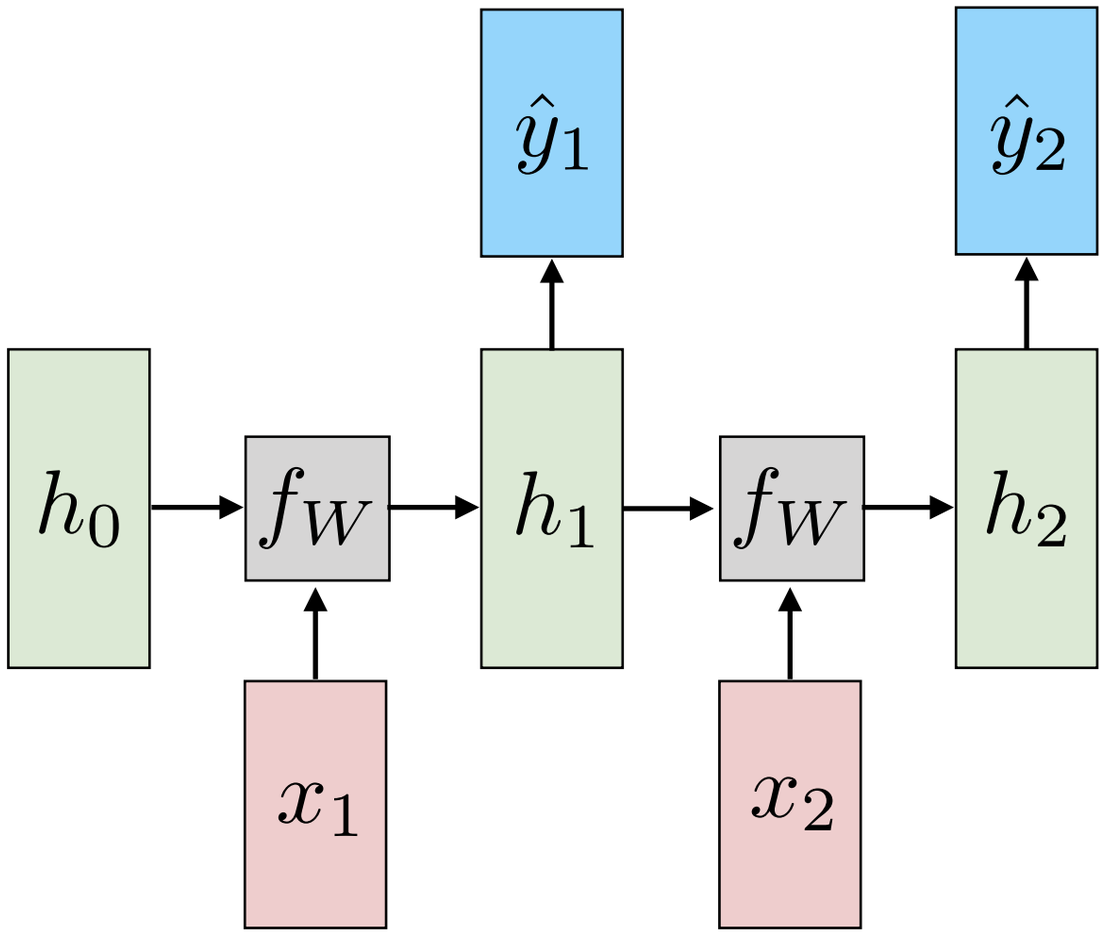
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution



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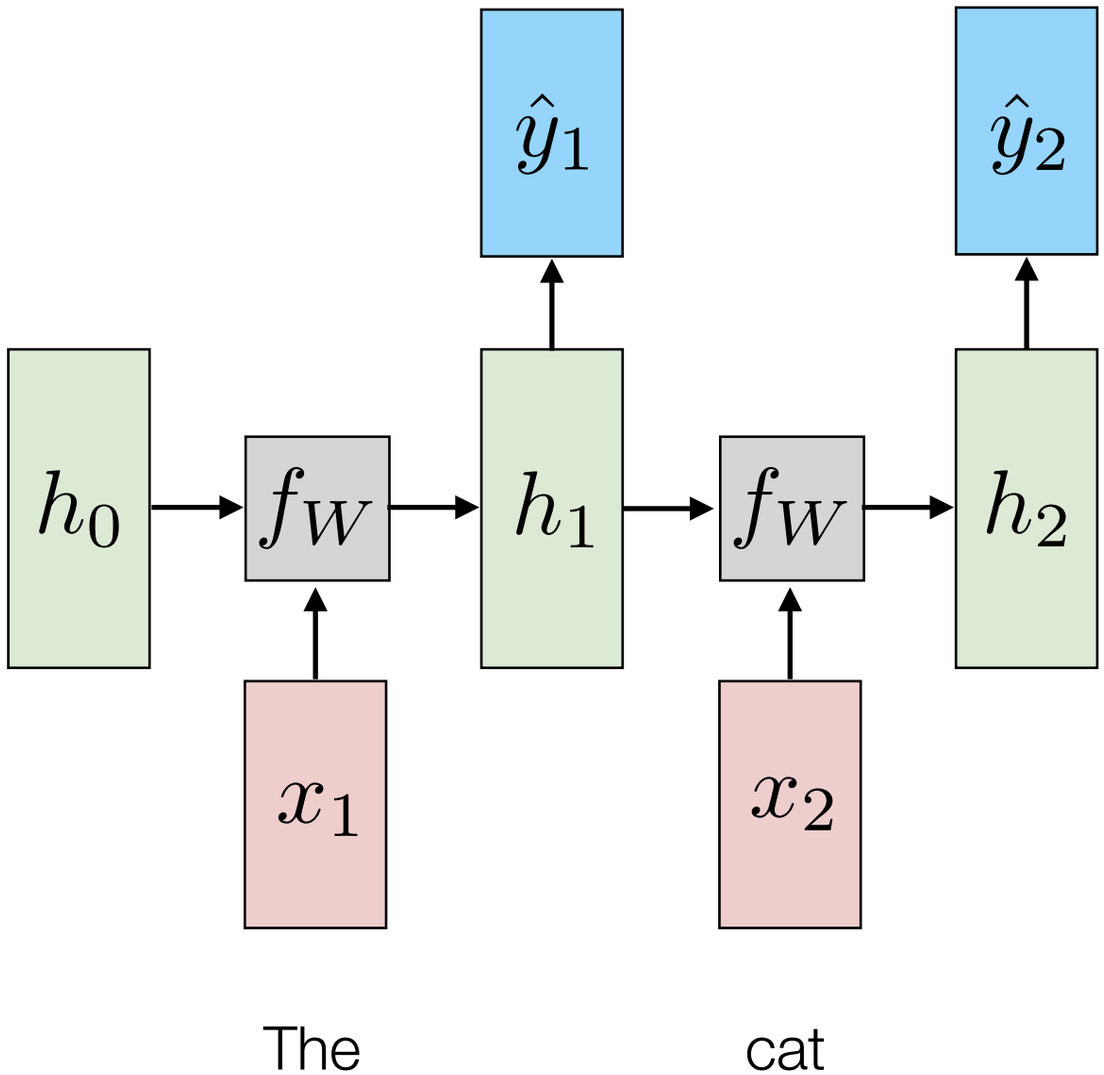

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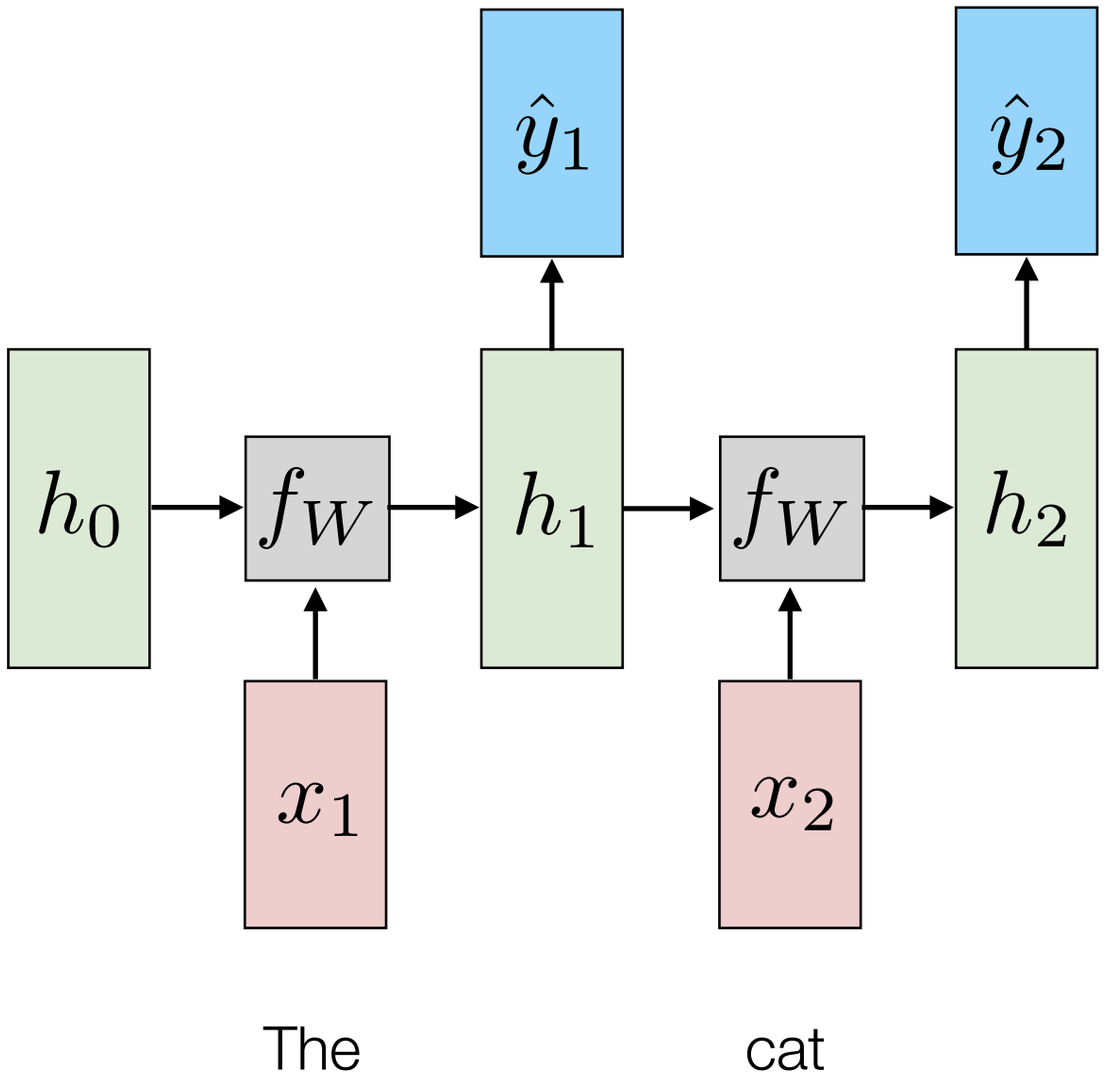
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution

$$\tilde{x}_3 \sim p(x_3) = \text{softmax}(\hat{y}_2)$$



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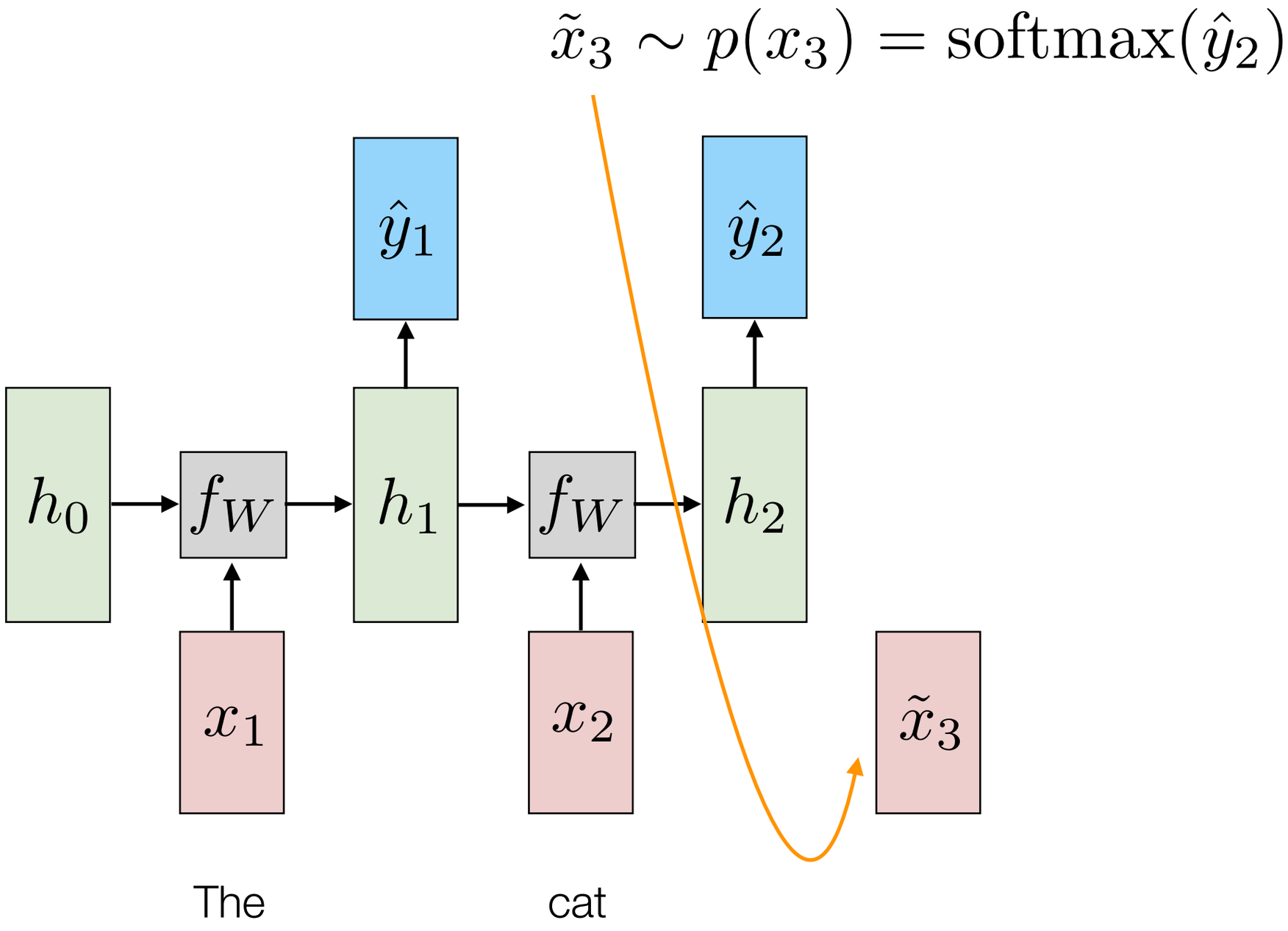
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution



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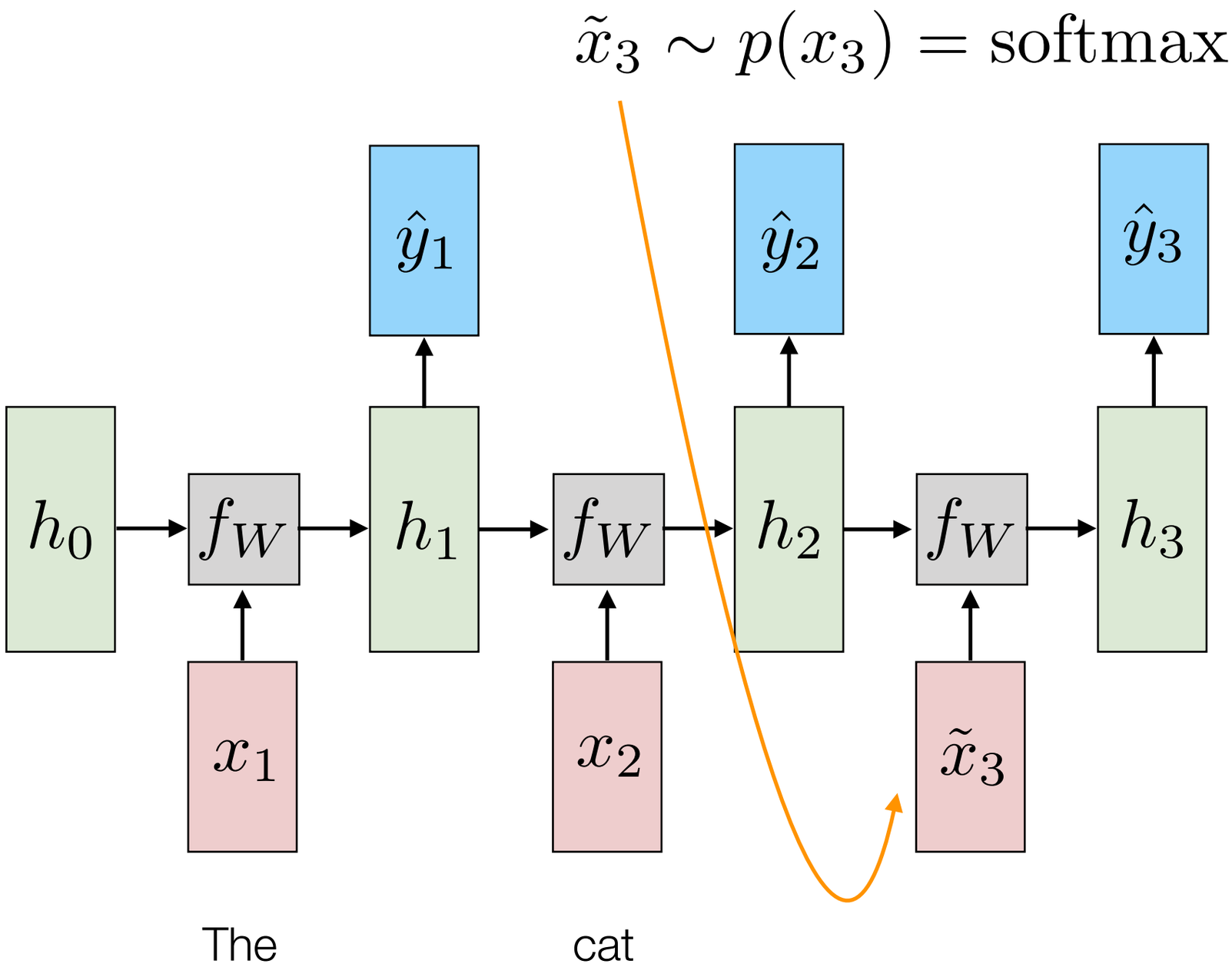
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution



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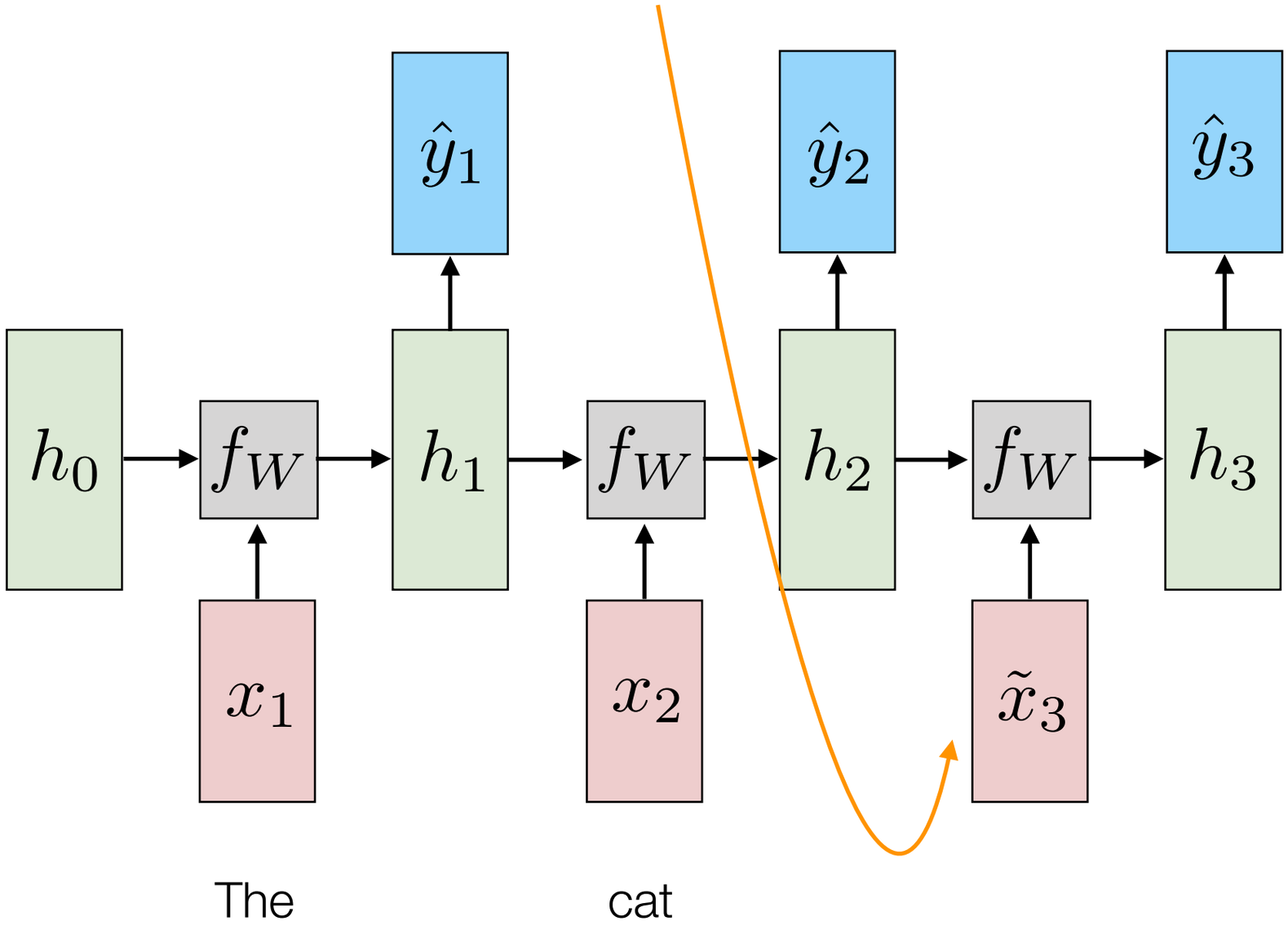
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution



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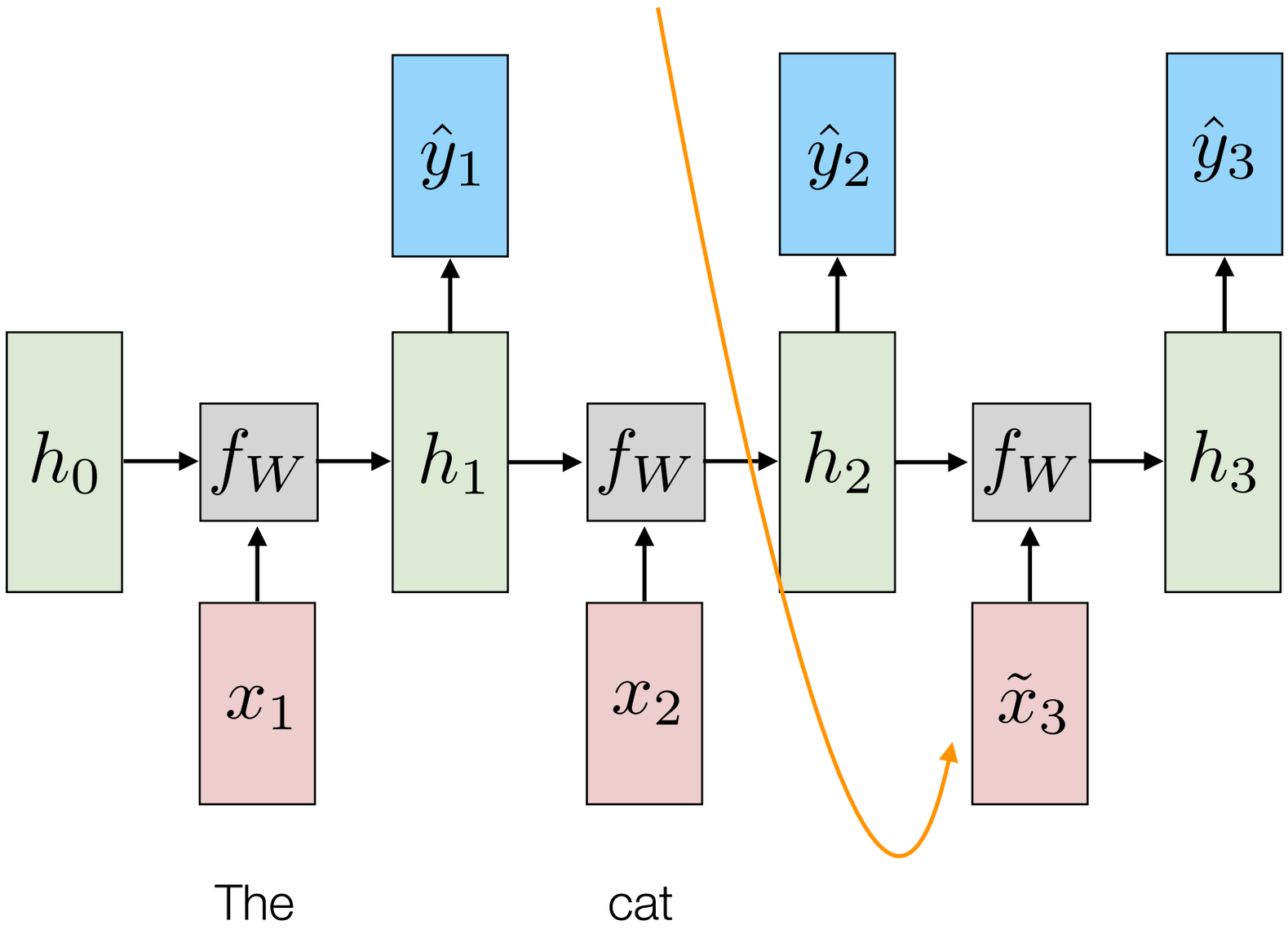
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution

$$\tilde{x}_4 \sim p(x_4) = \text{softmax}(\hat{y}_3)$$



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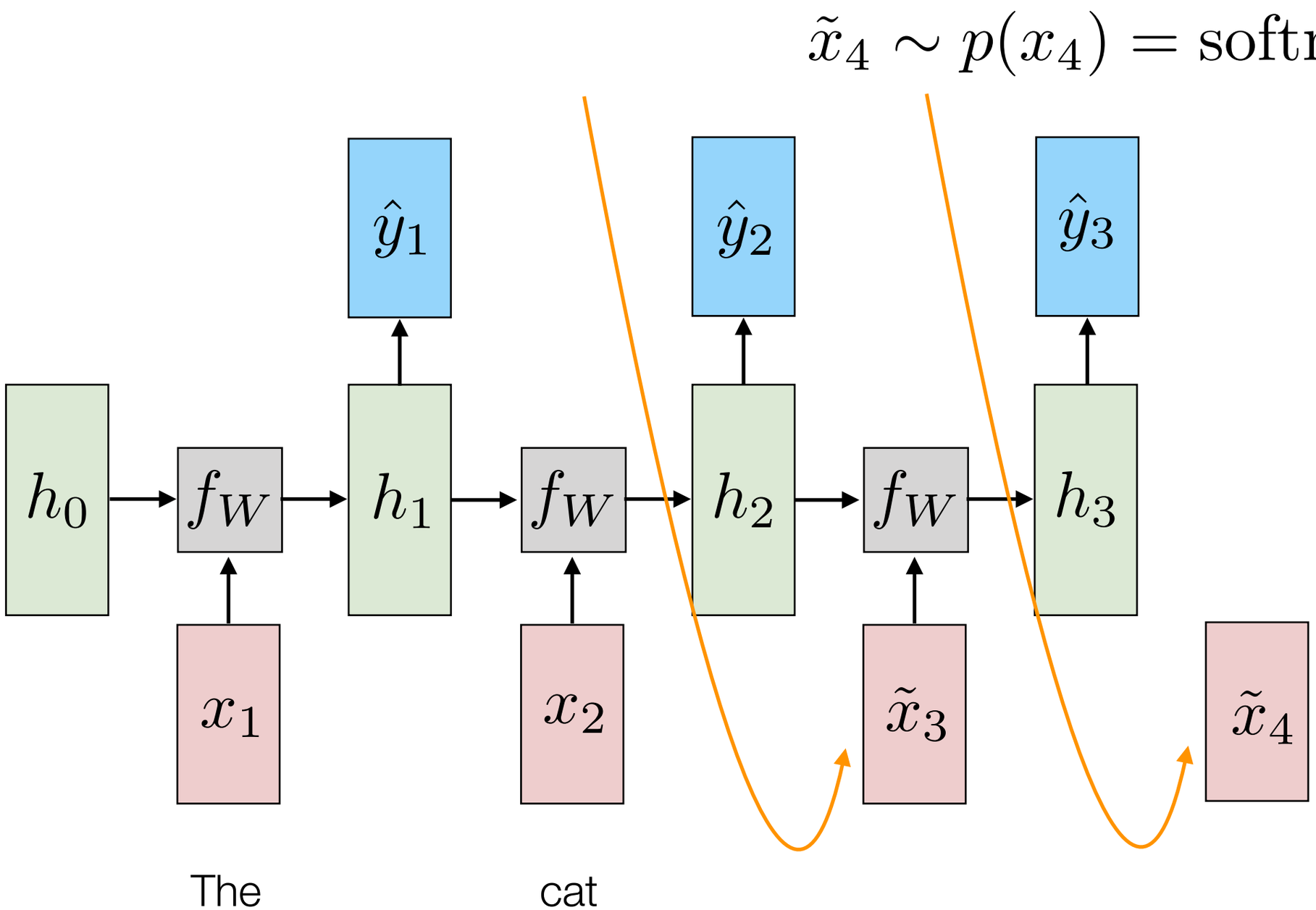
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution



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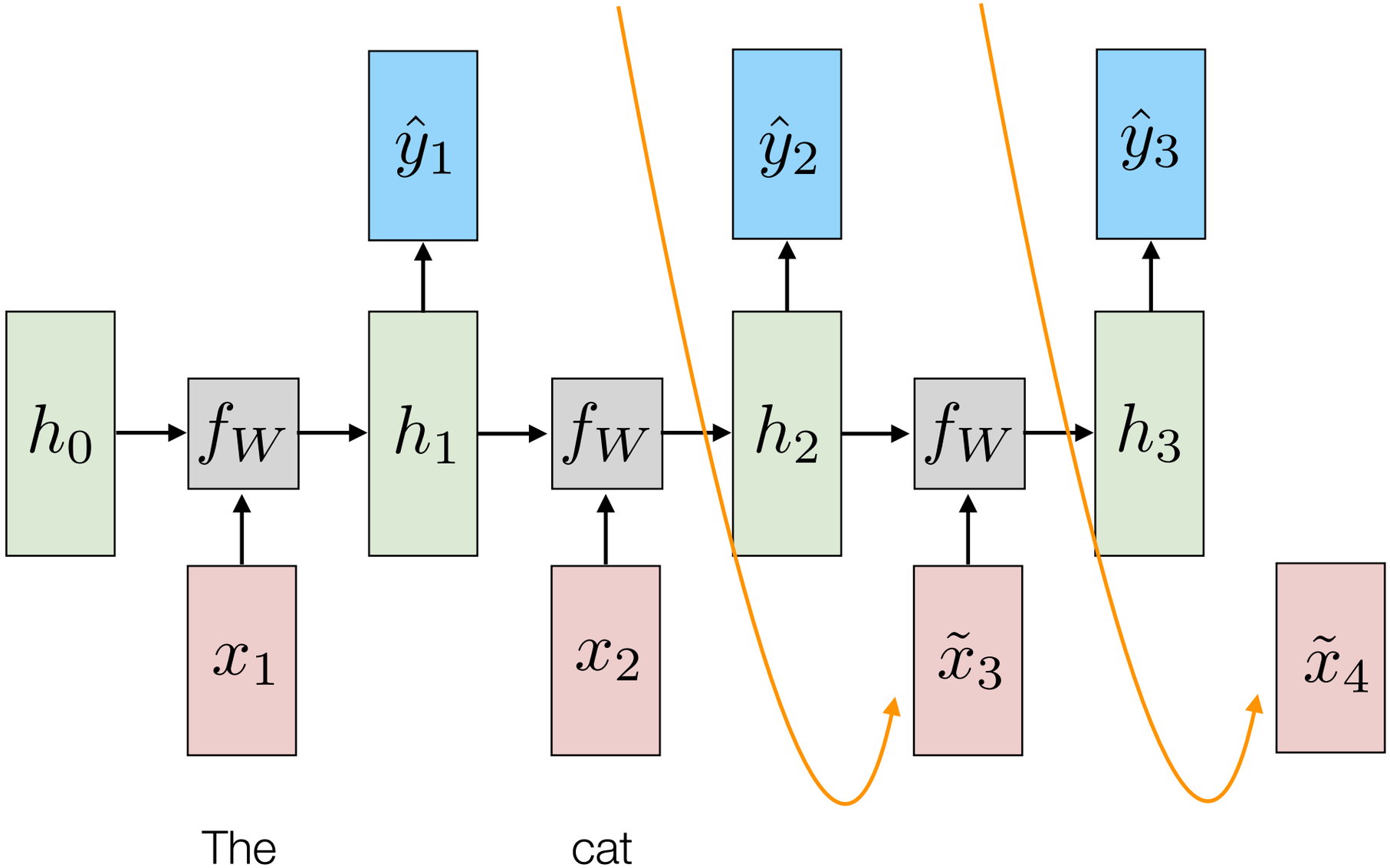
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution



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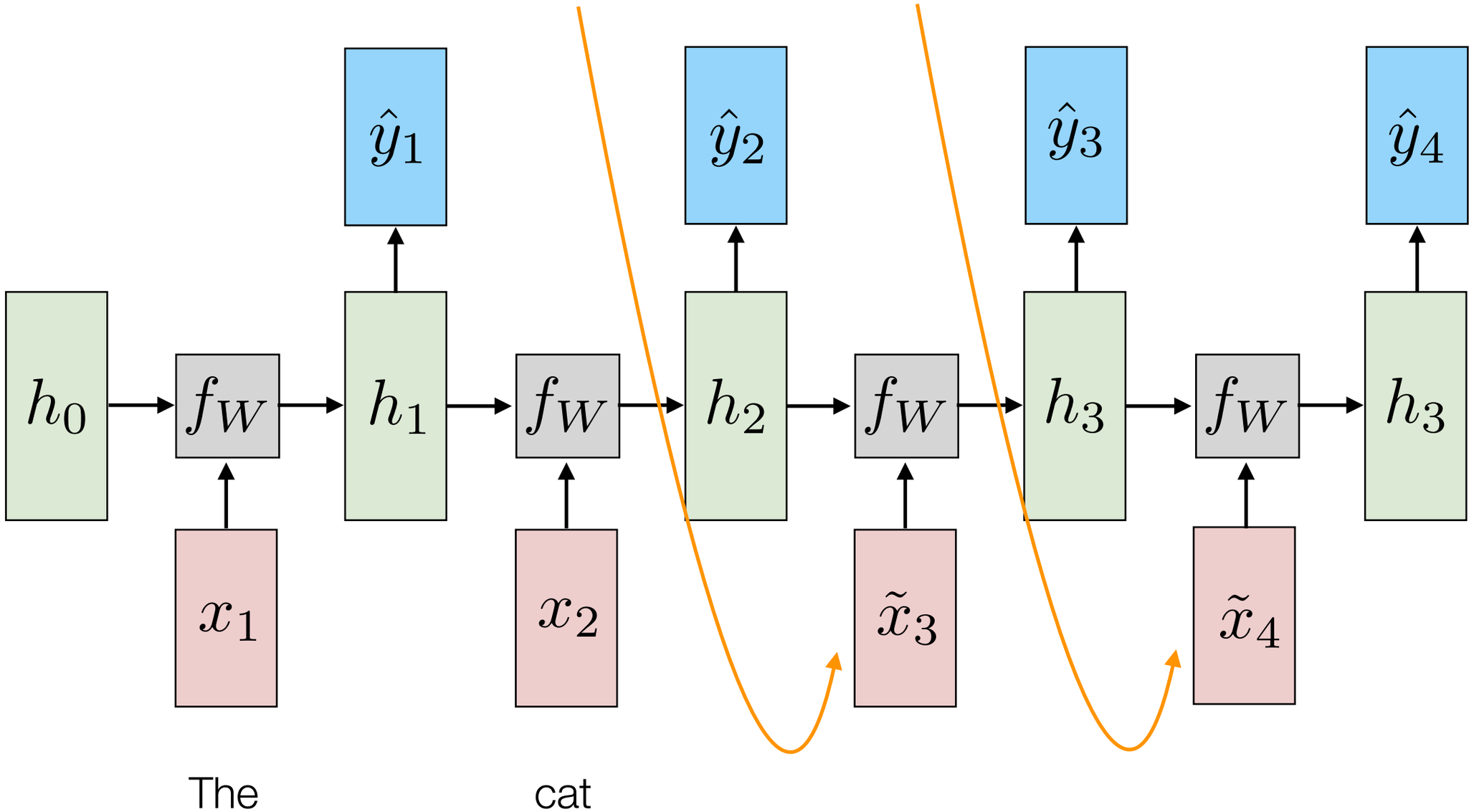
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution



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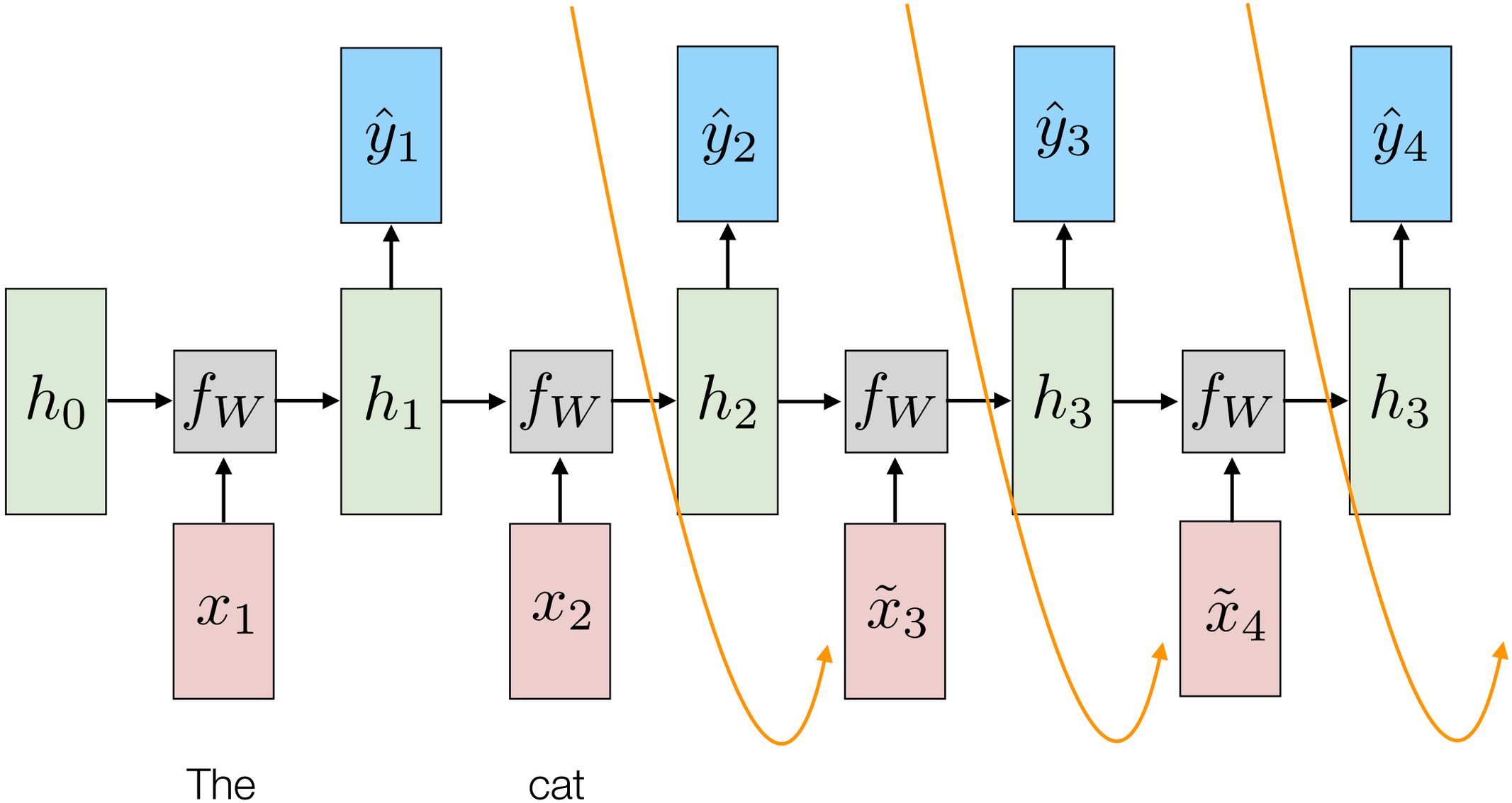
Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution

$$\tilde{x}_5 \sim p(x_5) = \text{softmax}(\hat{y}_4)$$



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Recurrent Neural Language Model (Test)

Sequentially Sample from Output Distribution

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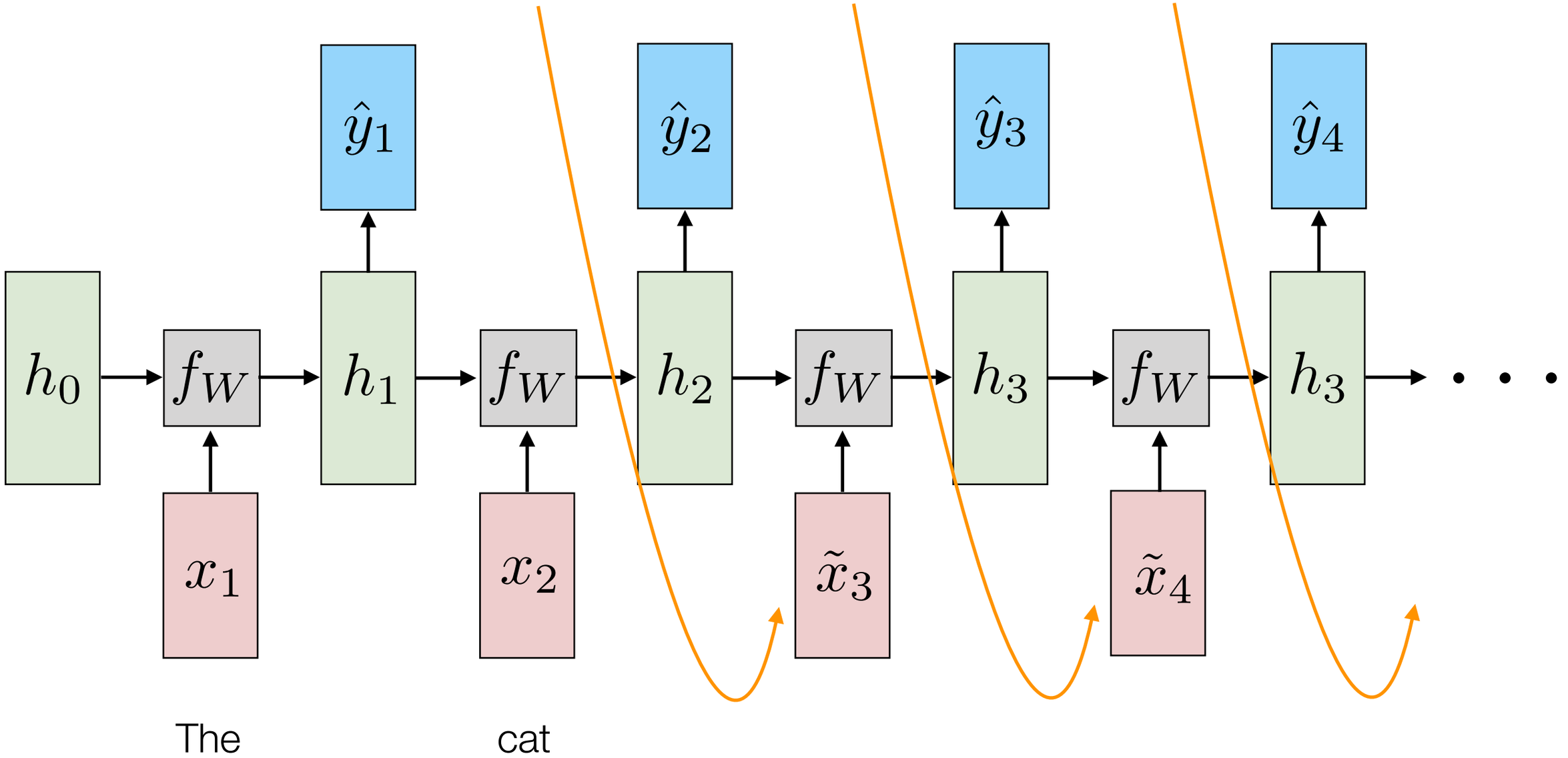

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Lab: RNNs for Time Series Prediction

Pablo M. Olmos pamartin@ing.uc3m.es

In this notebook, you will deploy a simple probabilistic model using a RNN to infer the probability distribution of a time-series. Given a set of signals $\mathcal{D} = \{X^1, X^2, \dots, X^N\}$ where

$$X^i = [X_0^i, X_1^i, \dots, X_T^i]$$

is the i-th signal, we will train a probabilistic model of the form

$$p(X|X_0) = \prod_{t=1}^{T} p(X_t|\mathbf{X}_{0:t-1})$$

where $\mathbf{X}_{0:t-1} = [X_0, X_1, \dots, X_{t-1}]$ and every conditional factor $p(X_t|\mathbf{X}_{0:t-1})$ corresponds to a Gaussian distribution with mean and variance obtained **from the state of a RNN** with input X_{t-1}. The RNN state $\mathbf{h}_{t-1}$ is a projection of the the signal up to time $t-2$, i.e, $\mathbf{X}_{0:t-2}$:

$$p(X_t|X_{t-1}, \mathbf{X}_{0:t-2}) = \mathcal{N}\left(f_{RNN}(X_{t-1}, \mathbf{h}_{t-1}), \sigma^2 + g_{RNN}(X_{t-1}, \mathbf{h}_{t-1})\right)$$

, where σ^2 is a constant basal variance that we treat as an hiper-parameter. During training, we approximate the expected loss at time t

$$\mathcal{L}_t = \mathbb{E}_{\hat{X}_t \sim p(X_t|X_{t-1}, \mathbf{X}_{0:t-2})}[(X_t - \hat{X}_t)^2]$$

using a single sample of $p(X_t|X_{t-1}, \mathbf{X}_{0:t-2})$. The global loss is accumulated during the whole signal length:

$$\mathcal{L} = \frac{1}{T}\sum_{t=1}^{T} \mathcal{L}_t$$